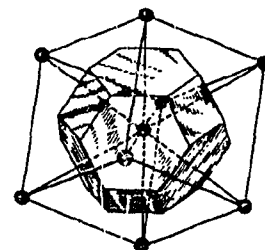


OAK RIDGE NATIONAL LABORATORY
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ORNL/RMIC-14

**DIRECTORY OF
CRYSTAL GROWTH AND SOLID STATE
MATERIALS PRODUCTION AND RESEARCH**

RESEARCH MATERIALS INFORMATION CENTER



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DIRECTORY OF CRYSTAL GROWTH AND SOLID STATE
MATERIALS PRODUCTION AND RESEARCH

Compiled by

T. F. Connolly
George C. Battle
Anne M. Keesee

Research Materials Information Center
Solid State Division

MARCH 1979

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The Research Materials Information Center, established in 1963 as part of the ORNL Solid State Division, provides information, on an exchange basis, on the properties, preparation, and availability of ultrapure inorganic research materials. In general, coverage is of all solid-state materials science except the manufacture and properties of devices and "engineering" properties of the materials covered. A sampling of the variety of specific subjects or materials covered includes ferroelectricity, superconductivity, dielectrics, semiconductors, metals, ceramics, crystal growth, crystal structure, photovoltaics, fast ionic conductors, methods of measurement, refractive indices, laser windows, specific heat, thermal expansion, impurity assay and effects, characterization methods, and materials for energy applications.

Technical inquiries and requests for further information on the Center should be addressed to:

T. F. Connolly
Research Materials Information Center
Oak Ridge National Laboratory
P. O. Box X, Building 2000
Oak Ridge, Tennessee 37830
Telephone (615) 574-5466
FTS 624-5466

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This directory lists only those who returned questionnaires distributed by the Research Materials Information Center during 1978. The directory includes, in addition to crystal growers, those preparing starting materials for crystal growth and ultrapure noncrystalline research specimens. It also includes responses from those characterizing, or otherwise studying, the properties of materials provided by others.

The international coverage of the directory is limited to those countries not covered by other recently published directories and is indicated in the table of contents immediately preceding the index of materials. Other countries are included in the following publications:

Sources of Single Crystals in the United Kingdom and Scandinavia, 1978 (for United Kingdom, Denmark, Norway, and Sweden).
Available from Mrs. B. M. R. Wanklyn, Clarendon Laboratory,
Parks Road, Oxford OX1 3PU, United Kingdom.

Information über Kristallzüchtung, 1976 (for Germany, Holland, and Switzerland).
Available from Prof. Dr. R. Nitsche, Kristallographisches
Institut der Universität Freiburg, 7800 Freiburg i. Br,
Hebelstrasse 25, West Germany and Prof. A. Raüber, Institut
für Angewandte Festkörperphysik der Fraunhofer-Gesellschaft,
7800 Freiburg i. Br, Eckerstrasse 4, West Germany.

Documentation sur les Synthèses Cristallines, 1976 (for Belgium, Spain, France, and Italy).
Available from Prof. A. M. Vergnoux, 42 rue St. Claire,
87000 Limoges, France.

Although most of the information in the OTAN/NATO list is also in this directory or one of the documents above, the list contains a very useful amount of availability information not found in the other compilations:

International List of Available Electronic Materials, 1976.
Available from Defence Research Group, Panel on Physics and
Electronics, North Atlantic Council, OTAN/NATO, 1110 Brussels,
Belgium.

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Classes or generic names of materials appear in the index (with a few exceptions) only when so given by the supplier of the information.

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ABBREVIATIONS FOR PREPARATION METHODS

Br	Bridgman
CVD	Chemical Vapor Deposition
Cz	Czochralski
EC	Electrocrystallization
EFG	Edge-Defined Film-Fed Growth
Flu	Flux
FZ	Float Zone
GE	Gas Epitaxy
Hyd	Hydrothermal
LE	Liquid Epitaxy
M	Melt
MBE	Molecular Beam Epitaxy
Sol	Solution
St	Stockbarger
Sub	Sublimation
THM	Traveling Heater Method
Ver	Verneuil
VLS	Vapor-Liquid-Solid
VP	Vapor Phase
Z	Zone Melting

1				
Name(s): M.C. ARST				
Inst. & Address: Consultant- 2601 Lighthouse Lane, Corona del Mar, CA. 92625 Telephone: 714-640-1242				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Si (silicon)	Cz Z M		Methods for single crystal growth and crystal characterization- Intrinsic and As, Ga, B, P, In and other doped silicon crystals, IR spectroscopy, IR detectors and misc. other applications.	--

2				
Name(s): Dr. M. A. Otooni				
Inst. & Address: 4 Fairview Drive, Glassboro, New Jersey 08028 Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
BULK COPPER SINGLE CRYSTAL. PREPARED BY A NEW TECHNIQUE IN CERAMIC TUBING.	MELTING	UP TO 150	BY CYCLIC MELTING	+
BULK COPPER SINGLE CRYSTAL. PREPARED BY 4 POINTS TIPS. EMPLOYING CZ TECHNIQUE.	NEW OTOONI	100	BY 4 POINTS TIP AND CYCLIC MELTING	+
THIN SINGLE CRYSTALS OF COPPER. PREPARED BY SEEDING THE SUBSTRATE SURFACE. A COMPLETELY NEW PROCEDURE TO PREPARE 180-300 Å THICK CRYSTAL.	NEW OTOONI	CONTINUOUS SHEET	BY SEEDING THE SUBSTRATE	+

3				
Name(s): ROCH R. MONCHAMP				
Inst. & Address: ADOLF MELLER COMPANY, PO BOX 6001, PROVIDENCE, RHODE ISLAND Telephone: (401) 331-3717				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Sapphire α -Al ₂ O ₃	M	200 ϕ x 130 high	99.999% Pure all orientations optical & electronic applications	Stock

4				
Name(s): Judith A. Osmer, A. B. Chase, Elsa Garmire				
Inst. & Address: The Aerospace Corporation, P.O. Box 92957, Los Angeles, CA 90009 Telephone: (213) 648-6973				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs	LE	10 x 17 x 1	Integrated Optics and Laser	-
AlB ₆ , CaB ₆ , BaB ₆ , LaB ₆	M		Electron emitter	-
YVO ₄	M and Sol	2 x 3 x 2	Laser	-
Garnets	Sol	7 x 4 x 4	Laser	-
CaY ₂ Mg ₂ GeO ₁₂ - CAMGAR	Sol	3 x 2 x 2	Laser	-

5 Name(s): ALFA DIVISION/VENTRON CORPORATION Inst. & Address: 152 Andover St., Danvers, MA 01923 Telephone: (617) 777-1970				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon monoxide	Sub		Vapor phase deposition	+
Titanium monoxide	Sub		Vapor phase deposition	+

6 Name(s): Frank J. Bruni, Group Leader, Crystal Growth Inst. & Address: Allied Chemical, Synthetic Crystal Products P. O. Box 2607, Charlotte, NC 28234 Telephone: 704-588-2340				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gadolinium Gallium Garnet (GGG)	C _z	85X85X250	Material available for sale as magnetic garnet substrates in boule and polished wafer form.	Yes
Yttrium Aluminum Garnet (YAG)	C _z	40X40X150	Neodymium doped for laser rod applications. Material available in fabricated laser rod form to custom sizes and shapes.	Yes
Sapphire	EFG*	2X100X1000	Material is produced for electronic substrates, windows and wear surfaces and is available in custom sizes and shapes.	Yes
Magnetic Garnet Films	LE	75mm Dia.	Films are commercially available to customer magnetic and physical specifications for magnetic bubble devices.	Yes
Silicon on Sapphire (SOS)	GE	75mm Dia.	Films are commercially available to customer electronic specifications.	Yes

7 Name(s): M. A. Gilleo, Manager, Optical Physics Department, Corporate Research Center Inst. & Address: Allied Chemical Corporation, P. O. Box 1021R, Morristown, NJ 07960 Telephone: (201) 455-4574				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
α -AlPO ₄ (berlinite)	Hyd	20x20x30	Research growth of material for piezo-electric applications with appropriate characterization. B.H.-T. Chai	
Magnetic garnet films	LE	25 mm diam x20 microns	Magnetic-bubble-domain films on gadolinium garnet substrates; stripe widths $\geq$ 0.5 microns; full characterization. Research growth only. D. M. Gualtieri	

8

Name(s): James B. McNeely, Manager, Crystal Products

Inst. & Address: Allied Chemical Corp., Corporate Development Center, P.O. Box 1021R,
Morristown, NJ 07960

Telephone: 201-455-4040

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gadolinium Gallium Garnet	Cz	76 mm dia. x 125-250 mm long	<111> Boules, As-sawn or Polished Wafers.	✓
Rare Earth Garnets	Cz	Various		Limited
Magnetic Garnet Films	LE	76 mm dia.	Magnetic property characterizations available.	✓
Lanthanum Beryllate	Cz	20 mm dia. x 100 mm long	Laser Rods - Neodymium - Doped Ref: Applied Phys. Letters, 27, No. 8, 15 Oct.1975, p. 444; J.Appl.Phys. 47, No. 4, 1976, p. 1496.	✓
Chrysoberyl (Alexandrite)	Cz	Various	Doped or Undoped - Ref. U.S. Patent 3,997,853.	Limited
Yttrium Aluminum Garnet	Cz	Various	Laser Rods - Neodymium - Doped.	✓
Beryllium Oxide	Flu	Various	Ref: R.C. Linare, ACCG-IV, Gaithersburg Meeting, 1978.	Limited

9

Name(s): Allusuisse Metals, Inc.

Inst. & Address: 2460 Lemoine Avenue, P. O. Box 1100,
Fort Lee, NJ 07024Branch:
15910 Ventura Blvd.
Encino, CA 91436Telephone: (201) 461-6660
(213) 986-5020

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gallium Metal, Gallium Oxide, Gallium Trichloride	Cz	Various	99.9% to 99.9999%+	Stock
Gallium Phosphide, GGG			GGG 3-inch single crystal boule and wafer Gallium Phosphide polycrystalline ingots	
High-purity Alumina			99.99%	
High-purity Aluminum			99.999%	
Red Phosphorus			99.9999%	
Indium			99.9999%	

10

Name(s): Howard R. Shanks

Inst. & Address: Ames Laboratory-DOE, Iowa State University, Ames, Iowa 50011

Telephone: (515) 294-6816

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tungsten Bronzes M_xWO_3 M = Li, Na, K, Rb, Cs, Tl, Rare Earths	EC	up to 10 x 10 x 10 depending on M	High purity various orientations	+
Indium Selenide InSe	Br	10 x 10 x 10	High purity	+
Niobium Selenide NbSe ₂	VP	1 x 3 x 3	High purity	+

11				
Name(s): A. J. Arko				
Inst. & Address: Argonne National Laboratory; Argonne, IL 60439			Telephone: (312) 972-5018	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
V_3Si , Ti_3Al , UA_2	Z	12 mm x 12 mm x 50 mm	Prepared in an argon atmosphere. Maximum resistance ratio = 30.	-
URh_3 , UIr_3 , URu_3 , Nb_3Ir	Z	6 mm x 6 mm x 50 mm	Electron beam zone melting in vacuum. Maximum resistance ratio = 300. Used to study the dHvA effect.	-
UGe_3	Flu	1 mm x 1 mm x 2 mm	Maximum resistance ratio = 500.	-
UA_2 , UGe_3 , Ti_3Au	Cz	3 mm x 3 mm x 10 mm	Poor quality	-

12				
Name(s): Norman L. Peterson, W. K. Chen				
Inst. & Address: Materials Science Division, Argonne National Lab., Argonne, IL 60439			Telephone: (312) 972-4955	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$<FeO>$, NiO , CoO MnO , $(Ni,Co)O$ $(Ni,Cr)O$	<u>Ver.</u>	10x10x10	99.99% pure Electrical conductivities, Cation and Anion diffusions Defect properties	
$CoFe_2O_4$	<u>Ver.</u>	3x3x3	99.99% pure Electrical conductivities, Cation diffusions, defect properties	
Fe_3O_4	<u>M</u>	10x10x10	99.99% pure Cation diffusions, defect properties	

13				
Name(s): Sherman Susman and David G. Hinks				
Inst. & Address: Argonne National Laboratory, Solid State Science Division 9700 S. Cass Avenue, Argonne, Illinois 60439 USA			Telephone: 972-5470	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KCN	<u>Cz</u>	15 x 15 x 30	High purity; zone refined	+
NaCN	<u>Cz</u>	15 x 15 x 30	High purity; zone refined	-
KCN:KBr solid solution	<u>Cz</u>	15 x 15 x 30	High purity; zone refined	+
Y_2O_3 (Yttria)	Float <u>z</u>	4 x 4 x 4	High purity; undoped	-
Dy_2O_3 (Dysprosia)	Float <u>z</u>	4 x 4 x 4	High purity; undoped	-
Er_2O_3 (Erbia)	Float <u>z</u>	4 x 4 x 4	High purity; undoped	-
Yb_2O_3 (Ytterbia)	Float <u>z</u>	4 x 4 x 4	High purity; undoped	-
KCl	<u>z</u> , <u>Cz</u>	20 x 20 x 50	Ultrahigh purity. Fractionally crystallized, halogen-treated and zone refined.	+
$Y(EtSO_4)_3$ (Yttrium ethylsulfate)	<u>Sol</u>	25 x 25 x 25	Pure and with Yb dopant	+
$CeSn_3$	<u>Cz</u>	10 x 10 x 30		+
$LiAl$	<u>Cz</u>	5 x 5 x 5		-
PrAg	<u>Cz</u>	10 x 10 x 10		+
NdAg	<u>Cz</u>	10 x 10 x 10		+

14				
Name(s): G. W. Crabtree				
Inst. & Address: Solid State Science Division Argonne National Lab., Argonne, ILL. 60439			Telephone: 312-972-5509	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Niobium	M,Z	1 x 1 x 3	RRR λ 2500, High temp, High vacuum anneal. Used for de Haas-van Alphen experiments.	-

15				
Name(s): Charles M. Falco and Robert T. Kampwirth				
Inst. & Address: Solid State Science Division, Argonne National Lab., Argonne Illinois 60439			Telephone: (312) 972-5482	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Nb ₃ Sn, Nb ₃ Ge, and NbN	VP	75x75x0.002	Superconductive properties of poly- crystalline sputtered films. TEM, SEM, and Rutherford backscattering analysis of films.	+

16				
Name(s): William S. Glaunsinger				
Inst. & Address: Arizona State University, Department of Chemistry Tempe, AZ 85281			Telephone: (602) 965-7155	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Pt Platinum	S	2x10 ⁻⁷ dia.	>99% pure; dopants - none, properties studied - electron spin resonance, magnetic susceptibility, chemisorption; characterization - electron micros- copy, X-ray; applications - catalysis. M. Bayard, R. F. Marzke, and W. S. Glaunsinger, Solid State Commun. 18, 1025 (1976).	+
VO ₂ vanadium dioxide	Sub	0.3x0.3x0.3	>99% pure; dopants - gallium; properties studied - electron spin resonance, thermodynamic, electrical; characterization - X-ray, electron microscopy, electron microprobe; applications - thermal switches.	-

17 Name(s): William S. Glaunsinger Inst. & Address: Arizona State University, Department of Chemistry Tempe, AZ 85281 Telephone: (602) 965-7155				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
M(NH ₃) ₆ , M=Ca, Sr, Ba, Eu, Yb Alkaline earth hexammines Li(NH ₃) ₄ -lithium tetraamine	M	powder	>99% pure, dopants - none; properties studied - magnetic resonance and susceptibility, neutron diffraction, electrical; characterization - chemical, neutron diffraction; applications - batteries, purification of ammonia. W. S. Glaunsinger, T. K. White, R. B. Von Dreele, D. A. Gordon, R. F. Marzke, A. L. Bowman, and J. L. Yarnell, Nature <u>271</u> , 414-417 (1978).	+
Ca ₂ MnO ₄ , Ca ₃ Mn ₂ O ₇ , Ca ₄ Mn ₃ O ₁₀ , CaMnO ₃ calcium-manganese oxides	HYD Ceramic	powder	>99% pure, dopants - none; properties studied - electron spin resonance, neutron diffraction, electrical, thermogravimetric; characterization - chemical, X-ray fluorescence, X-ray, neutron activation, electron microscopy; applications - catalysis, electrocatalysis.	+
MB ₆ , M = Eu, Yb, Gd metal hexaborides	Ceramic	powder	>99% pure; dopants - none; properties studied - electron spin resonance; characterization - X-ray; applications - high temperature electrodes. W. S. Glaunsinger, J. Chem. Phys. Solids <u>37</u> , 51-57 (1978).	-
M _x V ₂ O ₅ , M=Cu, Na, Cr vanadium bronzes	M	0.1×0.1×4	>99% pure; dopants - none; properties studied - electron spin resonance, optical, infrared, magnetic susceptibility; characterization - chemical, neutron activation, X-ray; applications-varister.	-
(Mo _x V _{1-x}) ₂ O ₅ molybdenum-vanadium oxides	Sub	0.1×0.1×0.4	99% pure, dopants - none; properties studied - electron spin resonance, optical, infrared, magnetic susceptibility; characterization - chemical, neutron activation, X-ray; applications - catalysis.	-

18 Name(s): Neal R. Armstrong Inst. & Address: Department of Chemistry, University of Arizona, Tucson, AZ 85721 Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Titanium Dioxide (Rutile) Strontium Titanate	M M	5 x 5 x 5 5 x 5 x 5	X-ray photoelectron and Auger electron spectroscopic study of surface composition of TiO ₂ , titanium oxides, SrTiO ₃ and above materials reduced in H ₂ and used for photoelectrolysis of water. (C. N. Sayers and N. R. Armstrong, Surface Science, in press).	-

19 Name(s): S. Sari Inst. & Address: Optical Sciences Center, University of Arizona, Tucson, AZ 85721 Telephone: (602) 626-3025				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon and Silicon Oxide Embedded in Silicon	VP	10 × 10 × 10 ⁻² mm	Phys. Rev. B <u>15</u> , 4817 (1977) AIP Conference Series #40, p. 317 J. Phys. Chem. Solids (in press, 1978) Physica B (in press, 1978)	-
Silver Oxide and Silver	EC	10 × 10 × a few mono- layers.	Surface plasmons and related effects	-

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Name(s): ATOMERGIC CHEMETALS CORP.

Inst. & Address: 100 Fairchild Ave., Plainview, N. Y. 11803

Telephone: (516) 822-8800

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
<u>SINGLE CRYSTALS:</u> Aluminum Aluminum Fluoride Aluminum Oxide (Ruby and Sapphire) Ammonium Hydrogen Phosphate (ADP) Anthracene Antimony Barium Fluoride Barium Oxide Barium Sodium Niobate (Banana) Barium Titanate Beryllium Bicrystals Bismuth Bismuth Germanium Oxide Bismuth Silicon Oxide Cadmium (con't)				

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MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cadmium Fluoride Cadmium Mercury Telluride Cadmium Selenide Cadmium Sulfide Cadmium Sulfoselenide Cadmium Telluride Calcium Carbonate (Gypsum) Calcium Fluoride Calcium Oxide Calcium Tungstate Carbon (Diamond) Cerium Dioxide Cesium Bromide Cesium Chloride Cesium Iodide Cesium Magnesium Fluoride Cesium Manganese Fluoride Chromium Chromium Fluoride (con't)				

22

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Telephone: (516) 822-8800

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Chromium Oxide Cobalt Cobalt Oxide Columbium Copper Dysprosium Erbium Europium Gadolinium Gadolinium Gallium Garnet Gadolinium Molybdate Gallium Antimonide Gallium Arsenide Gallium Fluoride Gallium Oxide Gallium Phosphide Garnets Germanium tri Glycine Sulfate Gold (con't)				

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MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Hafnium Hafnium Oxide Indium Indium Antimonide Indium Arsenide Indium Phosphide Iodic Acid Iridium Iron Iron Oxide KRS-5 Lanthanum Fluoride Lead Lead Molybdate Lead Oxide Lead Selenide Lead Telluride Lithium Bromide Lithium Chloride Lithium Fluoride (con't)				

24

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MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Lithium Iodate Lithium Niobate Lithium Sulfate Lithium Tantalate Magnesium Magnesium Aluminate (Spinel) Magnesium Fluoride Magnesium Oxide Manganese Fluoride Manganese Oxide Mercury Telluride Molybdenum Molybdenum Oxide Naphthalene Nickel Nickel Fluoride Nickel Oxide Palladium Platinum (con't)				

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MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Potassium Bromide Potassium Chloride Potassium Iodide Potassium Deuterophosphate Potassium Fluoride Potassium Hydrogen Phosphate Potassium Magnesium Fluoride Potassium Manganese Fluoride Proustite Quartz Rhenium Rhodium Rochelle Salt Rubidium Chloride Rubidium Bromide Rubidium Iodide Rubidium Magnesium Fluoride Rubidium Manganese Fluoride Ruthenium Scandium (con't)				

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MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon				
Silicon Carbide				
Silicon Dioxide				
Silver				
Silver Arsenic Sulfide (Proustite)				
Silver Bromide				
Silver Chloride				
Sodium Bromide				
Sodium Chloride				
Sodium Fluoride				
Sodium Iodide				
Spinel				
trans-Stilbene				
Strontium Barium Niobate				
Strontium Fluoride				
Strontium Oxide				
Strontium Titanate				
Tantalum				
Tellurium				
Tellurium Oxide (con't)				

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Thallium Bromide				
Thallium Bromide/Iodide (KRS-5)				
Thallium Chloride				
Thorium Oxide				
Tin				
Titanium				
Titanium Dioxide				
Tri Glycine Sulfate				
Tungsten				
Tungsten Oxide				
Uranium Dioxide				
Vanadium				
Vanadium Oxide				
Yttrium Aluminum Garnet (YAG)				
Yttrium Aluminum Perovskite				
Yttrium Gallium Garnet				
Yttrium Iron Garnet				
Yttrium Lithium Fluoride				
Yttrium Oxide (con't)				

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Inst. & Address: 100 Fairchild Ave., Plainview, N. Y. 11803

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MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Zinc Zinc Oxide Zinc Selenide Zinc Sulfide Zinc Telluride Zirconium Zirconium Oxide Ultra high purity elements and inorganic compounds of many types.				

29

Name(s): William Rolls

Inst. & Address: Barnes Engineering Company, 30 Commerce Rd., Stamford, Ct.

Telephone: (203) 348-5381

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Triglycine Sulphate (CH ₂ NH ₂ COOH) ₃ H ₂ SO ₄	<u>Sol</u>	a = 67.5 b = 80 c = 38	Undoped, single crystal grown with b-axis vertical, pyroelectric coefficient dielec- tric constant & loss tangent measured, fabricated into high sensitivity pyroelec- tric detectors. American Ceramic Society Bulletin Vol 46, No. 8, Aug 7 (1967)	-
Deuterated Triglycine Sulphate (CD ₂ NH ₂ COOD) ₃ D ₂ SO ₄	<u>Sol</u>	a = 51.5 b = 81.5 c = 35	Undoped, single crystal grown with b-axis vertical, pyroelectric coefficient dielec- tric constant & loss tangent measured, fabricated into high sensitivity pyroelec- tric detectors. American Ceramic Society Bulletin Vol 46, No. 8, Aug. 7 (1967)	

30

Name(s): William Rolls

Inst. & Address: Barnes Engineering Company, 30 Commerce Road, Stamford, Ct.

Telephone: (203) 348-5381

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Lead Tin Telluride Pb _x Sn _{1-x} Te x = 0.83, 0.79 or 0.75	VP	20 mm dia 40 mm long	Single crystal, undoped, (100) along growth axis, electrical characteristics, (e.g., mobility, carrier density) used to fabricate high sensitivity infrared detectors at 77°K, characterized for responsivity, quantum efficiency, 1/f noise and D*	-

31

Name(s): W.T. Pawlewicz

Inst. & Address: Battelle-Pacific Northwest Laboratory
P.O. Box 999, Richland, WA 99352

Telephone: (509)-942-2537

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Titanium Dioxide (TiO_2)	VP (Sputter Deposition)	50 x 50 x 0.0001 (thin films)	Basic materials study of optical properties vs. structure and stoichiometry and development of damage-resistant coatings for laser optics; reactive sputter deposition of 99.99% Ti metal in 99.999% Ar/ O_2 mixtures; properties studied include refractive index (350 - 1500 nm) and absorption coefficient near band edge; characterization includes phase composition (anatase/rutile), preferred crystallographic orientation and grain size by both x-ray diffraction and TEM, surface topography by SEM, and stoichiometry by optical absorption near band edge. W.T. Pawlewicz and H. Laegreid, "Electrical and Optical Properties of Sputter-Deposited Materials," in Proceedings of Society of Photo-Optical Instrumentation Engineers Technical Symposium East 1978, Washington, D.C. (March 1978), Volume 140, Optical Coatings-Applications and Utilization II.	+

32

Name(s): W.T. Pawlewicz

Inst. & Address: Battelle-Pacific Northwest Laboratory
P.O. Box 999, Richland, WA 99352

Telephone: (509)-942-2537

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Amorphous Silicon:Hydrogen (a-Si:H)	VP (Sputter Deposition)	20 x 20 x 0.001 (thin films)	Basic materials study to aid development of solar photovoltaic technology; study of electrical, optical and photoelectronic properties--their dependence on positional and compositional disorder, their range of variation and their controlled manipulation by sputtering; electronic-grade single-crystal source material sputtered in 99.9995% H_2 /Ar mixtures; some work with 0-1000 ppm P or B dopants; properties studied include resistivity vs. temperature (0 - 300°C), spectral dependence of photoconductivity (350 - 1500 nm), refractive index (350 - 1500 nm) and absorption coefficient near band edge; composition characterized by SIMS.	+

33

Name(s): C. A. Burrus

Inst. & Address: Bell Telephone Labs, Crawford Hill Lab, Holmdel, N.J. 07733 Telephone: (201) 949-2907

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Al_2O_3 (sapphire)	Z (CO_2 laser)	25-250 μm dia 5-30 cm length	<111>, <100> fibers	(-)
$\text{Cr}:\text{Al}_2\text{O}_3$ (ruby)	Z (CO_2 laser)	"	Laser quality fibers, 0.02% Cr. "Growth of Single-crystal sapphire-clad Ruby Fibers", C. A. Burrus and L. A. Coldren, Appl. Phys. Lett. <u>31</u> , 383 (1977). "Room-temperature Continuous Operation of a Ruby Fiber Laser", C. A. Burrus and J. Stone, J. Appl. Phys., <u>49</u> , 3118 (1978).	(-)
YAG and Nd:YAG (yttrium aluminum garnet)	Z (CO_2 laser)	"	<111>, <100> laser-quality fibers, 1% Nd "Self-contained LED-pumped Single-Crystal Nd:YAG Fiber Laser", J. Stone and C. A. Burrus, Fiber and Integrated Optics <u>2</u> , 19 (1978).	(-)

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Name(s): C. A. Burrus

Inst. & Address: Bell Telephone Labs, Crawford Hill Lab, Holmdel, N.J. 07733 Telephone: (201) 949-2907

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Nd}:\text{Y}_2\text{O}_3$ (neodymium doped yttria)	Z (CO_2 laser)	60-120 μm dia 5 cm length	0.5-3% Nd doped yttria fibers - Laser quality; fluorescence lifetime, fluorescence and absorption spectra, laser action "Nd:Y ₂ O ₃ Single-crystal Fiber Laser: Room- temperature CW Operation at 1.07 and 1.35 μm Wavelength", J. Stone and C. A. Burrus, J. Appl. Phys. <u>49</u> , 2281 (1978).	(-)
LiTaO_3 (lithium tantalate) LiNbO_3 (lithium niobate)	Z (CO_2 laser)	150-250 μm dia 10-15 cm length	A, B and C-axis Fiber Crystals No other information or publication	(-)

35 Name(s): Uri Cohen Inst. & Address: Bell Tel. Laboratories Murray Hill, N.J. 07974 Telephone: (201)582-5039				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Epitaxial-Ge	GE	20x10x0.01	CVD of organo-Ge compound, tetraethyl germ- anium (TEG). Found less than 20 ppm of C contamination. U. Cohen, Y. Avigal and M. Schieber, ICCG-III Meeting (abst. #D4-4), Marseilles, France, July 1971, and U. Cohen, M.S. Thesis, Jerusalem (1971).	-
Epitaxial-Si	EC	25x12x0.02	(111) orientation, 99.99% pure, p-type. U. Cohen, U.S. Pat. 3,983,012 (1976), and J. Elect. Mats. 6,607(1977).	-
Polycrystalline-Si	EC	25x12x0.04	Continuous and adherent films with up to 40-50 μ grain diameter, p-type, 99.99% pure. U. Cohen, pending U.S. Pat., and J. Elect. Mats. 6,607(1977).	-
Nb-Ge alloys	EC	25x12x0.04	Composition, crystal structure, and super- conductivity. U. Cohen, Ph.D. Thesis, Stanford (1977).	-
NbO and NbO ₂	EC	25x12x0.1	Composition, crystal structure, and super- conductivity. U. Cohen, Ph.D. Thesis, Stanford (1977).	-
Refractory metal (W, Mo, Nb, V) silicides	EC	25x12x0.05	Composition, crystal structure, and super- conductivity. U. Cohen, Ph.D. Thesis, Stanford (1977).	-

36 Name(s): Kurt Nassau Inst. & Address: Bell Laboratories, Murray Hill, NJ 07974 Telephone: (201) 582-2589				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
RE Iodates } TR Iodates }	Gel } sol }	5 max	Structure, Optic Props., Magnetic Props., etc.	-
SbSI	Flu	5x5x30	Structure, etc.	-
K ₂ Cd ₂ (SO ₄) ₃	M	5x5x10	Structure, ferroelastic transitions, etc.	-

37 Name(s): J. C. Bean Inst. & Address: Bell Laboratories, Murray Hill, N. J. 07974 Telephone: (201)582-3324				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon Molecular Beam Epitaxy	MBE	50x50x10 ⁻²	- (100) and (111) orientation - Hyperabrupt doping profiles from 10 ¹⁴ /cm ³ - 5x10 ¹⁸ /cm ³ - Dopants Al, As, Ga, Sb Publications: J. C. Bean, G. E. Becker, P. M. Petroff, T. E. Seidel, J. Appl. Phys. 48, 907 (1977). G. E. Becker, J. C. Bean, J. Appl. Phys. 48, 3395 (1977).	

38

Name(s): W. D. Johnston, Jr.

Inst. & Address: Bell Telephone Laboratories, Holmdel, NJ 07733

Telephone: 201-949-4250

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InP	GE	20x20x	on InP substrates	
$\text{In}_{1-x}\text{Ga}_x\text{As}_y\text{P}_{1-y}$	GE	10^{-2}	(100) - $1\frac{1}{2}^\circ$ + (110). HCl transport.	
GaAs	GE	40x40x	on GaAs (100) - $2-4^\circ$ + (110)	
$\text{Al}_x\text{Ga}_{1-x}\text{As}$	GE	10^{-2}	by $(\text{CH}_3)_3\text{Ga} + \text{AsH}_3 + (\text{CH}_3)_3\text{Al}$ reaction.	

39

Name(s): Daniel L. Rode

Inst. & Address: Bell Laboratories, Room 2C311, Murray Hill, N.J. 07974

Telephone: (201) 582-2498

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs, gallium-arsenide	Sol, LE	$10 \times 10 \times .01$	$10^{16}-10^{19}$, (100), (111), microwave devices, LEDs, laser diodes	Yes
AlGaAs, aluminum-gallium- arsenide	Sol, LE	$10 \times 10 \times .01$	$10^{16}-10^{19}$, (100), (111), microwave devices, LEDs, laser diodes	Yes
InP, indium-phosphide	Sol, LE	$10 \times 10 \times .01$	$10^{16}-10^{19}$, (100), (111), microwave devices, LEDs, laser diodes	Yes
InGaAsP, indium-gallium- arsenide-phosphide	Sol, LE	$10 \times 10 \times .01$	$10^{16}-10^{19}$, (100), (111), microwave devices, LEDs, laser diodes	Yes

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Name(s): J. H. Wernick

Inst. & Address: Bell Telephone Laboratories
600 Mountain Avenue, Murray Hill, New Jersey 07974

Telephone: (201) 582-2463

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
p, n-type, and seminsulating InP	CZ LEC	1"Dx2"	<111> and <100>, Zn, Cd, Sn dopants	-
$\text{V}_3\text{Si}, \text{V}_3\text{Ge}$	CZ Cold Hearth	1/4"Dx1"	<111> and <100>, Resistance ratios of 12 to 20	-
R E Garnets	Flu, CZ	All sizes up to 1"D		-
LiNbO_3 , LiTaO_3 , re Molybdates, ex: $\text{Tb}_2(\text{MoO}_4)_3$	CZ	All sizes up to 1"D		-
Transition Metal Dichalcogenides	VP			-
$\text{Pb}_4\text{Ge}_3\text{O}_{11}$	CZ	up to 1"D		-
I III VI_2 and II IV V_2 Chalcopyrites	M, VP			-

41 Name(s): Randall Caton Inst. & Address: Brookhaven National Laboratory, Upton, NY 11973 Telephone: 516-345-3502				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$Nb_{3+y}Al_xSi_{1-x-y}$ $0 \leq x \leq 1$ $0 \leq y \leq .6$	M Rapidly Quenched from $\sim 2000^\circ C$	$1\text{ mm} \times 1\text{ mm}$ $\times 1\text{ mm}$	Purity of starting materials: Nb 99.995%, Al 99.9% Si semiconductor grade. The samples are in the A-15 structure and characterized by x-ray diffraction, electron microprobe, and metallography. The superconducting transition temperature was measured as a function of silicon content. The samples are high temperature superconductors (with T_c as high as 18.6 K) and are desirable to study in regards to producing high temperature superconductors. <u>Publications:</u> R. Caton and A. R. Sweedler, J. Less Comm. Met. 60, 91 (1978); C. S. Pande, R. Caton, S. Moehlecke, J. Mat. Sci. 11, 2338 (1976).	Not available

42 Name(s): Jan Tauc Inst. & Address: Division of Engineering, Brown University, Providence, RI 02912 Telephone: (401) 853-2318				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Amorphous hydrogenated Si	Sputtering glow-discharge	$20 \times 20 \times 10^{-2}$ (films)	Electronic relaxation processes Photo-modulation experiments Optical and transport properties of Si:H doped with noble or transition metals	-
Metallic glasses: metal-metalloid, such as $Pd_{78}Ge_{22}$, $Pd_{78}Cu_6Si_{16}$ transition metal-transition metal, such as Nb:Ni, Ti:Ni	Splat cooling of melt	$20 \times 4 \times 1$	Structure by EXAFS (Phys. Rev. Lett. 40, 1282 (1978)) Photoemission (Phys. Rev. B13, 3284 (1976); Solid State Comm. 22, 471 (1977)) Optical properties	-
Amorphous metallic films $Au_{1-x}Si_x$, $0.12 \leq x \leq 1$	Sputtering	$20 \times 20 \times 10^{-3}$	Optical properties (Phys. Rev. Lett. 40, 1733 (1978))	-

43 Name(s): George R. Rossman Inst. & Address: Division of Geological and Planetary Sciences 170-25 California Institute of Technology, Pasadena, California 91125 Telephone: (213) 795-6811 Extension 2471				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Minerals, natural and synthetic. Silicates, phosphates, arsenates, iron and manganese oxides.			optical and infrared spectroscopy: anisotropy, color, site occupancy, H_2O content	

44 Name(s): DR. SHELDON SCHULTZ Inst. & Address: DEPARTMENT OF PHYSICS, B-019 UNIVERSITY OF CALIFORNIA, SAN DIEGO LA JOLLA CA 92093 Telephone (714) 452-4078				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cu, Ag, Au, Mg, Zn, Cd, In, Sn	BR	10 x 10 x 20	ULTRA HIGH PURITY OR LESS THAN 100 PPM LOCAL MOMENT DOPANTS. RESISTIVITY RATIO AND ELECTRON SPIN RESONANCE PROPERTIES STUDIED.	NEGOTIABLE

45 Name(s): G. W. Webb, Z. Fisk Inst. & Address: Institute for Pure and Applied Physical Sciences, University of California, San Diego, La Jolla, CA 92093 Telephone: (714) 452-4785				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Nb ₃ Sb	VP	15 x 10 x 5	DHVA, neutron scattering, others	+
Nb ₃ Sn	VP	4 x 2 x 2	DHVA, neutron scattering, others	+

46 Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720 Telephone: (415) 642-2985				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
MnCl ₂ Manganous Chloride	Br	3.3 cm dia sphere	PURITY: (wt. ppm) 5 Mg, 5 Si, 1 Cu. (emission spectroscopy) ORIENTATION: α crystal axis. PROPERTIES STUDIED: Magnetothermodynamics. FIELD RANGE: 0 to 100,000 G. TEMPERATURE RANGE: 1.3 to 4.2°K. 1. R. A. Butera and W. F. Giaque, J. Chem. Phys. <u>40</u> , 2379 (1964). 2. W. F. Giaque, G. E. Brodale, R. A. Fisher, and E. W. Hornung, J. Chem. Phys. <u>42</u> , 1 (1965). 3. W. F. Giaque, R. A. Fisher, E. W. Hornung, R. A. Butera, and G. E. Brodale, J. Chem. Phys. <u>42</u> , 9 (1965). 4. W. F. Giaque, E. W. Hornung, R. A. Fisher, and G. E. Brodale, J. Chem. Phys. <u>42</u> , 853 (1965). 5. W. F. Giaque, R. A. Fisher, E. W. Hornung, and G. E. Brodale, J. Chem. Phys. <u>43</u> , 4387 (1965).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\alpha\text{-MnCl}_2 \cdot 4\text{H}_2\text{O}$ Manganous Chloride Tetrahydrate	Sol	3.934 cm dia sphere	PURITY: (wt. ppm) 5 Mg, 5 Si, 1 Cu, 0.1 Fe. (emission spectroscopy) ORIENTATION: <i>c</i> , <i>b</i> , [100] crystal axis. PROPERTIES STUDIED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.4 to 4.2°K. 1. T. A. Reichert and W. F. Giaque, J. Chem. Phys. <u>50</u> , 4205 (1969). 2. W. F. Giaque, R. A. Fisher, G. E. Brodale, and E. W. Hornung, J. Chem. Phys. <u>52</u> , 2901 (1970). 3. W. F. Giaque, E. W. Hornung, G. E. Brodale, and R. A. Fisher, J. Chem. Phys. <u>52</u> , 3936 (1970). 4. W. F. Giaque, R. A. Fisher, E. W. Hornung, and G. E. Brodale, J. Chem. Phys. <u>53</u> , 1474 (1970).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{NiSiF}_6 \cdot 6\text{H}_2\text{O}$ Nickel Fluosilicate Hexahydrate	Sol	2.498 cm dia sphere	PURITY: (wt. ppm) 2.4 Na. (emission spectroscopy and neutron activation) ORIENTATION: <i>c</i> and <i>L c</i> axis. PROPERTIES STUDIED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.05 to 4.2°K. 1. E. W. Hornung, G. E. Brodale, R. A. Fisher, and W. F. Giaque, J. Chem. Phys. <u>45</u> , 614 (1966). 2. E. W. Hornung, R. A. Fisher, G. E. Brodale, and W. F. Giaque, J. Chem. Phys. <u>45</u> , 2011 (1966). 3. E. W. Hornung, R. A. Fisher, G. E. Brodale, and W. F. Giaque, J. Chem. Phys. <u>46</u> , 67 (1967). 4. W. F. Giaque, E. W. Hornung, G. E. Brodale, and R. A. Fisher, J. Chem. Phys. <u>46</u> , 1804 (1967). 5. W. F. Giaque, E. W. Hornung, R. A. Fisher, and G. E. Brodale, J. Chem. Phys. <u>47</u> , 2685 (1967).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
α -NiSO ₄ ·6H ₂ O Nickel Sulfate Hexahydrate	Sol	4.006 cm dia sphere	PURITY: (wt. ppm) 20 Cu, 5 Pb, 10 Fe, 10 Mn, 20 Co. (mass spectrograph and neutron activation) ORIENTATION: <i>c</i> , <i>a</i> , bisector <i>a</i> , <i>b</i> crystal axes. PROPERTIES STUDIED: Magnetothermodymics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.4 to 4.2°K. 1. R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giaque, J. Chem. Phys. <u>46</u> , 4945 (1967). 2. R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque, J. Chem. Phys. <u>48</u> , 1769 (1968). 3. R. A. Fisher and E. W. Hornung, J. Chem. Phys. <u>48</u> , 4284 (1968). 4. R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque, J. Chem. Phys. <u>49</u> , 4096 (1968). 5. R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giaque, J. Chem. Phys. <u>51</u> , 1959 (1969).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CuSO ₄ ·5H ₂ O Copper Sulfate Pentahydrate	Sol	3.700 cm dia sphere	PURITY: (wt. ppm) 20 Fe, 10 Ni and Co. (emission spectroscopy and neutron activation) ORIENTATION: α , β , γ magnetic axes. PROPERTIES STUDIED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.02 to 4.2°K. 1. T. H. Geballe and W. F. Giaque, J. Am. Chem. Soc. <u>74</u> , 3513 (1952). 2. R. J. Anderson and W. F. Giaque, J. Chem. Phys. <u>46</u> , 2413 (1967). 3. W. F. Giaque, R. A. Fisher, E. W. Hornung, and G. E. Brodale, J. Chem. Phys. <u>48</u> , 3728 (1968). 4. W. F. Giaque, E. W. Hornung, G. E. Brodale, and R. A. Fisher, J. Chem. Phys. <u>48</u> , 3906 (1968). 5. W. F. Giaque, G. E. Brodale, R. A. Fisher, and E. W. Hornung, J. Chem. Phys. <u>49</u> , 1848 (1968). 6. W. F. Giaque, R. A. Fisher, E. W. Hornung, and G. E. Brodale, J. Chem. Phys. <u>53</u> , 3733 (1970). 7. R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque, J. Chem. Phys. <u>63</u> , 4817 (1975).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giauque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{CuK}_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$ Copper Potassium Sulfate Hexahydrate	Sol	3.912 cm dia sphere	PURITY: Prepared from 99.9998% Cu metal. ORIENTATION: α , γ magnetic axes; b crystal axis. PROPERTIES STUDIED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.01 to 4.2°K. 1. J. R. Lemley and W. F. Giauque, J. Chem. . . . 51, 4650 (1969). 2. W. F. Giauque, G. E. Brodale, E. W. Hornung, and R. A. Fisher, J. Chem. Phys. 54, 273 (1971). 3. W. F. Giauque, R. A. Fisher, G. E. Brodale, and E. W. Hornung, J. Chem. Phys. 54, 4908 (1971). 4. W. F. Giauque, G. E. Brodale, E. W. Hornung, and R. A. Fisher, J. Chem. Phys. 55, 355 (1971). 5. W. F. Giauque, R. A. Fisher, E. W. Hornung, and G. E. Brodale, J. Chem. Phys. 55, 2859 (1971).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giauque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\beta\text{-Gd}_2(\text{MoO}_4)_3$ Gadolinium Molybdate	Gz	2.49 cm dia sphere	PURITY: (wt. ppm) 10 Na, 40 Al, 30 Ca, 5 Sc, 2V, 2 Cr, 5 Mn, 100 Fe, 5 Co, 150 Ni, 5 Cu, 5 Zn, 1 Sr, 150 Y, 1 Zr, 5 Nb, 10 La, 70 Ce, 15 Pr, 5 Nd, 1 Sm, 70 Eu, 50 Tb, 2 Dy, 2 Ho, 5 Hf. (emission spectroscopy and mass spectrograph) ORIENTATION: a , b , c^+ crystal axes. (polarized ferroelectric) PROPERTIES MEASURED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.08 to 4.2°K. 1. R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giauque, J. Chem. Phys. 56, 193 (1972). 2. E. W. Hornung, G. E. Brodale, R. A. Fisher, and W. F. Giauque, J. Chem. Phys. 56, 5007 (1972). 3. G. E. Brodale, R. A. Fisher, E. W. Hornung, and W. F. Giauque, J. Chem. Phys. 56, 6118 (1972). 4. R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giauque, J. Chem. Phys. 59, 5796 (1973). 5. R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giauque (To be published J. Chem. Phys.).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
β - $\text{Tb}_2(\text{MoO}_4)_3$ Terbium Molybdate	Cz	1.170 cm dia sphere	PURITY: (wt. ppm) 2 Al, 5 Si, 5 K, 20 Ca, 5 Cr, 1 Mn, 20 Fe, 2 Co, 10 Ni, 5 Cu, 5 Zn, 2 Cd, 1 Sn, 1 Pb, 5 Nd, 10 Dy, 5 Ho, 1 W. (emission spectroscopy) ORIENTATION: a^+ crystal axis. (polarized ferroelectric) PROPERTIES MEASURED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.4 to 4.2°K. 1. R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giaque, J. Chem. Phys. <u>63</u> , 1295 (1975).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Gd}_3\text{Ga}_5\text{O}_{12}$ Gadolinium Gallium Garnet	Cz	2.44 cm dia sphere	PURITY: (wt. ppm) 2 Mg, 5 Si, 6 Ca, 2 Cr, 5 Fe, 5 Cu, 30 Y, 10 Mo, 15 Eu, 10 Lu, 10 Hf, 2 Pb, 2 Bi. (emission spectroscopy and mass spectrograph) ORIENTATION: [100], [110], and [111] crystal axes. PROPERTIES MEASURED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.5 to 4.2°K. 1. R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque, J. Chem. Phys. <u>59</u> , 4652 (1973). 2. E. W. Hornung, R. A. Fisher, G. E. Brodale, and W. F. Giaque, J. Chem. Phys. <u>61</u> , 282 (1974). 3. G. E. Brodale, E. W. Hornung, R. A. Fisher, and W. F. Giaque, J. Chem. Phys. <u>62</u> , 4041 (1975).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giaque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Ce}_2\text{Mg}_3(\text{NO}_3)_{12} \cdot 24\text{H}_2\text{O}$ Cerous Magnesium Nitrate 24 Hydrate	Sol	4.023 cm dia sphere	PURITY: (wt. ppm) 2 V, 2 Cr, 20 Fe, 3 Co, 40 Ni, 1 Cu, 25 Zn, 15 Y, 1 Nb, 15 La, 25 Pr, 5 Nd. (mass spectrograph) ORIENTATION: a crystal axis. PROPERTIES MEASURED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.0006°K to 4.2°K. 1. R. A. Fisher, Rev. Sci. Instrum. <u>43</u> , 386 (1972). 2. W. F. Giaque, R. A. Fisher, E. W. Hornung, and G. E. Brodale, J. Chem. Phys. <u>58</u> , 2621 (1973). 3. R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giaque, J. Chem. Phys. <u>58</u> , 5584 (1973).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giauque

Inst. & Address: University of California, Low Temperature Laboratory, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Ce}_2\text{Zn}_3(\text{NO}_3)_{12} \cdot 24\text{H}_2\text{O}$ Cerous Zinc Nitrate 24 Hydrate	Sol	4.004 cm dia sphere	PURITY: (wt. ppm) 2 V, 2 Cr, 20 Fe, 3 Co, 40 Ni, 1 Cu, 25 Zn, 15 Y, 1 Nb, 15 La, 25 Pr, 5 Nd. (mass spectrograph) ORIENTATION: a crystal axis. PROPERTIES MEASURED: Magnetothermodynamics. FIELD RANGE: 0 to 90,000 G. TEMPERATURE RANGE: 0.0008°K to 4.2°K. 1. R. A. Fisher, Rev. Sci. Instrum. <u>46</u> , 155 (1975). 2. W. F. Giauque, R. A. Fisher, E. W. Hornung, and G. E. Brodale, J. Chem. Phys. <u>61</u> , 3495 (1974). 3. R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giauque, J. Chem. Phys. <u>62</u> , 555 (1975).	-

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Name(s): R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giauque

Inst. & Address: Low Temperature Laboratory, University of California, Berkeley, CA 94720

Telephone: (415) 642-2985

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Nd}(\text{C}_2\text{H}_5\text{SO}_4)_3 \cdot 9\text{H}_2\text{O}$ Neodymium Ethyl Sulfate 9 Hydrate	Sol	3.920 cm dia sphere	Purity: (wt. ppm) 50 Na (emission spectroscopy) Orientation: a and c crystal axes Properties measured: Magnetothermodynamics Field range: 0 to 90,000 G Temperature range: 0.01 to 4.2°K R. A. Fisher, Rev. Sci. Instrum. <u>47</u> , 1086 (1976) W. F. Giauque, R. A. Fisher, E. W. Hornung, and G. E. Brodale, J. Chem. Phys. <u>64</u> , 3330 (1976) R. A. Fisher, E. W. Hornung, G. E. Brodale, and W. F. Giauque, J. Chem. Phys. <u>65</u> , 4724 (1976) R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giauque, J. Chem. Phys. <u>66</u> , 2678 (1977) R. A. Fisher, G. E. Brodale, E. W. Hornung, and W. F. Giauque, J. Chem. Phys. <u>68</u> , 169 (1978)	-

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Name(s): Professor J. A. Pask

Inst. & Address: University of California, Department of Materials Science, and
Mineral Engineering, Hearst Mining Building, Berkeley, CA 94720

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
MgO	M	10x10x20	Purity 99; orientations <001>, <110>, <111> Mechanical behavior at elevated temps.	+
Mullite ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$)	Cz	10x40 boule	Purity nominal Mechanical behavior at elevated temps.	-

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Name(s): Hong N. Yeung				
Inst. & Address: Department of Pediatrics, California College of Medicine University of California, Irvine, Ca. 92717			Telephone: (714) 525-5162	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Antimony	---	3x3x10	Study its electrochemical potential and its pH and oxygen dependence	-

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Name(s): Dr. R.F. Bunshah				
Inst. & Address: 6532 Boelter Hall, Materials Dept., University of California, LA, CA 90024			Telephone: (213) 825-2210	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ti Polycrystalline	VP	150 x 150x5	Total Impurity Content 60 ppm	-
Be Polycrystalline	VP	" " "	Total Impurity Content 30 ppm	-
Ni Polycrystalline	VP	" " "	Not Analyzed	-
TiC Polycrystalline	VP	150x150x.01	Not Analyzed	-
Al ₂ O ₃ Polycrystalline	VP	" " "	Not Analyzed	-
ZrC Polycrystalline	VP	" " "	Not Analyzed	-
HfC Polycrystalline	VP	" " "	Not Analyzed	-
HfN Polycrystalline	VP	" " "	Not Analyzed	-
TiN Polycrystalline	VP	" " "	Not Analyzed	-
NbC Polycrystalline	VP	" " "	Not Analyzed	-
Y ₂ O ₃ Polycrystalline	VP	" " "	Not Analyzed	-
Ni-20Cr Polycrystalline	VP	150x150x5	Not Analyzed	-

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Name(s): Prof. L. G. DeShazer				
Inst. & Address: Dept. of Physics, University of Southern California, Los Angeles, CA 90007			Telephone: (213) 741-6195	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
YAG	Cz	5x5x30	Optical and thermal properties, Nd dopant	-
YVO ₄	Cz	5x5x20	Optical properties, rare-earth dopants J. Appl. Phys. <u>47</u> , 232 (1976)	+
La ₂ Be ₂ O ₅	Cz	10x10x10	Optical and thermal properties, Nd dopant	-
YAlO ₃	Cz	5x5x50	Optical and thermal properties, rare-earth dopants, laser applications IEEE J. Quan. Elect. <u>QE-11</u> , 874 (1975)	+
Ca ₅ F(PO ₄) ₃	Cz	5x5x50	Optical and thermal properties, laser applications	+
Y ₂ O ₃	Ver	5x5x10	Optical properties, laser applications	+

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Name(s): M. Gershenzon

Inst. & Address: Materials Science Department
University of Southern California, Los Angeles, CA 90007

Telephone: (213) 741-6215

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaI	VP	20 x 10 x 2	(0120) single crystals on sapphire, 4 x 10 ¹⁸ cm ⁻³ native shallow donor, mobility ~120 cm ² /V-sec.	

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Name(s): F. A. Krüger

Inst. & Address: Department of Materials Science, University of Southern Calif. Telephone: (213) 741-6224
University Park, Los Angeles, California 90007

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe, cadmium telluride	EC	amorphous or poly- crystall- ine films	Pure and doped with In (n-type) or As (p-type). R. M. Panicker, M. Knaster and F. A. Krüger, J. Electrochem. Soc. <u>125</u> [4] 566 (1978).	

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Name(s): H. P. Ong, J. Savage

Inst. & Address: Physics Department & Materials Science Department, University of Southern
California, Los Angeles, California 90007

Telephone: (213) 741-6214

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NbSe ₃	Direct reaction	8 x 1 x 0.05	Single crystals grow with longest dimension along b-axis and shortest along a*. Linear chain crystal structure ¹ with conductivity anisotropy ² of ~ 20. Metallic behavior at all temperatures with RRR between 80 and 180. Two phase transitions ³ at T ₁ (142 K) and T ₂ (58 K). Strong non-Ohmic behavior below T ₁ and T ₂ suggests sliding conductivity from charge-density-wave condensate. Hall ⁵ , magnetoresistance ² , Shubnikov-de- Haas ⁶ , thermopower ⁷ , Young's modulus ⁸ and microwave ⁴ studies have been made. Superconducting under pressure (T _c ~ 2 K at 10 kbar) ⁹ . 1. A. Meerschaut and J. Rouxel, J. Less Common Metals 39, 197 (1975). 2. N. P. Ong and J. W. Brill, Phys. Rev. B (to appear) 3. J. Chaussy et al., Solid State Commun. <u>20</u> , 759 (1976). 4. N. P. Ong and P. Monceau, Phys. Rev. B <u>16</u> , 3443 (1977). 5. N. P. Ong and P. Monceau, Solid State Commun. <u>26</u> , 487 (1978). 6. P. Monceau, Solid State Commun. <u>24</u> , 331 (1977). 7. J. Kwak, R. Greene, P. Chaikin and N. P. Ong, unpublished. 8. J. W. Brill and N. P. Ong, Solid State Commun. <u>25</u> , 1075 (1978). 9. P. Monceau, J. Peyrard, J. Richard and P. Molinie, Phys. Rev. Lett. <u>39</u> , 160 (1977).	

65				
Name(s): SEKERKA, ROBERT F., PROFESSOR AND HEAD				
Inst. & Address: Department of Metallurgy & Materials Science Carnegie-Mellon University, Pittsburgh, Pa. 15213			Telephone: (412) 578-2700	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ice	N	Few	For study of anisotropic growth kinetics and morphological stability.	--
Oxides and oxide-metal composites	M	30X 30X 100	To check modeling of internal centrifugal zone growth.	--

66				
Name(s): CERES Corporation				
Inst. & Address: 411 Waverly Oaks Park, Waltham, MA 02154			Telephone: 617-899-5522	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cubic-Zirconia (ZrO_2) crystals (Y_2O_3 - stabilized)	M	25x25x80	99.99% pure - Random orientation - Undoped or doped with various rare earth or transition metals	Research & Production Quantities

67				
Name(s): Jun. Ito*, A. Grandin de L'Eprevier, O.J. Kleppa**				
Inst. & Address: University of Chicago, The James Franck Institute 5640 S. Ellis Ave., Chicago, IL 60637			Telephone: (312) 753-8212	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$MgSiO_3$ Orthoenstatite	flu	13 x 6 x 3	Pure orthorhombic form obtained by multiple cooling. Am. Conf. on Crystal Growth IV, July 1978.	-
$MgSiO_3$ Orthoenstatite	flu	5 x 1.5 x 1.5	Pure or doped with Ca, Ni, Mn, Cd, Cu, Zn, Cr, up to 20%. Mixed doping or higher concentration on request. Can. Mineral 15, 321-338, (1977) Geophysical Research Letter 2, No. 12, 533-536, (1975).	+
$Li_{x'} Sc_{x'} Mg_{2-2x'} Si_2 O_6$ Protoenstatite	flu	9 x 3 x 3	Stabilized Li/Sc Protopyroxene Am. Mineral 62, 1252-1257, (1977).	+
$MgSiO_3$ Enstatite IV	flu	1.5 x .5 x .3	New protopyroxene of higher temp. stabilized Li/Sc Proc. Japan Acad. 53, Ser. B, 60-63 (1977)	-
$LiScSiO_4$ Lithium Scandium Olivine	flu	2 x 1 x 1	Am. Mineral 62, 356-361, (1977)	-
$CaMgSi_2 O_6$ Diopside	flu	4 x 3 x 3	Pure end member or slightly Mg rich Diopside VO^{++} impurity < 0.05 wt% $V_2 O_5$	+
*Jun Ito is now deceased - Crystal growth discontinued. **O. J. Kleppa is in charge of crystal collection.				

68				
Name(s): R. Shankar Subramanian and Robert Cole				
Inst. & Address: Department of Chemical Engineering Clarkson College of Technology, Potsdam, New York 13676			Telephone: (315) 268-6648 (315) 268-2302	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Low-melting glasses such as sodium borate	M	5-25 DIA	Will study surface-tension driven flows, bubble migration, and bubble centering in microgravity environment aboard space shuttle flights. Also will study similar phenomena in ground-based laboratory.	-

69				
Name(s): William R. Wilcox				
Inst. & Address: Clarkson College of Technology, Potsdam, NY 13676			Telephone: (315) 268-6650	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Camphor, camphene, succinonitrile, CBr ₄ , naphthalene	Z, Br	10x100	Used for studies of particle rejection and of bubble generation during solidification. Reported in J. Crystal Growth, Separation Science & Technology, and Proceedings of the 15th Aerospace Sciences Meeting (1978).	-

70				
Name(s): M. J. Skove and E. P. Stillwell				
Inst. & Address: Department of Physics, Clemson University, Clemson, S. C. 29631			Telephone: (803) 656-3417	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tin, Indium, Bismuth, Lead, Aluminum and various alloys of these metals, all in "whisker" form	Growth From the solid state	0.002x0.002 x 10	All of these are from source materials at least 99.999%. Tin is available isotopically pure.	

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Name(s): Alan Carlson, Lee Shiozawa				
Inst. & Address: Cleveland Crystals, Inc. Box 17157, Euclid, Ohio 44117			Telephone: (216) 486-6100	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdS Cadmium Sulfide	sub	30x30x25	5-9's pure, as grown or S-compensated, doped on special order.	stocked
CdSe Cadmium Selenide	sub	40x30x25	5-9's pure, as grown or Se-compensated, doped on special order.	stocked
CdTe Cadmium Telluride	sub	(15) ³	5-9's pure, as grown, also polycrystalline	stocked
ZnSe Zinc Selenide	sub	5x5x10	5-9's pure, as grown	
ZnTe Zinc Telluride	sub	(15)	5-9's pure, as grown, any orientation available; cubic materials may have (111) twin planes parallel to major surface.	stocked
BaTiO ₃	Flu	1x1x0.5	size, thickness and freedom from flaws vary from piece to piece. Ferroelectric, Inquiries welcomed	limited supply

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Name(s): Allan Carlson, Lee Shiozawa

Inst. & Address: Cleveland Crystals, Inc. Box 17157, Euclid, Ohio 44117

Telephone: (216) 486-6100

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KH_2PO_4 , KDP	sol	(100) ³	> 99.8% pure, all orientations, piezoelectric, and electro-optic, SHG; available doped with Cr^{3+} & Na^+	stocked
KD_2PO_4 , KD*P	sol	(100) ³	> 99.8% pure, all orientations, electro-optic, piezoelectric, SHG stocked in 99%, 90% grades, other levels on special order	stocked
$\text{NH}_4\text{H}_2\text{PO}_4$, ADP	sol	(75) ³	> 99% pure, all orientations, Electro-optic, piezoelectric, SHG, available doped with Cr^{3+}	stocked
$\text{NH}_4\text{H}_2\text{AsO}_4$, ADA	sol	(25) ³	> 99% pure, all orientations, electro-optic, piezoelectric, SHG	stocked
KH_2AsO_4 , KDA	sol	25x10x10	> 99% pure, electro-optic, piezoelectric, SHG W.R. Cook Jr., J. Appl. Physics, 38, 1637-42 (1967) J. Inorg Nucl. Chem 25, 844-7 (1967)	stocked
$\text{KB}_5\text{O}_8 \cdot 4\text{H}_2\text{O}$, KBS	sol	(25) ³	> 99% pure, all orientations, SHG, piezoelectric. C.F. Dewey Jr., W.R. Cook Jr., R.T. Hodgson & J.J. Wynne Appl. Phys. Lett. 26 714-16 (1975)	stocked
Triglycine Sulfate, TGS	sol	(25) ³	> 99% pure, ferroelectric, pyroelectric	stocked
LiIO_3	sol	(20) ³	> 98% pure, SHG, pyroelectric, piezoelectric	stocked
+ many other noncentrosymmetric crystals	sol	variable	purity & orientations available, & quantities are limited We invite inquiries	stocked
NaClO_3	sol	25x25x3	> 99.5% pure, (100) orientation, piezoelectric	limited stock
NaNO_3	sol	15x15x2	> 99% pure, rhombohedron surface, calcite substitute	stocked
$\text{Li}_2\text{SO}_4 \cdot \text{H}_2\text{O}$	sol	(25) ³	> 99% pure, all orientations, pyroelectric, piezoelectric	limited stock
K_2 Tartrate $\cdot \frac{1}{2}\text{H}_2\text{O}$	sol	(25) ³	> 98% pure, all orientations, piezoelectric	stocked
Rochelle salt	sol	(25) ³	> 98% pure, all orientations, piezoelectric, ferroelectric	limited stock

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Name(s): David L. Olson, D. K. Matlock, G. R. Edwards, and W. A. Averill

Inst. & Address: Department of Metallurgical Engineering
Colorado School of Mines Golden, CO 80401

Telephone: (303) 279-0300

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Lithium			The compatibility of ferrous alloy for the containment of liquid lithium is being established. Weight loss and grain boundary penetration kinetics of ferrous alloys as a function of temperature, stress and lithium purity is being determined. Role of interstitial contaminants on the corrosion behavior and the effective use of gettering materials is being studied. Lithium metal extraction techniques are also being investigated.	

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Name(s): S. Geller

Inst. & Address: Dept. of Elec. Engrg., Univ. of Colorado, Boulder, CO 80309

Telephone: (303) 492-7157

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$(C_5H_5NH)Ag_5I_6$: Pyridinium hexaiodo- pentaargentate	<u>Gel</u> <u>Sol</u> (Poly)	1 x 1 x 5	Solid electrolyte: Crystal structure, silver ion site distribution, conductivity, phase transitions between -30° and $125^\circ C$. S. Geller, Science <u>17b</u> , 1016 (1972), S. Geller and B. B. Owens, J. Phys. Chem. Solids <u>33</u> , 1241 (1972); phase transitions below $-30^\circ C$, T. Hibma and S. Geller, J. Solid State Chem. <u>20</u> , 209 (1977) (also T. Hibma, Phys. Rev. <u>B15</u> , 5797 (1977)).	-
$(C_5H_5NH)_5Ag_{18}I_{23}$: Pentapyridinium 18-silver iodide	<u>Gel</u> (Poly)	1 x 1 x 5	Solid electrolyte: Crystal structure and conductivity, S. Geller and P. M. Skarstad, Phys. Rev. Letters, <u>1484</u> (1974); S. Geller, P. M. Skarstad and S. A. Wilber, J. Electrochem. Soc. <u>122</u> , 332 (1975).	-
$RbAg_4I_5$: Rubidium silver iodide	<u>M</u>		Solid electrolyte: Crystal structures of β - and γ - $RbAg_4I_5$, S. Geller, Phys. Rev. <u>B14</u> , 4345 (1976).	-
$Ag_{26}^{18}W_4O_{16}$: 26-Silver 18-iodide tetratingstate	<u>M</u> (Poly) <u>Cz</u>	 7 x 4 x 110	Solid electrolyte: Crystal structure and conductivity, L. Y. Y. Chan and S. Geller, J. Solid State Chem. <u>21</u> , 331 (1977). [110], [010], [001], [100] Large crystals now being grown.	-
$(C_5H_5NH)_2Cu_5Br_7$: Dipyrindinium heptabromopentacuprate	<u>Br</u>	6 diam 20 long	Solid electrolyte: Crystal structure, L. Y. Y. Chan, S. Geller and P. M. Skarstad, J. Solid State Chem. <u>25</u> , 85 (1978). For growth see P. M. Skarstad and H. S. Parker, J. Crystal Growth, in press.	-

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Name(s): S. Geller

Inst. & Address: Dept. of Elec. Engrg., Univ. of Colorado, Boulder, CO 80309

Telephone: (303) 492-7157

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$Ag_8W_4O_{16}$: Silver tetratingstate (High temperature phase)	<u>Cz</u>	13 diam 40 long	$\sim[110]$, Crystal structure, polarization, P. M. Skarstad and S. Geller, Mat. Res. Bull. <u>10</u> , 79 (1975); high temperature Raman spectroscopy, A. Turković et al., Mat. Res. Bull. <u>12</u> , 189 (1977).	-
$(C_4N_2H_{12})Ag_{10}I_{12} \cdot 4C_3H_7NO$: Piperazinium 10-silver 12-iodide $\cdot$ 4-dimethyl- formamide	<u>Sol</u>	0.1 x 0.1 x 10	[001], conductivity, crystal structure; S. Geller, L. Y. Y. Chan and G. F. Ruse, Mat. Res. Bull. <u>13</u> , 339 (1978). (See also June 1978 issue)	-
$SrAg_2I_4 \cdot 8H_2O$, Strontium Tetraiodo- diargentate Octahydrate	<u>Sol</u>	2 x 2 x 15	[001], Crystal structure, S. Geller and T. O. Dudley, J. Solid State Chem. to appear <u>26</u> , #3 (Nov. 1978).	-
$RGaO_3$, $R \equiv Sm$ to Er Perovskite-like rare earth gallium oxides	<u>M</u>		Prepared at $1650^\circ - 2000^\circ C$. Lattice constants determined, S. Geller, P. J. Curlander and G. F. Ruse, Mat. Res. Bull. <u>9</u> , 637 (1974).	-
$YGaO_3$: High temperature modification of yttrium gallium oxide. Also isostructural $HoGaO_3$ and $ErGaO_3$	<u>M</u>	0.1 x 0.1 x 0.1	Crystal structure of hexagonal $YGaO_3$ quenched from $1950^\circ C$, S. Geller, J. B. Jeffries and P. J. Curlander, Acta Cryst. <u>B31</u> , 2770 (1975).	-

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Name(s): S. Geller

Inst. & Address: Dept. of Elec. Engrg., Univ. of Colorado, Boulder, CO 80309

Telephone: (303) 492-7157

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NdGaO ₃ : Neodymium gallium oxide	Cz	8 diam 35 long	[100], [010], ultimately for substrate use; G. F. Ruse and S. Geller, J. Cryst. Growth, <u>29</u> , 305 (1975).	-
YIG(Si): Yttrium iron garnet doped with silicon	Flu	largest ~ 10 mm	0.12 Si ⁴⁺ per formula unit: magnetic circular dichroism determination of effects of O ₂ and N ₂ annealing; B. Antonini, et al. Appl. Phys. Lett. <u>22</u> , 700 (1975); 0.12 Si ⁴⁺ - 0.3 Si ⁴⁺ per formula unit: Saturation of Fe ²⁺ ion concentration, P. Paroli and S. Geller, J. Appl. Phys. <u>48</u> , 1364 (1977).	-

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Name(s): Edward Kostiner
Institute of Materials Science
University of Connecticut

Inst. & Address: Storrs, CT 06268

Telephone: 203-486-4615

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Apatites: Ca ₁₀ (PO ₄) ₆ (OH) ₂ Ca ₁₀ (PO ₄) ₆ Cl ₂	Hyd Hyd & Flu	2 x 2 x 5 3 x 3 x 8	High purity and stoichiometry; defect structures by ESR	
Hydroxyphosphates and arsenates: M ₂ ²⁺ (PO ₄) _X , X = OH, F, Cl	Hyd & Flu	1 x 1 x 1	Single crystal structural studies (XRD)	
M ₅ ²⁺ (PO ₄) ₂ (OH) ₄ and other stoichiometries	Hyd	1 x 1 x 1	" " " "	
SiO ₂ (¹⁷ O enriched)	Hyd	2 x 2 x 3		
GeO ₂ (pure and doped)	Hyd & Flu	2 x 2 x 3	ESR studies of radiation-induced defects	

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Name(s): B. Chamberland

Inst. & Address: University of Connecticut, Department of Chemistry, Storrs, CT 06268

Telephone: (203) 486-2743

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ba ₃ Pt ₂ O ₇	Flu	.5x.5x1.0	STRUCTURE, MAGNETIC, THERMAL PROPS, CONDUCTIVITY J. Solid State Chem., <u>21</u> , 217 (1977)	-
BaIrO ₃	Flu	.5x.5x2.0	STRUCTURE, MAGNETIC, CONDUCTIVITY (to be published)	+
CrO ₂	H.P.	.5x.5x2.0	STRUCTURE, MAGNETIC, CONDUCTIVITY, STABILITY Mater. Res. Bull., <u>2</u> , 827 (1967)	+
VO ₂ (H.P.)	H.P.	.5x.5x2.0	STRUCTURE, MAGNETIC, CONDUCTIVITY (TRANSITION TEMP.) J. Solid State Chem., <u>7</u> , 377 (1973)	+
SrRh ₂ O ₄	Flu	.01x.5x.5	STRUCTURE (to be published)	-
LiVO ₂ (H.P.)	H.P.	.5x.5x.5	STRUCTURE, MAGNETIC, CONDUCTIVITY (to be published)	-
BaCrO ₃ (several polytypes)	H.P.	1 x 1 x 2	STRUCTURE, MAGNETIC, CONDUCTIVITY Inorg. Chem., <u>8</u> , 236 (1969)	+
NiCrO ₃	H.P.	.5x.5x1	STRUCTURE, MAGNETIC J. Appl. Phys., <u>40</u> , 434 (1969)	+
SrCrO ₃	H.P.	.5x.5x1	STRUCTURE, MAGNETIC, CONDUCTIVITY Solid State Commun., <u>5</u> , 663 (1967)	+
CaCrO ₃	H.P.	.5x.5x.5	STRUCTURE, MAGNETIC, CONDUCTIVITY J. Solid State Chem., <u>3</u> , 529 (1971)	-
SrVO ₃	H.P.	.5x.5x1	STRUCTURE, MAGNETIC, CONDUCTIVITY J. Solid State Chem., <u>3</u> , 243 (1971)	-
Ba ₃ Re _{2-x} O ₉ H.P. = high pressure-high temperature grain growth.	H.P.	.5x.5x1	STRUCTURE, MAGNETIC, IR Inorg. Chem., <u>17</u> , 699 (1978)	-

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Name(s): Theo Z. Kattamis

Telephone: 203-4862816

Inst. & Address: Department of Metallurgy, Institute of Materials Science,
University of Connecticut, Storrs, CT 06268

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ni-Al-Ta	M	5 dia x 200 long	Dendritic monocrystals. Study of the interaction between composition of the alloy, microsegregation and dendritic coarsening (Ostwald Ripening) during growth.	

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Name(s): Bernard F. Addis

Telephone: 256-5001

Inst. & Address: Materials Preparation Facility, Cornell University, Ithaca, NY 14853
Bard Hall

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tungsten single crystals	Z	1/16" - 1/2" Diam. 8" long	99.999% purity or better oriented $\pm 2^\circ$	+
Nb, V, Ta, Ti, Mo, Re	Z	Up to 1/2" Diam. 8" long	99.999% purity - oriented $\pm 2^\circ$	+
Ni, Pt, Pd, Ir	Z	3/8" Diam. x 3" long	99.999% purity	+
Cu, Zn, Ag, Au, Al	M, Br	1" Diam. x 4" long	99.999% purity	+
In, Sn	M, Br	1" Diam. x 4" long	99.999% purity	
Bicrystals of Nb, V, Ta, W, Mo & Re	Z + EB welding	1/8 x 1/2 x 1-1/2	99.999% purity	
Alloy metal Crystals Grown, Arc cast and Zone Refined -	Z	1/4 x 3"		

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Name(s): CRYSTAL GROWING FACILITY, MATERIALS SCIENCE CENTER, Gerhard Schmidt, Facility Manager
 Inst. & Address: CORNELL UNIVERSITY, ITHACA, N.Y. 14853 Telephone: 607-256-3340

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CaF ₂ - Calcium Fluoride	Cz	25x25x80	All Starting Materials are highest purity available.	
CsBr - Cesium Bromide	Cz	20x20x80		
CsCl - Cesium Chloride	Cz	20x20x80	Most of these crystals are grown in the 100, 110 or 111 direction and can be doped with a wide variety and concentration of materials.	
CsF - Cesium Fluoride	Cz	20x20x80		
CsI - Cesium Iodide	Cz	20x20x80		
KBr - Potassium Bromide	Cz	30x30x100		
KCl - Potassium Chloride	Cz	30x30x100		
KF - Potassium Fluoride	Cz	20x20x80		
LiBr - Lithium Bromide	Cz	20x20x80		
LiCl - Lithium Chloride	Cz	20x20x80		
LiF - Lithium Fluoride	Cz	30x30x100		
MgF ₂ - Magnesium Fluoride	Cz	25x25x80		
MnF ₂ - Manganese Fluoride	Cz	20x20x60		
NaBr - Sodium Bromide	Cz	25x25x80		
NaCl - Sodium Chloride	Cz	30x30x100		
NaF - Sodium Fluoride	Cz	30x30x80		
NaI - Sodium Iodide	Cz	20x20x80		
RbBr - Rubidium Bromide	Cz	20x20x80		
RbCl - Rubidium Chloride	Cz	20x20x80		
RbI - Rubidium Iodide	Cz	20x20x80		
KCoF ₃ - Potassium Cobalt Fluor.	Cz	20x20x50		
KMnF ₃ - Potassium Manganese Fl.	Cz	20x20x50		
RbCoF ₃ - Rubidium Cobalt Fluoride	Cz	15x15x50		
RbMnF ₃ - Rubidium Manganese Fluo.	Cz	20x20x75		

82

Name(s): Crystal Optics Research, Inc.
 Inst. & Address: P.O. Box 2756, Escondido, CA. 92025 Telephone: 714-741-7080

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Aluminum Oxide	CZ	35 mm dia. 40 cm lg.	Purity: 99.99% Orientation: 0°, 60°, 90° Dopant: Cr	stock
Yttrium Aluminum Garnet	CZ	30 mm dia. 40 cm. lg.	Purity: 99.95% Orientation: (100), (111) Dopants: Cr, Nd	stock

83 Name(s): Frederick Schmid and Chandra P. Khattak Inst. & Address: Crystal Systems, Inc., Shetland Industrial Park 35 Congress St., Salem, MA 01970 (617) Telephone: 745-0088				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Sapphire	M (HEM)	320 mm diameter x 130 mm high	Method: Heat Exchanger Method (HEM) of Casting by Directional Solidification Purity: 99.999+% Orientation: (10T0), (11Z0), (0001) Characterization: No light scattering; Transmission, Dislocation Density Applications: Infrared, Optical Publications: F. Schmid and D. Viechnicki, J.Am.Ceram.Soc. 53, 528 (1970); F. Schmid and D. Viechnicki, Solid State Technology 16, 45 (1973); D. Viechnicki and F. Schmid, J.Crystal Growth 26, 162 (1974).	Yes
Silicon	M (HEM)	152 mm diameter x 70 mm high	Method: Heat Exchanger Method (HEM) of Casting by Directional Solidification Purity: Semi-conductor grade Orientation: (100) and (111) Dopants: Boron Properties: Crystallinity, dislocation density, solar cell efficiency, minority carrier lifetime, diffusion length Application: Photovoltaic Publications: C. P. Khattak and F. Schmid, "Casting Large Silicon Crystals in Clear Silica Crucibles," Am.Ceram.Soc.Bull. (in press); C.P.Khattak and F. Schmid, "Directional Solidification of Crack-free Silicon Ingots by Heat Exchanger Method, IEEE Photovoltaic Specialists Conf., Wash.DC (1978)	Yes

84 Name(s): Lawrence J. Castelli, Sales Manager Crystal Technology, Incorporated Inst. & Address: 2510 Old Middlefield Way Mountain View, CA 94043 Telephone: (415) 961-9311 TWX: 910-379-6625				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiNbO ₃ , lithium niobate	Cz	75x75x300	for optical and acoustic applications	stock
LiTaO ₃ , lithium tantalate	Cz	35x35x 75	for optical and acoustic applications	stock
TeO ₂ , tellurium dioxide	Cz	50x50x 75	for acousto-optical applications	stock
Bi ₄ Ge ₃ O ₁₂ , bismuth germanate	Cz	60x60x100	for X-ray scintillator applications	stock
Bi ₁₂ GeO ₂₀ , bismuth germanium oxide	Cz	30x30x250	for optical and acoustic applications	stock
Bi ₁₂ SiO ₂₀ , bismuth silicon oxide	Cz	30x30x250	for optical and acoustic applications	stock
SiO ₂ , quartz	Sol	-	for acoustic applications	stock

85 Name(s): W.H. Robinson Inst. & Address: Crysteco, Inc. 180 East Main Street, Wilmington, OH 45177 Telephone: 513-382-5555				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon	Cz	100 mm dia x 1000 mm long	Semiconductor grade; 6PPba (up to 40 ohm-cm P-type or 12 ohm-cm N-type); zero dislocation density; 1-1-1, 1-0-0 and 1-1-0 orientations	+

86

Name(s): H. Windawi

Inst. & Address: Center for Catalytic Science and Technology, Univ. of Del.
Newark, Delaware 19711

Telephone: (302) 738-2543

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Zirconia-supported nickel Ni/ZrO ₂	SOL	3 X 3 X 5	2 to 5% Ni, ZrO ₂ remainder. Generally pellets of irregular shapes are produced and are used as a methanation catalyst. Characterization by AES, ESCA, SEM, EDS, Laser Raman, EXAFS and X-ray diffraction. UHP materials used.	-

87

Name(s): ROBERT G. HUNSPERGER

Inst. & Address: DEPARTMENT OF ELECTRICAL ENGINEERING
UNIVERSITY OF DELAWARE, NEWARK, DELAWARE 19711

Telephone: (302) 738-2405

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ga _(1-x) Al _x As Gallium Aluminum Arsenide	LE	10x10x0.1	Growth of multilayer structures using slide-bar technique on (100) GaAs substrates - applications, opto-electronic devices and optical integrated circuits (OIC's)	
Zn ₃ P ₂ Zinc Phosphide	MBE VP	10x10x0.1	Growth of thin films - applications, solar cells and optoelectronic devices	

88

Name(s): J. D. Meakin

Inst. & Address: Institute of Energy Conversion, University of Delaware,
One Pike Creek Center, Wilmington, DE 19808

Telephone: 302-995-7155

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Zinc phosphide, Zn ₃ P ₂	VP	8 x 8 x 20	99.9995 + %. Random orientation. Optical absorption, electronic properties. Photovoltaic absorber.	
CdS	Sub.	60 cm ² x .03 mm thick	evaporated films with preferred c-axis orientation $\perp$ to substrate; $0.1 < \rho < 100 \Omega\text{-cm}$.	
Zn _x Cd _{1-x} S (0 ≤ x ≤ 0.4)	Sub.	60 cm ² x .03 mm thick	evaporated films with preferred c-axis orientation $\perp$ to substrate; $0.1 < \rho < 10^4 \Omega\text{-cm}$.	

89

Name(s): Dr. Leon D. Crossman, Dr. Jim McCormick

Inst. & Address: Dow Corning Corporation, 12334 Geddes Rd., Hemlock,
Michigan 48626

Telephone: (517)642-5201

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon (Si)	CVD	200 mm x 1000 mm	Polycrystalline Silicon for Electronics Industry	+
	CZ	125 mm x Dia.	Single Crystal Ingot and Wafers for Electronics Industry	+
	FZ	100 mm x Dia.	" "	"
	CVD	135 mm x 2150 mm	Polysilicon Diffusion Tubes	+

90

Name(s): Rex Trammell

Inst. & Address: EG&G ORTEC
100 Midland Road, Oak Ridge, TN 37830 U.S.A.

Telephone: (615) 482-4411

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Germanium	Cz	50x50x200	Suitable for lithium compensated radiation detectors.	
Germanium	Cz	50x50x100	Impurity concentration <1 part in 10^{12} , suitable for high purity germanium radiation detectors.	

EM Laboratories, Inc.
500 Executive Boulevard
Elmsford, New York U.S.A. 10523
Att: Electrooptics Division
Phone 914/592-4660

associate of
E. Merck, Darmstadt, Germany
Postfach 4119
D-6100 Darmstadt
Phone 06151-721

MATERIALS FOR ELECTRONIC AND OPTIC APPLICATIONS

Materials used to prepare synthetic single crystals for optical and electronic applications are subject to stringent purity requirements and E. Merck, Darmstadt, has drawn upon extensive experience in producing chemicals of extraordinary purity to assemble a program of materials specifically designed for this application. These chemicals are marketed under the tradename Optipur.^R

EM's optical program features chemicals additionally designated as "FO", which are specially suited for the production of fiber optics. These substances rank among the purest materials commercially available; transition metal impurities which might affect fiber attenuation are analyzed and guaranteed in the range of several parts per billion. Materials designated Optipur HP, specially suited for hot pressing applications, are also available.

Other specially designed and applications tested materials are Patinal^R chemicals for vacuum deposition and Suprapur^R reagents of the highest purity.

Cited below is a partial listing of the many high purity chemicals available. Further information and brochures may be obtained from the local EM representative.

Acids	Lithium compounds	Sodium bromide
Ammonium aluminum sulfate	Magnesium fluoride	Sodium carbonate
Ammonium titanyl oxalate	Manganese fluoride	Sodium chloride
Arsenic (III) Chloride	Mercury	Sodium fluoride
di-Arsenic trioxide	Mercury compounds	Sodium iodide
Barium compounds	Magnesium nitrate	Sodium nitrate FO
Bismuth oxide	Niobium oxide	di-Sodium tetraborate
Boric Acid	Phosphorous oxide trichloride FO	Solvents
di-boron trioxide	Phosphorous tribromide	Strontium compounds
Boron tribromide	Phosphorous trichloride	Tetraethyl orthosilicate FO
Cadmium selenide	Potassium bromide	Thallium bromide
Cadmium sulfide	Potassium carbonate FO	Thallium chloride
Calcium carbonate	Potassium chloride	Thallium iodide
Calcium fluoride	Potassium dihydrogen arsenate	Titanium dioxide
Cesium bromide	Potassium fluoride	Triglycine sulfate
Cesium chloride	Potassium iodide	Zinc selenide
Cesium iodide	Potassium nitrate FO	Zinc sulfide
Chromium	Potassium sodium tartrate	Zirconium dioxide
Chromium compounds	Rubidium compounds	
Germanium tetrachloride FO	Silicon dioxide FO	
Lead (II) Fluoride	Silver chloride	
Lead (II) oxide FO	Silver peroxide	

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Name(s): J. E. Powderly, Contracts Manager

Inst. & Address: Eagle-Picher Industries, Inc.
P.O. Box 1090, Miami, OK 74354

Telephone: (918)542-1801

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdS Cadmium Sulfide	VP	30x30x30	99.999% purity of synthesized II-VI powders.	+
CdSe Cadmium Selenide	M	20x20x10	Orientation available for single crystals.	+
ZnS Zinc Sulfide	Br	15x15x5	Almost any dopant usable in the melt/	+
ZnSe Zinc Selenide	Sub	15x15x5	Bridgman crystals. Applications as wave plates, substrates, IR windows, solar cells, etc.	+
CdO Cadmium Oxide		<2mm ³		+
CdTe Cadmium Telluride		10x10x5		+
ZnO Zinc Oxide		<2mm ³		+
ZnTe Zinc Telluride		<3mm ³		+
B Boron	CZ	5mm dia x 30 mm l (poly)	Purity to 99.999%; beta rhombohedral. Rods and crystalline powder available. Also isotopic Boron-10, Boron-11 products.	+
InP Indium Phosphide				-
GaAs Gallium Arsenide	Br		Research on improved methods of crystal growth.	-

93 Name(s): Dr. Charles F. Pulvarl Inst. & Address: Electrocrystal Corporation, Inc. - 2014 Taylor St., N.E. Washington, D. C. 20018 Telephone: 202/526-2400				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
HI-V Compounds Oxides Arsenides Nitrates Trinitrotoluene (TNT) Pentaerythritol Tetranitrate (PETN) Niobates Tantalates	Vapor Melt Melt Water Water Water Molten Salt Molten Salt	Few lbs. Few lbs. Few lbs. Few lbs. Few lbs. Few lbs. Few lbs. Few lbs.	The various types of crystals come in different dimensions some of which can be grown as requested. Some others, especially the high temperature types are usually smaller in size mainly because the small demand on such crystals.	

94 Name(s): G. Perkowitz Inst. & Address: Physics Department, Emory University, Atlanta, Georgia 30322 U.S.A. Telephone: (404) 329-6584				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
The following materials are studied (see comments) GaAs, InAs, InGaAsP, PbTe, PbSe, PbSnTe, PbSnSe, HgTe, HgMnTe		Can deal with sizes from 3mm dia. to 10 mm dia.	These semiconductors and semiconductor alloys are studied in bulk and epitaxial form by far infrared spectroscopy for basic information and as a characterization method. We will consider making such measurements on any interesting samples loaned or given us by crystal growers and will supply the results to the grower. All the listed materials and any related ones are of potential interest. We can determine lattice and free carrier parameters, and on epitaxial samples can probe the interface region.	Not avail- able.

95 Name(s): Juergen F. Schroeter - E. C. Winkelmann Engelhard Industries Inst. & Address: 700 Blair Road Carteret, New Jersey 07008 Telephone: (201) 321-5721				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
. Iridium and Platinum Crucibles	--	180 m/m OD 350 m/m High 3.5 m/m thick	Crystal growing quality with controlled impurity levels	4 weeks
. Platinum, Palladium, Iridium, Rhodium Targets for sputtering		various	Commercial purity and high purity, vacuum melted	6 weeks
. Platinum, Palladium, Iridium, Rhodium, strip, tape, foil or wire		various	Commercial purity and high purity, vacuum melted and low oxygen content	2 weeks
. Platinum 5% Gold seed holders		custom fabricated	Commercial purity	4 weeks

96

Name(s): Lawrence B. Ebert
 Inst. & Address: Exxon Corporate Research Laboratory
 P. O. Box 45
 Linden, New Jersey 07036

Telephone: 201 - 474-3930

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$C_{8-10}AsF_5$	VLS	10 X 10 X 1	XRD(T), ^{19}F NMR (T) ESR (T), Raman, susceptibility	-
$C_{8.7}XeOF_4$, $C_{19}XeF_6$, $C_{13}UF_6$	VLS	0.1 X 0.1 X 0.005	XRD, ^{19}F NMR, ESR	-
$C_{13}CrO_3$, $C_{27}CrO_2Cl_2$	VLS	0.1 X 0.1 X 0.005	XRD, ESR, susceptibility	-

97

Name(s): L.H. Peirce
 Inst. & Address: Department of Physics
 Florida State University, Tallahassee, Fla. 32306

Telephone: (904) 644-6496

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Rubidium Tungsten Bronze Rb_xWO_3	VP	3X3X5	Grown for superconductivity studies	-
Cesium Tungsten Bronze Cs_xWO_3	VP	3X3X5	Grown for Superconductivity studies	-

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Name(s): Prof. Stanley S. Ballard
 Inst. & Address: Dept. of Physics, University of Florida, Gainesville, Fla. 32611

Telephone: 904/392-0487

MATERIALS (Formula & Technical or Common Name)	STUDIED	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Optical Materials, especially the newer ones of use in Infrared Instrumenta- tion (Samples are furnished by the manu- facturers of the optical materials.)		DNA	Solid cylinders 13 mm long; diameter is not critical (6 mm is suggested)	Can determine coefficient of linear thermal expansion in the temperature range 10 K to 500 K.	DNA
" " "		DNA	Disk-shaped samples with plane- parallel faces. Max. diameter 25 mm; max. thickness 1.5 mm (much thinner samples are preferred)	Can measure low-frequency dielectric constant in the temperature range 300-500 K. (Similar measurements, from 60 K to 300 K, are made by Dr. J. S. Browder, Physics Dept., Jacksonville University, Jacksonville, Florida 32211.)	DNA

99

Name(s): SANGHAMITRA SEN, W. R. WILCOX

Inst. & Address: GENERAL ELECTRIC CO.,
175 CURTNER AVENUE, SAN JOSE, CALIF. 95125

Telephone: (408) 925-6498

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{In}_x \text{Ga}_{1-x} \text{Sb}$ Indium Gallium Antimonide	Br	100 mm long 10 mm diam.	<111> , Effects of Magnetic Field, Annealing, on Grain and Twin boundaries. (1) Crystal Growth, 43 (1978) 526 (2) Mat. Res. Bull. Vol. 13, 293, 1978 (3) Met. Trans. A, Vol. 9, 462, 1978	

100

Name(s): Glen A. Slack and Tom F. McNelly

Inst. & Address: General Electric R&D Center, Schenectady, New York 12301

Telephone: (518) 385-8601

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
AlN, aluminum nitride	Sub.	10x3x3	high purity, low oxygen content, undoped; thermal conductivity, thermal expansion, optical properties	-

101

Name(s): Wayne Lo

Inst. & Address: General Motors Research Laboratories
Warren, MI 48090

Telephone: 313/575-2974

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. Lead-Tin-Telluride ($\text{Pb}_{1-x}\text{Sn}_x\text{Te}$) 2. Lead-Sulfide-Selenide ($\text{PbS}_{1-x}\text{Se}_x$)	Sub	10X10X10	1) High purity (6'9 materials used) 2) Cubic structure, (100) orientation 3) No dopant. Carrier concentration and type generated from stoichiometry adjustment 4) Semiconductor lasers 5) Publications: a) W. Lo, G. P. Montgomery, Jr., D. E. Swets, J. Appl. Phys. <u>47</u> , 267 (1976). b) W. Lo, J. Electron Materl. <u>6</u> , 39 (1977) Same as above	

102 Name(s): THE HARSHAW CHEMICAL CO. Inst. & Address: Crystal & Electronic Products Dept. 6801 Cochran Rd., Solon, Ohio 44139 Telephone: 216-248-7400				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
BaClF Barium Chloro Fluoride	Performance specifications will determine growth method, dimensions, and purity. Harshaw growth capabilities include: M, Sol, Flu, VP, Cz, Br, & St.	Ingot size has not to date been a limitation.	Scintillator $\tau = .5\mu s$, 385nm, 4.51g/cm ³	+
BaF ₂ Barium Fluoride			Optics, transmits .135 to 15 μm	+
Bi ₄ Ge ₃ O ₁₂ BGO			Epitaxial growth base	
CdF ₂ Cadmium Fluoride			Scintillator $\tau = .63\mu s$, 325 & 625nm, 4.88g/cm ³	+
CaF ₂ Calcium Fluoride			Scintillator $\tau = .30\mu s$, 480nm, 7.13g/cm ³	+
CaF ₂ Polytran ^R Calcium Fluoride			Scintillator/Optic transmits .206 to 11 μm	+
CaF ₂ (Eu) Europium doped Calcium Fluoride			Optics, transmits .125 to 12 μm	+
CsBr Cesium Bromide			Optics, transmits .125 to 12 μm	+
CsF Cesium Fluoride			Polycrystalline forged optic	
CsI Cesium Iodide			Scintillator $\tau = 1.1\mu s$, 435nm, 3.18g/cm ³	+
CsI Cesium Iodide powder			Optics, transmits .210 to 50 μm	+
CsI(Na) Sodium Doped Cesium Iodide			Scintillator $\tau = .005-.15\mu s$, 335-390nm, 3.59g/cm ³	+
			Optics, transmits .235 to 60 μm	+
			IR pellets	+
			Scintillator $\tau = .65-1.0\mu s$, 420nm, 4.51g/cm ³	+

103 Name(s): The Harshaw Chemical Co. Inst. & Address: Crystal & Electronic Products Dept. 6801 Cochran Rd., Solon, Ohio 44139 Telephone: 216-248-7400				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CsI(Tl) Thallium doped Cesium Iodide	Performance specifications will determine growth method, dimensions, and purity. Harshaw growth capabilities include: M, Sol, Flu, VP, Cz, Br, & St.	Ingot size has not to date been a limitation.	Scintillator, $\tau = 1.1\mu s$, 420-570nm, 4.51g/cm ³	+
PbF ₂ Lead Fluoride - β			Optics, transmits .29 to 11.6 μm	+
LiF Lithium Fluoride			Optics, transmits .11 to 7.0 μm , X-ray	+
LiI(Eu) Europium doped Lithium Iodide			Diffraction plates	
⁶ LiI(Eu) Europium doped 6Lithium Iodide			Scintillator, $\tau = 1.4\mu s$, 440nm, 4.06g/cm ³	+
⁷ LiI(Eu) Europium doped 7Lithium Iodide			Scintillator, $\tau = 1.4\mu s$, 440nm, 4.06g/cm ³	+
LiNbO ₃ Lithium Niobate			Scintillator, $\tau = 1.4\mu s$, 440nm, 4.06g/cm ³	+
MgF ₂ Magnesium Fluoride			Electro-optic	+
MgF ₂ Hot pressed Magnesium Fluoride			Optics, transmits .11 to 7.5 μm	+
MgTiO ₃ Magnesium Titanate			Ceramic IR optic, 2 to 7 μm	+
KBr Potassium Bromide			Microwave ceramic	+
KBr Potassium Bromide powder			Optics, transmits .205 to 25 μm	+
			IR pellets	+

104 The Harshaw Chemical Co. Name(s): Crystal & Electronic Products Dept. 6801 Cochran Rd., Solon, Ohio 44139 Inst. & Address: Telephone: 216-248-7400				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KBr Potassium Bromide Wickstick ^R Elements	Performance specifications will determine growth method, dimensions, and purity. Harshaw growth capabilities include: N, Sol, Flu, VP, Cz, Br and St.	Ingot size has not to date been a limitation	Wick separation and IR spectra	+
KCl Potassium Chloride			Optics, transmits .180 to 20μm	+
KCl Polytran ^R Potassium Chloride			Optics, transmits .180 to 20μm	+
KF Potassium Fluoride			Optics, transmits .160 to 15μm	+
KI Potassium Iodide			Optics, transmits .25 to 40μm	+
Kl(Tl) Thallium doped Potassium Iodide			Scintillator τ = .24μs, 410nm, 3.13g/cm ³	+
RbBr Rubidium Bromide			Optics, transmits .21 to 40μm	+
RbCl Rubidium Chloride			Optics, transmits .21 to 40μm	+
RbI Rubidium Iodide			Optics, transmits .24 to 50μm	+
AgBr Silver Bromide			Optics, transmits .45 to 35μm, sheet and formed optics	+
AgCl Silver Chloride			Optics, transmits .40 to 30μm, sheet and formed optics	+
NaBr Sodium Bromide			Optics, transmits .220 to 23μm	+

105 The Harshaw Chemical Co. Name(s): Crystal & Electronic Products Dept. 6801 Cochran Rd. Solon, Ohio 44139 Inst. & Address: Telephone: 216-248-7400				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NaCl Sodium Chloride	Performance specifications will determine growth method, dimensions, and purity. Harshaw growth capabilities include: N, Sol, Flu, VP, Cz, Br, and St.	Ingot size has not to date been a limitation	Optics, transmits .172 to 17μm, x-ray diffraction plates	+
NaCl Polytran ^R Sodium Chloride			Optics, transmits, .172 to 17μm, 2000psi yield + Polycrystalline forged optic	+
NaF Sodium Fluoride			Optics, transmits .13 to 12μm	+
NaI Sodium Iodide			Scintillator, optic transmits .25 to 25μm	+
NaI(Tl) Thallium doped Sodium Iodide			Scintillator τ = .25μs, 420nm, 3.67g/cm ³	+
NaI(Tl) Polyscin ^R Thallium doped Sodium Iodide			Scintillator rods, plates and unusual unusual geometries	+
NaNO ₃ Sodium Nitrate			Optics, transmits .35 to 3μm	+
SrF ₂ Strontium Fluoride			Optics, transmits .13 to 11μm	+
TlBr Thallium Bromide			Optics, transmits .45 to 45μm	+
TlBr Thallium Bromide powder			IR pellets	+
TlCl Thallium Chloride			Optics, transmits .40 to 30μm	+
TlCl(Be, I) Beryllium and Iodine doped Thallium Chloride			Scintillator τ = 20μs, 465nm, 7.00g/cm ³	+
Tl(Br, I) KRS-5			Optics, transmits .6 to 40μm	+

106 Name(s): The Harshaw Chemical Co. Inst. & Address: Crystal & Electronic Products Dept. 6801 Cochran Rd., Solon, Ohio 44139 Telephone: 216-248-7400				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tl(Cl,Br) KRS-6	Performance specifications will determine growth method, dimensions, and purity. Harshaw growth capabilities include: M, Sol, Flu, VP, Cz, Br, & St.	Ingot size has not to date been a limitation	Optics, transmits .4 to 32 μ m	+
(Ba,Ca)F ₂ T-12			Polyphase optic, transmits 2 to 12 μ m	+
YIG Yttrium Iron Garnet			Microwave ferrite ceramic	+

107 Name(s): C. W. Chu Inst. & Address: Physics Department, University of Houston, Houston, TX 77004 Telephone: (713) 749-2842				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
V ₃ Si (single crystal)	Flux	4x0.9x0.5	Electrical transport and superconducting properties studied	-
Nb ₃ Sn (single crystal)	Flux	"	Electrical transport and superconducting properties studied	-
V ₃ Si, Nb ₃ Ge (films)			Electrical transport and superconducting properties studied	-
ReRh ₄ B ₄ , ReMo ₆ Se ₈	M	4 x 3 x 3	Superconducting and magnetic properties studied	-
CuCl	M, VLD Flux	10 x 5 x 5	Transport, thermal and magnetic	+
V ₂ D	Z	7 x 1 x 1	Transport properties	

108				
Name(s): C. L. Anderson, J. A. Roth, G. S. Picus				
Inst. & Address: Hughes Research Laboratories, Malibu CA 90265			Telephone: (213) 456-6411	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Si	Metal- free solid- phase Epitaxy (SPE)	10x10x0.001	Grown by solid phase epitaxy of amorphous Si deposited on crystalline Si substrates in ultra-high vacuum. (100) only. Ref.: Appl. Phys. Lett. <u>31</u> , 689 (1977)	Research
Ge on GaAs	Metal- free SPE	10x10x0.001	Grown by solid-phase epitaxy of amorphous Ge deposited on crystalline GaAs substrates in ultra-high vacuum. (100) only. Ref. on technique only: Appl. Phys. Lett. <u>31</u> , 689 (1977)	Research

109				
Name(s): G.S. Kamath, G.S. Picus				
Inst. & Address: Hughes Research Laboratories 3011 Malibu Canyon Road, Malibu CA 90265			Telephone: (213) 456-6411	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs	Sol	50x50x2	n,p, pure (10^{13} cm^{-3}), hi ρ	Res.
(AlGa)As	Sol	50x50x2	n,p, pure (10^{13} cm^{-3}), (Photovoltaic Conf.'78)	Res.
InP	Sol	20x20x1	n, high purity ($\mu_{c_{250c}} = 4500 \text{ cm}^2 \text{ v}^{-1} \text{ sec}^{-1}$)	Res.

110				
Name(s): M. Robinson, R.C. Pastor, M. Braunstein, J.A. Harrington, G.S. Picus				
Inst. & Address: Hughes Research Laboratories 3011 Malibu Canyon Road, Malibu CA 90265			Telephone: (213) 456-6411	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiF, NaF	CZ, Br	80		Res.
MgF ₂ , CaF ₂ , SrF ₂ , BaF ₂ , LaF ₃	CZ, Br	60	Melting points as a function of purity. Hardness, press forging, optical properties studied. Matl's grown in various reactive atmospheres; for example HF, CF ₄ , and BF ₃ . CaF ₂ absorption coefficient at 3.8 μm = 10^{-4} cm^{-1} ; SrF ₂ absorption coefficient at 3.8 μm = 10^{-4} cm^{-1} .	Res.
PrF ₃ NdF ₃	CZ	50		Res.
GdF ₃ (Nd ³⁺ doped)	CZ	10	Melting point, hardness	Res.
YF ₃	CZ	3	Melting point	Res.
HoF ₃	CZ	80	Laser quality. Hardness of about 8 on MOH scale. Melting point optical properties.	Res.
ErF ₃	CZ	15	Optical properties	Res.
TbF ₃	CZ	60		
ThF ₄	CZ	60	Hardness, optical properties	Res.
KCl	Br	10x10x100	Absorption coefficient at 5 μm = 10^{-6} cm^{-1}	+

111				
Name(s): K. Zanio, L. Fraas, G.S. Picus				
Inst. & Address: Hughes Research Laboratories 3011 Malibu Canyon Road, Malibu CA 90265			Telephone: (213) 456-6411	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Indium phosphide	VP	10x10x10 ⁻³	Epitaxy on InP, CdS and GaAs Thin film solar cells, microwave devices infrared detectors J. Electron Mat. <u>7</u> , 211 (1978)	Res Progra
Germanium	VP	10x10x10 ⁻³	Epitaxy on Ge and GaAs High efficiency solar cells J. Vac. Sci. Technol. <u>15</u> , 119 (1978)	Res. Progra

112				
Name(s): Dr. Issai Lefkowitz				
Inst. & Address: Department of Physics, Hunter College, CUNY, New York 10031 Department of Physics, Univ. of N. C., Chapel Hill, NY 17415 US Army Research Office, P. O. Box 12211, Research Triangle Park, NC 27709				
			Telephone: (919) 549-0641	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Alkali Metal Tungsten	<u>E C</u>	5 x 5 x 5		+ some- thing
Bronzes	<u>V P</u>			
Cu Cl	<u>V P</u> <u>M</u> Gel	5 x 5 x 5	amorphous & single crystal	+ some- thing

113				
Name(s): M. W. Shafer				
Inst. & Address: IBM T. J. Watson Research Center, Box 218, Yorktown Heights, NY				
			Telephone: 914-945-1007	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
RuO ₂	VP Sub	6 x 4 x 2	Doped with Ir, Mo, & Wo Electrochem ESCA, Transport Properties Measured	
IrO ₂	Sub	6 x 2 x .5	Electrochemical and Transport Measurements	
MoO ₂	VP	5 x 5 x 3	" " "	
WO ₂	VP	4 x 4 x 3	" " "	
Mo ₄ O ₁₁	VP	15x10x1/2	" " "	
Mo ₈ O ₂₃	VP	3 x 2 x 2	" " "	
Mo ₉ O ₂₆	VP	3 x 2 x 1	" " "	
AgI	M Sol	3 x 3 x 2	Raman and Transport Measurements (doped with CuI)	
TaS ₃ TaSe ₃	VP	8 x .5 x .2	Raman Measurements	

114				
Name(s): R. James Kirkpatrick				
Inst. & Address: Dept. of Geology, University of Illinois, Urbana, IL 61801			Telephone: (217) 333-7414	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{CaMgSi}_2\text{O}_6 - \text{CaAl}_2\text{SiO}_6$ Pyroxene	M	3mm (Fibrillar)	Rate of Crystal Growth and Phases Present at Large Undercoolings (Am. J. Sci., 274, p. 215, 1974).	+
$\text{CaMgSi}_2\text{O}_6$ (Diopside) and $\text{CaAl}_2\text{Si}_2\text{O}_8$ (Anorthite)	M	1mm max	Rate of Crystal Growth at Small Undercoolings (J. Geophys. Res. 81, p. 5715, 1976).	-
$\text{CaAl}_2\text{Si}_2\text{O}_8 - \text{NaAlSi}_3\text{O}_8$ (Feldspar)	M	1mm max and Fibrillar	Rate of Crystal Growth at Small and Large Undercoolings (J. Geophys. Res., in press).	-
Natural Basalt	M	1mm max	Rates of Nucleation and Growth in Lava Flows (Geol. Soc. Am. Bull., 88, p. 78, 1977).	+

115				
Name(s): R.A. Milano, M. Feng, T. Lepkowski, L. Zinkiewicz, G.E. Stillman				
Inst. & Address: 155 EEB, University of Illinois, Dept. of Elect. Eng., Urbana, IL 61801			Telephone: (217) 333-3097	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1) GaAs	LE	9.52x12.7x.01	Both high purity ($N_A - N_D \sim 2 \times 10^{14}$) and heavily doped material have been grown. Dopants-Te, Sn, Ge. All work to date have been on <100> oriented substrates. Hall effect and C-V measurements are used to judge quality of material. Applications include near IR detectors and emitters.	X
2) $\text{Al}_x\text{Ga}_{1-x}\text{As}$ (.3 < x < .6)	LE	9.52x12.7x.01	Only Sn-doped material with $1 \times 10^{15} < n < 5 \times 10^{16}$ has been examined for use as heterobarriers.	X
3) InP	LE	9.52x6.35x.358	High purity layers ($n \sim 1 \times 10^{15}$) and pin mesa diodes have been grown on both (100) and (111) oriented substrates.	X
4) $\text{In}_x\text{Ga}_{1-x}\text{As}_y\text{P}_{1-y}$	LE	9.52x6.35x.35	Varying compositions of this material have been grown to cover the 1.1 μ 1.3 μ spectral region. The highest purity to date is $\sim 1 \times 10^{16}$. Pin photo-detectors with reverse breakdown voltages of 50-60V have been fabricated. For further information see M. Feng et.al. Applied Physics Letters, June 1, 1978	X
5) GaAs, InP, InGaAsP	VE		A vapor phase reactor using AsH_3 and PH_3 has been constructed and will be used to grow all of the materials listed	

116				
Name(s): C. T. Sah				
Inst. & Address: Solid State Electronics Laboratory, University of Illinois, Urbana, IL 61801			Telephone: (217) 333-7133	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon	Cz	50x15(LxD)	Bi-doped, In-doped, P & Bi double doped, Al-doped,	-
Silicon	Z	various	Al-doped, P-doped, B-doped	-

117

Name(s): M. E. Salamon

Telephone: (217) 333-6.86

Inst. & Address: Department of Physics, University of Illinois, Urbana, IL 61801

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
RbAg ₄ I ₅ , NH ₄ Ag ₄ I ₅ , KAg ₄ I ₅	Sol	5x5x5	Phys. Rev. <u>17</u> , 269 (1978)	
FeCl ₂ , CoCl ₂ (anhydrous)	M - Br	10x10x2	Multicritical point studies	

118

Name(s): Ralph O. Simmons

Telephone: (217) 333-3760

Inst. & Address: Department of Physics and Materials Research Laboratory
University of Illinois at Urbana-Champaign, Urbana, Ill. 61801

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
³ He, ⁴ He	M	1.6x1.6x15	In ³ He, ⁴ He content less than 300 ppm. Growth at constant pressure, in bcc and hcp phases. Thermal vacancy content measured by temperature dependence of x-ray lattice parameter. J. Cryst. Growth <u>42</u> , 370 (1977)	-
CH ₄ , CD ₄	M	6x6x35	typically 5 to 20 ppm non-isotopic impurities, solid-solid phase transitions, lattice parameters, and thermal expansion studied by x-ray diffraction. Ref. same as He above.	-
Ar, Kr, Xe	M	1x1x15	Growth at constant pressure, to 300 MPa. Study of melting density, (dP/dT), and thermal vacancy properties by x-ray diffraction. phys. stat. solidi (a) <u>43</u> , 611 (1977)	-
H ₂	M	1x1x2	Study of crystal hcp to fcc and fcc to hcp transitions and of molecular ordering versus ortho-para H ₂ composition, at temperatures to 0.1 K, by x-ray diffraction. Prof. 15th Int. Conf. on Low Temp. Physics Grenoble (paper by J. V. Gates et al) 1978.	-

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Name(s): Paul J. Kruth

Telephone: 412-443-5624

Inst. & Address: International Monocrystals, 2596 Elgro Rd., Gibsonia, PA 15044

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Al ₂ O ₃ (Aluminum Oxide)	Ver	12 x 50	(0001), (1010), (1120); 6-9's pure High-temperature windows; unpolished	from stock
CaCO ₃ (Calcite)	Natural	40x40x25	(0001) or natural cleavage faces	"
CdS (Cadmium Sulfide)	VP		High-purity raw material & single crystal pieces	"
GaAs (Gallium Arsenide)		12 dia x 1 thick	Limited quantities; undoped	"
Pb (Lead)	M	75x75x12	100, 110, 111; 5-9's pure; gamma ray, spectrometry plates	"
MgO (Magnesium Oxide)	Arc	10x10x10	100, 110, 111, -- Pure only; high-temp. substrates; infrared & upper UV windows	"
SiO ₂ (Quartz)	Natural and synth.	25x50x12	0001, 1010, 1120; windows, unpolished	"
Other metal crystals: Al, Zn, Sn, Ag, W, Bi, Mg			Limited quantities; limited sizes	"

120				
Name(s):		S. Daigneault/R. Aldrich		
Inst. & Address:		Itek Corp., 10 Maguire Road, Lexington, MA. 02173		Telephone: (617) 272-2207
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Bi}_{12}\text{SiO}_{20}$	Cz	45mm diam.	High optical quality (low strain, core free) crystals grown along (111) and (110) axis for use in Itek's PROM.	
FeO_2	Cz	25 x 25 x 50	High optical quality crystals for use in Tunable Acousto Optic Filters, grown along (011) and (001) axis.	

121				
Name(s):		Dr. Robert B. Somoano		
Inst. & Address:		Jet Propulsion Laboratory 4800 Oak Grove Drive Pasadena, California 91103		Telephone: (213) 354-2213
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{TTF}_{12}\text{SECN}_7$ (Tetrathiafulvalene) $_{12}^{2-}$ (selenacyanate) $_7$	Sol	5 x 0.1 x 0.01	Crystals grown in the form of long needles, with highly reflecting faces; the crystals are organic metals and the electrical, magnetic, optical, and structural properties have been measured.	+
$\text{TTF}_{12}(\text{SCN})_7$ (Tetrathiatetracene) $_{12}^{2-}$ (thiocyanate) $_7$	Sol	5 x 0.1 x 0.01	Ref. R. B. Somoano, A. Gupta, V. Hadek, M. Novotny, M. Jones, T. Datta, R. Deck, and A. M. Hermann, Phys. Rev., B <u>15</u> , 595 (1977).	
TTT_2I_3 (Tetrathiatetracene) $_2$ (iodide) $_3$	Sol	5 x 0.1 x 0.01	Crystals grown in the form of long needles, with highly reflecting faces; the crystals are organic metals and the electrical, magnetic, optical, and structural properties have been measured. Ref. R. B. Somoano, S. P. S. Yen, V. Hadek, S. K. Khanna, M. Novotny, T. Datta, A. M. Hermann, and J. A. Woollam, Phys. Rev. B (1978).	+

122

Name(s): Charles Feldman

Inst. & Address: The Johns Hopkins University, Applied Physics Laboratory
Johns Hopkins Road, Laurel, Maryland 20810

Telephone: 953-7100, Ex. 7100

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
B Boron	Sub	25.4x8.4 x 10 ⁻³	Amorphous films on fused silica. SIMS analysis, electrical, optical measurements. Effect of C & H on electrical conductivity. See: "Amorphous Boron Films" in Boron and Refractory Boride - Ed. V. F. Matkovich, Springer Verlag 1977, p. 581	
Si Silicon	Sub	25.4x8.4 x 10 ⁻²	Polycrystalline and amorphous films on sapphire, glass, fused silica. SIMS analysis, electrical properties studied. Boron doped films. Studies of device applications. See: Evaporated Polycrystalline Silica Films for Photovoltage Application - Grain Size Effects, J. Electron Materials 7, 309 (1978)	

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Name(s): Lambda/Airtron, division of Litton Systems, Inc.

Dr. Roger F. Belt

Inst. & Address: 200 E. Hanover Avenue
Morris Plains, New Jersey 07950

Telephone: (201) 539-5500

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Y ₃ Al ₅ O ₁₂ , yttrium aluminum garnet	Cz	40x40x200	99.99%, [111] [100] rare earths	+
Gd ₃ Ga ₅ O ₁₂ , gadolinium gallium garnet	Cz	50x50x200	rare earths	+
YAlO ₃ , yttrium orthoaluminate	Cz	25x25x150	99.99%, [010] [001] rare earths	+
Y ₃ Fe ₅ O ₁₂ , yttrium iron garnet	Flu	25x25x25	99.99%, Ga	+
CaLa ₄ (SiO ₄) ₃ O, calcium lanthanum silicate	Cz	25x25x150	99.9%, [001] rare earths	+
ZnO, zinc oxide	Hyd	50x50x15	99.99%, [1010] [0001]	+
AlPO ₄ , aluminum phosphate	Hyd	10x10x10	99.99%	+
KTiOPO ₄ , potassium titanyl phosphate	Hyd	5x5x5	99.9%	+
ZrO ₂ , zirconium dioxide	Z	20x25x25	99.9%, Ca, Y, Mg	+
LiYF ₄ , lithium yttrium fluoride	Cz	5x5x150	99.9%, rare earths	+
YF ₃ , yttrium fluoride	Cz	5x5x100	99.9%, rare earths	+
KY ₃ F ₁₀ , potassium yttrium fluoride	Cz	10x10x100	99.9%, rare earths	X

124 Name(s): Robert Goldstein Inst. & Address: Lasermetrics, Inc. 111 Galway Place, Teaneck, New Jersey 07666 Telephone: 201-837-9090				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$(\text{NH}_4)_2\text{H}_2\text{PO}_4$ - ammonium phosphate (ADP)	Sol	150x150x150	All optical quality single crystals 	+
$(\text{ND}_4)_2\text{D}_2\text{PO}_4$ - deuterated ammonium phosphate (AD*P)	"	50x50x75		+
KH_2PO_4 - potassium phosphate (KDP)	"	200x200x220		+
KD_2PO_4 - deuterated potassium phosphate (KD*P)	"	100x100x175		+
$(\text{NH}_4)_2\text{H}_2\text{AsO}_4$ - ammonium arsenate (ADA)	"	30x30x50		+
$(\text{ND}_4)_2\text{D}_2\text{AsO}_4$ - deuterated ammonium arsenate (AD*A)	"	30x30x50		+
CsD_2AsO_4 - cesium dideuterium arsenate (CD*A)	"	25x25x50		-
LiIO_3 - lithium iodate	"	30x30x75		+
$\text{LiCHO}_2 \cdot \text{H}_2\text{O}$ - lithium formate monohydrate	"	50x50x80		+

125 Name(s): Robert Goldstein Inst. & Address: Lasermetrics, Inc. 111 Galway Place, Teaneck, New Jersey 07666 Telephone: 201-837-9090				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NaClO_3 - sodium chlorate	Sol	25x25x25	All optical quality single crystals 	+
NaBrO_3 - sodium bromate	"	25x25x25		+
NaNO_2 - sodium nitrite	"	20x20x40		+
$\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ - nickel sulfate	"	50x50x50		+
$(\text{NH}_2\text{CH}_2\text{COOH})_3\text{H}_2\text{SO}_4$ - triglycine sulfate (TGS)	"	75x75x75		+
$\text{NH}_4\text{OOC}(\text{COONH}_4) \cdot \text{H}_2\text{O}$ - ammonium oxalate monohydrate (AMO)	"	20x20x40		+
$\text{HOOC}_6\text{H}_4\text{COOK}$ - potassium acid phthalate (KAP)	"	75x75x100		+
$\text{Ba}(\text{NO}_2)_2$ - barium nitrite	"	10x10x10		+
Hg_2Cl_2 - mercurous chloride	Sub	10^3		-

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Name(s): J. Washburn, K. Seshan, B. Chin				
Inst. & Address: MMRD, Dept. of Materials Science, Lawrence Berkeley Lab, Univ. of Calif., Berkeley, California 94720				
Telephone: 415-843-2740				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Single crystals of CdS Cadmium Sulphide Cd(Zn)S Cadmium Zinc Sulphide *Grown by Hot Wall Vapor Deposition Process	VP*	1cmx1cm X10-30µm thick	Purity: Not determined. Expected two 9's. pure Orientation: Grown heteroepitaxially on single crystal on (111) and 14° off (111), conducting and semi-insulating GaAs substrates on both Ga and As faces. Dopants: none added. Autodoped n type. Properties: heterojunction with Cu _x S. Hall measurements of mobility, carrier concentration and resistivity. Characterization: SEM-EDAX, TEM, XRD Applications: Heterojunction Solar Cells. Publications: UCID Reports 4044, 4009, 3940 and 3979.	+

127				
Name(s): Eugene E. Haller, G. Scott Hubbard, William L. Hansen				
Inst. & Address: Department of Instrument Techniques, Building 70A, Room 4473, Lawrence Berkeley Laboratory, Berkeley, CA 94720				
Telephone: (415) 843-2740 Exts. 5632 and 5294				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. Ultrapure Polycrystalline Germanium 2. Ultrapure Single Crystal Germanium	Z Cz	35w x 29h x 600λ 45 dia x 200λ	Net electricaly active impurity concentration $\sim 10^{10}$ cm ⁻³ , measured by Hall effect and conductivity. ¹ Residual impurities B, Al, Ga, P, determined by photoelectric spectroscopy (PES). ² [100], [111], [113] oriented crystals grown. Material used for radiation detectors ³ and studies of solid state interactions in an ultrapure medium. ^{4,5,6} 1. IEEE Trans. Nucl. Sci., NS-23, No. 1, 81 (1976) 2. IEEE Trans. Nucl. Sci., NS-22, No. 1, 127 (1975) 3. IEEE Trans. Nucl. Sci., NS-19, No. 1, 265 (1972) 4. Rad. Eff. in Semiconductors 1976, Inst. Phys. Conf. Ser. No. 31, 309 (1977). 5. IEEE Trans. Nucl. Sci., NS-24, No. 1, 48 (1977) 6. Phys. Rev. Lett., Vol. 40, No. 9 584 (1978) +Material may not be sold, small research samples available.	(+)+

128				
Name(s): Guy A. Armantrout				
Inst. & Address: Lawrence Livermore Laboratory, P. O. Box 5504, Livermore, CA 94550 Telephone: 415/422-8616				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cu ₂ S	Reac. Sputt.	0.5 µm Films	Part of Heterojunction Cu ₂ S-CdS Solar Cells	+
Germanium	Cz	7 cm 7 cm 7 cm	High Purity Detector Grade Germanium	+
CdS	Evap. and Sputt.	50 µm Films	Heavily doped CdS for Solar Cells	+

129				
Name(s): Richard L. Landingham				
Inst. & Address: Lawrence Livermore Laboratory, P. O. Box 808, Livermore, CA 94550 Telephone: 415-422-8022				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Sialon Family of Materials	VP	38 dia - 13	Amorphous and crystalline deposits for protective coatings on reaction bonded silicon nitride turbine components (ceramic helical expander) UCRL-77905, UCRL-52390, and UCID-17805.	-

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Name(s): Dr. Herbert W. Newkirk				
Inst. & Address: Lawrence Livermore Laboratory, P. O. Box 808, Livermore, CA 94550 Telephone: (415)-422-9539				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdS Cadmium Sulfide	VP	25		+
CdTe Cadmium Telluride	VP	25		+
AlSb Aluminum Antimonide	M	25		+
BeF ₂ Beryllium Fluoride	Flu & M	25		+
BeO Beryllium Oxide	Flu	10		+
Si Silicon	M	4 in.		+
Ge Germanium	M	4 in.		+
CuS Copper Sulfide	VP	25		+

131				
Name(s): M. J. Weber, W. L. Smith, D. Milam				
Inst. & Address: Lawrence Livermore Laboratory, Box 5508, L-465, Livermore, CA 94550 Telephone: 415/422-8209				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Alkaline-earth fluorides (MgF ₂ , BaF ₂ , PbF ₂ , CaF ₂) Alkali-halides (LiF, NaF, NaCl) Beryllium fluoride crystals & glasses (BeF ₂) Silicates (glasses) Phosphates (glasses) Fluorophosphates (glasses)			Spectroscopic and Nonlinear Optical Properties	

132 Name(s): Leybold-Heraeus Vacuum Products Inc. Inst. & Address: 200 Seco Road, Monroeville, PA 15146 Telephone: 412 856-9030				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Laboratory and Production Crystal Growers	Z Cz Br Kyropoulos	2440 mm x 2440 mm x 8530 mm	Leybold-Heraeus offers upon request, Cz and Hot Zone Units with crucible capacities up to 45 Kgm and diameters up to 127 mm.	

133 Name(s): N. P. Barnes, M. S. Piltch Inst. & Address: Los Alamos Scientific Laboratory P.O. Box 1663 MS 564, Los Alamos, NM 87545 Telephone: (505) 667-7102				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe, Cadmium Telluride	BR	10 x 30 x .2	Temperature dependent sellmeier coefficients and coherence length for infrared second harmonic generation. Journal of the Optical Soc. of America <u>67</u> 628 (1977)	+

134 Name(s): Roger Eckhardt, Lawrence B. Edgett Inst. & Address: Los Alamos Scientific Laboratory P.O. Box 1663, Los Alamos, NM 87545 Telephone: (505) 667-7102				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
AgGaSe ₂	Br	14 mm dia 50 mm length	"The Second Phase Precipitate in AgGaSe ₂ " Eckhardt & Edgett, presented at ICCG5, July 1977	none
CdGeAs ₂	Br	-----	Extensive cracking, low transparency	none

135 Name(s): M. S. Piltch, Roger Eckhardt, Ronald Hinsley Inst. & Address: Los Alamos Scientific Laboratory Box 1663 MS 564, Los Alamos, NM 87545 Telephone: (505) 667-7102				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe, Cadmium Telluride	Br	10 x 30 x .5	Intracavity second harmonic generation of CO ₂ laser radiation. Proceedings of Electro-Optics 77 Conf., Anaheim, CA October 1977.	+
ZnSe, Zinc Selenide	VP	10 x 12 x 2	Measurement of nonlinear optical coefficient and second harmonic coherence length for the non-bire- fringent single crystal. Optics Communications <u>22</u> , 239 (1977).	-

136

Name(s): M. S. Pillich, C. Tallman, J. Rink

Inst. & Address: Los Alamos Scientific Laboratory, P. O. Box 1663, MS 564
Los Alamos, NM 87545

Telephone: (505) 667-4629

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdGeAs ₂ , Cadmium Germanium Arsenide	Br	4 x 10 x 12	Nonlinear infrared pulsed difference frequency mixing using CO ₂ and CO lasers. Optics Communications <u>15</u> 112 (1975).	+

137

Name(s): M. S. Pillich, C. D. Cantrell, R. C. Sze

Inst. & Address: Los Alamos Scientific Laboratory, P. O. Box 1663, MS 564
Los Alamos, NM 87545

Telephone: (505) 667-4629

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe, Cadmium Telluride	Br	9 x 28 x .9	Use of oriented non-birefringent single crystal plates to generate infrared second harmonic radiation of 10 micrometer CO ₂ laser wavelengths. Journal of Applied Physics <u>47</u> 3514 (1976).	+

138

Name(s): William J. McCreary

Inst. & Address: CMB-6, Los Alamos Scientific Laboratory, Los Alamos, NM 87545 USA

Telephone: 505-667-5192

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Mo ₂ C, Molybdenum dicarbide	CVD	<< 1 mm to >> 1 mm	FCC, Crystallite size ~ 40 Å, A ₀ = 4.183 ± 0.008 Å, tensile properties. Hollow spherical laser fusion targets.	-

139

Name(s): Robert W. Springer and Duane S. Catlett

Inst. & Address: CMB-6, Los Alamos Scientific Laboratory, Los Alamos, NM 87545 USA

Telephone: (505) 667-4258

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Aluminum-Al _x O _y Composites	VP	125x125x.05	5 n's pure aluminum is evaporated in a vacuum. During deposition, a precise amount of oxygen is introduced for a short time, thus giving a periodic structure to the material. The period determines the mechanical properties. Characterization: Tensile test, Auger Electron Spectroscopy and Depth Profiling, Scanning Electron Microscopy, Residual Gas Analysis. Paper: Structure and Mechanical Properties of Al- Al _x O _y Vacuum Deposited Laminates submitted to the proceedings of the Third International Metallurgical Coatings Conference. April 1978.	-

140				
Name(s): H. Piller				
Inst. & Address: Dept. of Phys. & Astr., LSU, Baton Rouge, LA 70803			Telephone: (504)388-8477	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$Pb_{1-x}Sn_xTe$ ($x=0.2$)		5x10x1	Electroreflectance and Faraday Rotation in $Pb_{1-x}Sn_xTe$, IL NUOVO CIMENTO, <u>39B</u> , 392 (1977) Single crystals from Dr. R. A. Chapman of Texas Instruments res. labs. in Dallas, Tex.	

141				
Name(s): G. H. Dierksen, T. Gabor				
Inst. & Address: 3M Company, Central Research Laboratories - 201-1E St. Paul, Minnesota 55101			Telephone: (612)733-2497	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cadmium Sulfide Single crystal seeded	VP	cylinder d=35; h=35	Spectrographic Anal., approximately 1 ppm total impurities. Laser quality. J. of Cryst. Growth 43 (1978) 572	(-)

142				
Name(s): Paul R. Camp				
Inst. & Address: University of Maine, Bennett Hall, Orono, Maine 04473			Telephone: (207)581-7745	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ice and zoned refined water	M	56 mm x 56 mm x 600 mm	Purity - C_1 axis	+

143				
Name(s): S.D. Mittleman				
Inst. & Address: Dept. of Electrical Engineering, University of Maine, Orono, Maine 04473			Telephone: (207) 581-7516	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$CuInS_2$ - Copper Indium Sulfide	Br and thin films	10mm dia x 40 mm long	Doped both n- and p-type. Used in solar cell applications. S.S. Comm. <u>22</u> 659 (1977) Proc. 12th Photovoltaic Specialists Conf. <u>12</u> 534, 1977. J.A. Physics <u>48</u> 3878 (1977)	
$CuInSe_2$ - Copper Indium Selenide	Br and thin films	10mm dia x-40mm long.	n- and p-type for solar cell applications. Proc. 12th Photovoltaic Specialist Conf. <u>12</u> 534(1977). Electrical Properties of Schottky Barrier Solar Cells Utilizing Single Crystals of $CuInSe_2$.	

144				
Name(s): S. D. Mittleman and J.F.Vetelino				
Inst. & Address: Department of Electrical Engineering, Barrows Hall, University of Maine Orono, Maine 04473			Telephone: (207) 581-7516	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
AlPO_4 - Berlinite		4x6x10mm	Theoretical Studies and Experimental Verification of Surface Acoustic Wave (SAW) Properties. Esp. Temperature Compensated Cuts Proc. IEEE Ultrasonics Symposium, Sept. 28, 1976 J.A.P. <u>48</u> 887(1977). Appl. Phys. Letters, <u>33</u> , 117 (1978).	

145				
Name(s): W. A. Beck, T. S. Sun, N. E. Byer				
Inst. & Address: Martin Marietta Laboratories, 1450 S. Rolling Rd., Balto., Md. 21227			Telephone: 301/247-0700	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
PbTe and $\text{Pb}_{.8}\text{Sn}_{.2}\text{Te}$	GE (MBE)	10x10x0.01	(111) Orientation; P-type, stoichiometrically adjusted; <u>Properties studied:</u> Hall mobility; conductivity; x-ray diffraction; properties of sputtered & oxi- dized surfaces by Auger, x-ray and ultraviolet electron spectroscopy; Schottky junction proper- ties; effect of electron beam on the sticking co- efficient of film to substrate. <u>Application:</u> Infrared photodiodes <u>Publications:</u> "Oxygen Uptake on Epitaxial PbTe (111) Surfaces," T.S. Sun, N.E. Byer, J.M. Chen, J. Vac. Sci. and Technol., <u>15</u> , 585 (1978). "Oxygen Uptake on Epitaxial $(\text{PbSn})\text{Te}$ (111) Surfaces," T.S. Sun, S.P. Buchner, N.E. Byer and J.M. Chen, To be published in J. Vac. Sci. & Technol. "Electron Beam Delineation of $(\text{Pb},\text{Sn})\text{Te}$ Arrays," W.A. Beck, S.P. Buchner and N.E. Byer. IRIS Spec. Group Meeting, Detector & Imaging Groups, Annapolis, Md., June 15, 1978.	-

146				
Name(s): Robert E. Fahey				
Inst. & Address: Lincoln Laboratory, M.I.T., Lexington, MA 02173			Telephone: 617-862-5500 Ext. 491	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
MgF ₂ -- magnesium fluoride	Thermal	25 (diameter) x 85 (length)	Growth axis of unseeded boules 30° from c-axis. Dopants: Ni ²⁺ , Co ²⁺ Tunable source of coherent infrared radiation Journal of Crystal Growth <u>42</u> , 569 (1977)	Limited
Ba ₂ TiSi ₂ O ₈ -- fresnoite	Cz	15 (diameter) x 40 (length)	Surface acoustic wave properties Appl. Phys. Lett. <u>32</u> , 203 (1978)	Limited

147				
Name(s): S. H. Groves				
Inst. & Address: M.I.T. Lincoln Laboratory, Lexington, Mass. 02173			Telephone: (617)-862-5500 Ext 5836	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
In P	LE	10	Undoped, $N_d - N_a = 2 \times 10^{15} \text{ cm}^{-3}$ $N_d/N_a = 6$	
M _{1-x} Ga _x As _y P _{1-y}	LE	10	Undoped, $N_d - N_a = 2 \times 10^{15} \text{ cm}^{-3}$ $N_d/N_a = 2$	
PbTe, PbSnTe	LE	10	DH diode Lasers	
PbTe, PbSnTe	VP	20	T _g doped, substrate material	

148				
Name(s): John Melngailis, R. E. Fahey and E. Stern				
Inst. & Address: MIT Lincoln Laboratory, Lexington, Mass. 02173			Telephone: 617-862-5500 X5487 or X359	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ba ₂ Si ₂ TiO ₈ Fresnoite	M	40x15x15	Single crystal, no intentional dopants, surface acoustic wave propagation studied, see Appl. Phys. Lett. <u>32</u> p. 203 15 Feb. 1978	(-)

149				
Name(s): Alan J. Strauss				
Inst. & Address: Lincoln Laboratory, M.I.T., Lexington, MA 02173			Telephone: (617) 862-5500 Ext. 481	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe- cadmium telluride	Br	20x20x50	In-doped $\sim 1 \times 10^{17} \text{ cm}^{-3}$ for conversion to semi-insulating. Electrooptic modulators.	small quantities for re- search

150 Name(s): August F. Witt Inst. & Address: Room 13-4134, MIT, Cambridge, Mass. 02139 Telephone: (617)253-5303				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InSb	LE, Cz, Br	20 x 150	Intrinsic to heavily-doped (p & n)	limited
Ge	Cz, Br	20 x 150	Intrinsic to heavily-doped (p & n)	limited
GaAs	LE	0.2	Intrinsic to heavily-doped (p & n)	limited

151 Name(s): Paul Frieberthausen Inst. & Address: McDonnell Douglas Astronautics Company 5301 Bolsa Ave., Huntington Beach, CA 92647 Telephone: (714) 896-1940				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gallium Arsenide, GaAs	VP	diameter to 75 MM	n and P dopants, Hall effect measurements. Carrier concentration, range from 10^{10} to 10^{18} cm ⁻³ .	+

152 Name(s): Gerald L. Pollack Inst. & Address: Physics Department, Michigan State University, East Lansing, MI 48824 Telephone: (517) 355-9709				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ar, Argon	VLS, Z Modified Br	1x1x5	For measurement of thermal conductivity, sublimation pressure, etc.	

153 Name(s): Robert Gerson, P. S. Nayar, E. B. Hale and C. M. Cooper Inst. & Address: University of Missouri - Rolla, Physics Dept., Rolla, MO 65401 Telephone: 341-4793				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SrTiO ₃ (strontium titante)	Ver	5 X 20 X 4X10 ⁻⁴	Boron doping by ion implantation to produce n-type semiconductors for implant doses of 10^{16} to 10^{17} cm ⁻² Resistance varies from 10^6 to 10^3 ohms/square. Conductivity, Seebeck effect, and Hall effect as a function of temperature. Samples are sliced from boule obtained from N L Industries with no deliberate doping. Bull. Am. Phys. Soc. <u>23</u> (1978) 367.	+

154				
Name(s): P. S. Nayar				
Inst. & Address: Dept. Physics, Univ. Missouri-Rolla Rolla, MO 65401			Telephone: 341-4793	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Thallium Selenide (TlSe)	Br, Z	50mm (l) 6mm (d)	1 ohm-Cm to 50 ohm-Cm P-type Conductivity, Hall effect, Seebeck effect, doping and band structure. Low temperature conductivity. Low temperature infrared detector. J. Phys. Soc. Japan 23, 144 (1967) J. Appl. Phys. 39, 359, 4465 (1968) J. Appl. Phys. 40, 123 (1969) J. Phys. Chem. Solids 31, 3167 (1970) Infrared Phys. 14, 31 (1974) J. Opt. Soc. Am. 65, 831 (1975) Appl. Opt. 16, 2942 (1977)	-

155				
Name(s): Monocrystals Company				
Inst. & Address: 1721 Sherwood Blvd., Cleveland, Ohio 44117			Person to contact: Virgil E. Straughan Telephone: 486-2959	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SINGLE CRYSTALS OF METALS AND ALLOYS FROM STOCK OR ON CUSTOM ORDER BASIS				
Antimony	Br	37 x 150 long	49s and 59s, major orientations	
Bismuth	Br	25 d. x 50	59s	
Cadmium	Br	18 d. x 150	59s	
Cobalt	Br	25 d. x 75	49s, random orientation only	
Copper	Br	50 d. x 150	49s and 59s, major orientations	
Indium	Br	12 d. x 100	59s, random orientation only	
Iron	strain anneal	9 d. x 75	39s, oriented discs	
Lead	Br	18 d. x 100	59s	
Nickel	Br	37 d. x 75	Carbonyl Ni, oriented shapes, major orientations	
Silver	Br	25 d. x 150	59s, oriented shapes, major orientations	
Tin	Br	18 d. x 100	49s, 59s, 69s, oriented shapes, major orientations	
Zinc	Br	37 d. x 100	49s, 59s, oriented specimens	
Alloys: CoFe, NiCo, NiFe, NiCu, SiFe, CuZn, and others upon request				

156				
Name(s): Kedar P. Gupta,* Robert O. Gregory and Melvin Rossnick				
Inst. & Address: *Monsanto Company - P. O. Box 8, St. Peters, Mo. 63376 USA			Telephone: 314-434-1698	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon, Semiconductor Grade	Z	50x50x400	Limitations in Using Kiloherzt Radio Frequencies for Float-Zone Silicon Crystals, to be published in J. of Crystal Growth	+

157 Name(s): Nicholas C. Lias, Gen. Mgr.; Joseph F. Balascio, Res. & Dev. Inst. & Address: Motorola, Inc., Quartz Operations, P.O. Box 279, Carlisle, PA 17013 Telephone: (717) 249-1456				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SiO ₂ (alpha quartz)	Hyd	86.4x177.8 x 63.5 X x Y x Z	<u>Purity</u> Acoustic, electronic and optical grades <u>Orientation</u> 0°X, 2°X, 5°X, 35.25°X (AT), 42.75°X (ST) 22°Z, (for TTC cut) <u>Dopants</u> Quotes upon request <u>Properties Examined</u> Optical and piezoelectric <u>Applications</u> Electronic oscillators and filters, SAW devices, narrow bandpass optical filters, fiber optics	+ + +

158 Name(s): Aslan Baghdadi, Richard Gurtler Inst. & Address: Motorola Inc., 5005 E. McDowell Road A-110 Phoenix, Arizona Telephone: 602-244-5921 85008				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon Ribbon	Z	0.1x75x∞	Large grained ribbon suitable for use as a substrate for solar cells.	

159 Name(s): Wm. Ingle, S. W. Thompson, R. Chaney Inst. & Address: Motorola, Inc., 5005 E. McDowell Road, Phoenix, Arizona 85008 Telephone: (602) 244-3127				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Amorphous or Polycrystalline silicon	VP	N.A.	Low cost, high purity process for production of solar silicon; Ingle U.S. Patent #4,070,444; J.P.L./D.O.E. contract #954442.	

160 Name(s): H. Ming Liaw Inst. & Address: Motorola Inc. Semiconductor Research & Development Labs 5005 E. McDowell Road Phoenix, Az 85008 U.S.A. Telephone: (602)244-6328				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Si Single Crystals	Cz	100x100x 650	Purity: 10^{15} atm/cc to 10^{19} atm/cc Orientation: $\langle 100 \rangle$ $\langle 111 \rangle$ Dopants: B, P, As, Sb. Properties Studies: Doping Uniformity Crystalline Defects, Impurity Concentrations Characterization: Chemical Etching, IR Ab- sorption, X-ray Topographs, Spreading Resistance, Diffusion Length, Diode Character- istics Applications: Semiconductor Devices Publication: (1) "Phosphorus Donor Homogeni- zation of Czochralski Crystals by Thermal Neutron Irradiation Overdoping" Semicon- ductor Silicon, Eds H.R. Huff & G. Sirtl, Electrochem. Soc. p 116, (1977). (2) "Influence of Heat Treatment of Silicon Wafers on Infrared Absorption" to be published on Tropical Conf. on Characterization Tech- niques, Electrochem. Soc. 1978	+

161 Name(s): Robert R. Rutherford Inst. & Address: Motorola, Inc., PO Box 20932, Phoenix, AZ 85036 Telephone: 602-244-4133				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon	CZ	51mm 76mm 100mm 125mm	Crystal, Sawed Wafers, Etched Wafers, Polished Wafers 111, 100 Orientations P/Boron, N/Phosphorous, Arsenic, & Antimony	+
	G.E.		Single Layers, Multiple Layers Graded Layers Thickness From 2 to 100 Microns - Complete Resistivity Range	+

162 Name(s): J. G. Hust or R. E. Michaelis Inst. & Address: National Bureau of Standards Boulder, Colo 80303 National Bureau of Standards Washington, D.C. Telephone: 303-499-1000 x3733				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
High purity metals (platinum, gold zinc, aluminum, copper) also numerous physical property standard reference materials	M	wires, blocks, rods	Purities generally about 99.999% or better characterized by chemical composition, low temperature electrical resistivity and thermo- physical properties.	+

163

Name(s): Leslie E. Smith

Inst. & Address: Institute for Materials Research

Telephone: (301) 921-3321

National Bureau of Standards, Gaithersburg, MD 20234

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability +
Polyethylene SRM 1482			Narrow molecular weight distribution linear polyethylene certified for weight and number average molecular weight, and limiting viscosity number. Weight average molecular weight is 13,600.	Through office of Standard Reference Materials National Bureau of Standards, Washington, DC 20234.
Polyethylene SRM 1483			Narrow molecular weight distribution linear polyethylene certified for weight and number average molecular weight, and limiting viscosity number. Weight average molecular weight is 32,100.	
Polyethylene SRM 1484			Narrow molecular weight distribution linear polyethylene certified for weight and number average molecular weight, and limiting viscosity number. Weight average molecular weight is 119,600.	

164

Name(s): Leslie E. Smith

Inst. & Address: Institute for Materials Research

Telephone: (301) 921-3321

National Bureau of Standards, Gaithersburg, MD 20234

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability +
Polystyrene Standard Reference Material (SRM) 705			Narrow molecular weight distribution polystyrene certified for weight and number average molecular weight and limiting viscosity number. Weight average molecular weight is 185,000.	Through Office of Standard Reference Materials National Bureau of Standards, Washington, DC 20234.
Polystyrene SRM 706			Broad molecular weight distribution polystyrene certified for weight and number average molecular weight and limiting viscosity number. Weight average molecular weight is 290,000.	
Polyethylene SRM 1475			Broad molecular weight distribution linear polyethylene certified for molecular weight distribution, melt flow and density. Weight average molecular weight is 52,000.	
Polyethylene SRM 1476			Branched polyethylene certified for limiting viscosity number, melt flow and density.	

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Name(s): W. Murray Bullis, Chief, Electron Devices Division

Inst. & Address: National Bureau of Standards, Washington, DC 20234

Telephone: (301) 921-3786

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Semiconductor Device Grade Silicon			The Electron Devices Division develops and evaluates measurement methods, data, and associated technology for the characterization of electron devices and the materials, processes, and equipment used in their manufacture. Emphasis is on characterization of single-crystal silicon. Selected standard reference materials are under development. Results are announced in <i>Semiconductor Measurement Technology Progress Briefs</i> which can be obtained on request from the Division. Recent publications on silicon characterization methods include: Li, S. S., <i>Semiconductor Measurement Technology: The Dopant Density and Temperature Dependence of Electron Mobility and Resistivity in N-Type Silicon</i>, NBS Spec. Publ. 400-33 (March 1977). Buehler, M. G., and Thurber, W. R., Measurement of the Resistivity of a Thin Square Sample with a Square Four-Probe Array, <i>Solid-State Electronics</i> <u>20</u>, 403-406 (May 1977). Ehrstein, J. R., Effect of Specimen Preparation on the Calibration and Interpretation of Spreading Resistance Measurements, <i>Semiconductor Silicon/1977</i>, H. R. Huff and E. Sirtl, Eds., pp. 377-386 (Electrochemical Society, Princeton, NJ, May 1977).	

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Name(s): W. Murray Bullis, Chief, Electron Devices Division

Inst. & Address: National Bureau of Standards, Washington, D C 20234

Telephone: (301) 921-3786

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
			Li, S. S., and Thurber, W. R., The Dopant Density and Temperature Dependence of Electron Mobility and Resistivity in n-Type Silicon, <i>Solid-State Electronics</i> <u>20</u>, 609-616 (July 1977). Mattis, R. L., and Buehler, M. G., A New Method for Calculating Background Dopant Density from p-n Junction Capacitance-Voltage Measurements, <i>J. Electrochem. Soc.</i> <u>124</u>, 1918-1923 (December 1977). Buehler, M. G., Dopant Profiles Determined from Enhancement Mode MOSFET D-C Measurements, <i>Appl. Phys. Letters</i> <u>31</u>, 848-850 (15 December 1977). Mayo, S., and Evans, W. H., Thermodynamic Considerations in the Use of Polysilicon Oxidation Tubes for Clean SiO₂ Films Preparation, <i>J. Electrochem. Soc.</i> <u>125</u>, 106-110 (January 1978). Thurber, W. R., and Buehler, M. G., <i>Semiconductor Measurement Technology: Microelectronic Test Pattern</i> NBS-4, NBS Spec. Publ. 400-32 (April 1978). Blackburn, D. L., Photovoltaic Technique for Measuring Resistivity Variations of High Resistivity Silicon Slices, <i>NBS J. Res.</i> <u>83</u>, 265-271 (May - June 1978).	

167				
Name(s): W. Murray Bullis, Chief, Electron Devices Division				
Inst. & Address: National Bureau of Standards, Washington, DC 20234			Telephone: (301) 921-3786	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
			Koyama, R. Y., Detection of Deep Levels in High Power Semiconductor Materials and Devices, <i>NBS J. Res.</i> 83, 273-281 (May - June 1978). Myers, D. R., and Phillips, W. E., Existence of an Isotope Shift for the Sulfur Deep Level in Silicon, <i>Appl. Phys. Letters</i> 32, 756-758 (1 June 1978). Koyama, R. Y., Phillips, W. E., Myers, D. R., Liu, Y. M., and Dietrich, H. B., The Energy Levels and the Defect Signature of Sulfur-Implanted Silicon by ITCAP and TSC Measurements, <i>Solid-State Electronics</i> 21, 953-955 (1978).	

168				
Name(s): David A. Ditmars				
Inst. & Address: National Bureau of Standards - Washington, D.C. 20234			Telephone: 301-921-2786	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tungsten	--	--	single crystal, 99.994% purity, random orientation, Heat capacity 300 - 2600 K	-

169				
Name(s): Dr. Masao Kuriyama				
Inst. & Address: National Bureau of Standards, Room A153, Bldg. 223, Washington, D. C. 20234			Telephone: (301) 921-2986	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Aluminum, Copper, Nickel	Cz	100 mm, 25 mmφ	Characterized by x-ray double crystal topography (ACT). Dislocation densities are smaller than 2000 lines/cm ² , for example, M. Kuriyama et. al., <i>J. Cryst. Growth</i> 43 (1978) 287	0+

170				
Name(s): C. J. Eckhardt				
Inst. & Address: Department of Chemistry, University of Nebraska-Lincoln, Lincoln, NE 68588			Telephone: 402-472-2734	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Anthracene	Z	4 x 4 x 10	Piezoreflexion spectra studied	
TCNQ	Sub-sol	1 x 4 x 4	Reflection spectroscopy	-
Charge transfer complexes (non-conducting)	Sub-sol VP	Varied	Piezoreflexion spectra	-

171				
Name(s): D. J. Sellmyer and R. D. Kirby				
Inst. & Address: Department of Physics and Astronomy, University of Nebraska, Lincoln, NE 68588 Telephone: (402)472-2407				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
HoGa ₂	<u>Br</u>	1 x 1 x 3	Resistance ratio = 120	+
V ₅ S ₈	<u>VP</u>	2 x 3 x 4		+
V ₅ Se ₈	<u>VP</u>	2 x 3 x 4		+
NbSe ₃	<u>VP</u>	1 x 1 x 4		+
TiSe ₂	<u>VP</u>	1 x 5 x 5		+
NbSe ₂	<u>VP</u>	1 x 5 x 5		+
Zr ₄₀ Cu ₆₀	<u>M</u>	1 x 10 x 10	splat cooled (amorphous)	+
R ₆₅ Co ₃₅	<u>M</u>	1 x 10 x 10	splat cooled, (amorphous) R = rare earth	+

172				
Name(s): H. M. Dess				
Inst. & Address: NL Industries, Inc., Commercial Crystals Dept. 100 Chevalier Avenue, South Amboy, New Jersey 08879 Telephone: 201-679-7187				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SrTiO ₃ - Strontium Titanate	Ver	25 mm dia. 40-50 mm long	99.95% SrTiO ₃ , typical purity. Random orientation unless otherwise specified by customer. Crystals can be grown doped to order, best effort basis. Applications: optical components, dielectric material, solar energy research.	
TiO ₂ - Rutile	Ver	25 mm dia. 40-50 mm long	99.95% TiO ₂ , typical purity. C-axis approximately parallel to length. Crystals can be grown doped to order, best effort basis. Applications: optical components dielectric material, solar energy research.	
Brochure available on request.				

173				
Name(s): Dr. Ali Abul-Fadl				
Inst. & Address: North Carolina A & T State University Department of Electrical Engineering, Greensboro, N.C. 27411 Telephone: (919)379-7760				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs (single crystal)	LPE	1.2X1.2 X 1-20 μm	10 ¹⁵ -10 ¹⁸ (100), Si, Ge, Hall effect, resistivity	+
GaAs (polycrystal)	LPE & CCLPE*	1.2X1.2X 1-5μm	10 ¹⁶ -10 ¹⁸ , undoped, Si, Ge, Hall effect, Resistivity; for low cost solar cells	+
GaAlAs homojunction	LPE	1.2 X1.2	10 ¹⁶ -10 ¹⁸ (100), Si, Ge, Mg, Hall effect, Resistivity, photoluminescence, for high concentration solar cells	+
* CCLPE-Current Controlled Liquid Phase Epitaxy				

174				
Name(s): J. J. Blauzani				
Inst. & Address: Dept. of Chemistry, State University Raleigh, N.C. 27650			Telephone: (919-737-2996)	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CBr_4 Carbon tetrabromide	Sub.	3x.6x.6 cm	J. Crstal Growth 43,648(1978)	

175				
Name(s): Charles B. Childs				
Inst. & Address: Department of Physics and Astronomy: University of North Carolina at Chapel Hill; Chapel Hill, North Carolina 27514			Telephone: (919) 933-5002	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silver Bromide Silver Chloride	Br	20 x 20 x 200	Purity: less than 1×10^{-6} mole fraction cationic impurities. Dopants: available.	Yes Yes
Silver Bromide Silver Chloride	Cz	20 x 300	Purity: less than 1×10^{-6} mole fraction cationic impurities Dopants: available.	Yes Yes

176				
Name(s): Henn H. Soonpaa				
Inst. & Address: University of North Dakota, Physics Dept., Grand Forks, North Dakota 58202			Telephone: 701-777-2911	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Bi}_{1-x}\text{Te}_x$; $\frac{1}{3} < x < \frac{1}{2}$; Tetradymite; x = 0.48 Bismuth Tellurium Sulfide	M	2x10x40	Thin crystals of x = 0.48 compound can be in intrinsic conductivity range. Bulk crystals are semi-metallic. Rhombohedral axis is perpendicular to the cleavage plane and the direction of growth. Have been used in studies of thermoelectric phenomena, quantum size effects, finite lattice effects, and two-dimensional transport phenomena. 1) D. Harker, Z. Kristallogr. 89 175 (1934) 2) H. H. Soonpaa, Technical Report No. 1940. Mechanical Division, General Mills, Inc. (1960) A D - 235354 3) A. C. Glatz, Am. Mineralogist 52, 161 (1967) 4) L. Pauling, Am. Mineralogist 60, 994 (1975) 5) K. R. Wilhelm and H. D. Bale, J. Phys. Chem. Solids 36, 624 (1975) 6) H. H. Soonpaa, in Physics of Semiconductors, Proc. of the 13th International Conference Rome, 1976. Edited by F. G. Fumi. North-Holland Publishing Co. Amsterdam and New York 1976. p. 364. 7) H. H. Soonpaa, Thin Crystal as a 2D system. Proc. of the 2nd International Conference on the Electronic Properties of 2D Systems. Berchtesgaden 1977. Edited by J. F. Koch and G. Landwehr. Schmitt and Meyer, Würzburg 1977. p. 128.	+

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Name(s): Carlos E. Bamberger

Inst. & Address: Oak Ridge National Laboratory, P. O. Box X, Oak Ridge, TN 37830

Telephone: (615) 574-4944
FTS 624-4944

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
BaCu ₂ O ₂	M	Powder	Preparation from Cu ²⁺ +Ba(OH) ₂ for H ₂ production cycles	-
Cerium (III) Titanates	M or Sol	Powder	Preparation for H ₂ production cycles	-
Copper (II) Hydroxiodides	Sol	Powder	Preparation and identification (Inorg. Chim. Acta <u>27</u> , 275 (1978))	-
Lead (II-IV) Oxyiodides	M & Sol	Powder	Identification and chemical reactions J. Inorg. Nucl. Chem. (in print)	-
Potassium Chromate (V) - K ₃ CrO ₄	M	Powder	Preparation, Enthalpy of formation Can. J. Chem. <u>56</u> , 446 (1978)	-
Strontium Chromate (IV), Sr ₂ CrO ₄	M?	Powder	Preparation and reactions for H ₂ production cycles	-
Strontium Hydroxi Chromate (V), Sr ₅ (CrO ₄) ₃ OH	M?	Powder	Preparation and reactions for H ₂ production cycles. ChemTech., May, 330 (1976)	-

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Name(s): L. A. Boatner

Inst. & Address: Solid State Division, Bldg. 2000
Oak Ridge National Laboratory, Oak Ridge, TN 37830Telephone: (615) 574-5492
FTS 624-5492

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KTaO ₃	Flu	20x10x10	Pure and doped with iron-group elements. Electronic and electromechanical properties are studied as well as the paramagnetic resonance of impurity ions. L. A. Boatner, Abdul-Hamid Kayal, and U. T. Höchli, Helv. Phys. Acta <u>50</u> , 167-169 (1977); U. T. Höchli and L. A. Boatner, J. Phys. C:Solid State Phys. <u>10</u> , 4319 (1977).	+
KTa _{1-x} Nb _x O ₃	Flu	10x10x10	Both pure and iron-doped crystals are grown. Quantum mechanical effects in ferroelectricity are investigated. U. T. Höchli, H. E. Weibel, and L. A. Boatner, Phys. Rev. Letters <u>39</u> , 1158 (1977); L. A. Boatner, U. T. Höchli, and H. E. Weibel, Helv. Phys. Acta <u>50</u> , 620 (1977). Photorefractive effects for optical data storage applications are also studied.	+
K _{1-x} Na _x TaO ₃	Flu	8x8x8	Ferroelectricity at low temperatures is being investigated. Possible applications include photoelectrochemical cell electrodes. L. A. Boatner and U. T. Höchli, Bull. Am. Phys. Soc. <u>23</u> , 197 (1978).	+
K _{1-x} Li _x TaO ₃	Flu	8x8x8		+

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Name(s): L. A. Boatner, M. M. Abraham and Y. Chen

Inst. & Address: Solid State Division, Oak Ridge National Laboratory, P.O. Box X,
Oak Ridge, Tennessee 37830Telephone: (615) 574-5492
FTS 624-5492

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
MgO	Arc Fusion	20x20x10	Both pure and doped single crystals are grown and a wide variety of studies have been performed using these materials (e.g. L. A. Boatner, R. W. Reynolds, M. M. Abraham, and Y. Chen, Phys. Rev. B-16, 86 (1977); M. M. Abraham, Y. Chen, L. A. Boatner, and R. W. Reynolds, Phys. Rev. Letters 37, 849 (1976).	Limited (+)
CaO	"	10x10x5		
SrO	"	10x10x5		
Y ₂ O ₃	Arc Fusion	10x10x8	The growth process can be adapted to produce either stoichiometric or nonstoichiometric crystals.	-
CeO ₂	Arc Fusion	5x5x5	The resulting material is generally highly reduced.	+

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Name(s): L. A. Boatner and W. E. Brundage

Inst. & Address: Solid State Division, Oak Ridge National Laboratory, P. O. Box X,
Oak Ridge, Tennessee 37830Telephone: (615) 574-5492
FTS 624-5492

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
WC	Flu	2 x 2 x 1	A cobalt flux is employed.	+
SrCl ₂	Br	30x10x10	Doped with rare-earth, iron-group, and actinide impurities (e.g. L. A. Boatner, R. W. Reynolds, C. B. Finch, and M. M. Abraham, Phys. Rev. B-13, 953 (1976).	+
PbS	Br	4 x 4 x 4		
CsCl	So1 Br	3 x 3 x 3 5 x 5 x 5		
WO ₃ , MgO, MoO ₂ , SiO ₂ , ZnO, etc.	(Sub) (VP)	10 ⁻⁴	Small oxide particle suspensions for light scattering investigations.	+

180 a				
Name(s): Yung-Kwang Chang				
Inst. & Address: Solid State Division, Oak Ridge National Laboratory, P. O. Box X, Oak Ridge, Tennessee 37830			Telephone: (615) 574-5496 FTS 624-5496	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ti	Z	10 dia. X 100	1) In general, all starting materials were choices of best purity available commercially. Initial casting (arc and/or RF) was done under purified inert atmospheres (argon or helium). Frequently the starting materials were further purified by zone-melting with gas treatment (usually oxygen) or in a high vacuum (10^{-9} torr). 2) Many crystals were grown with seeded orientations. 3) Characterization techniques include chemical etching (for revealing grain boundaries and dislocations), x-ray Laue back reflection, x-ray topography, resistivity, vacuum fusion mass spectrometry, combustion-thermal conductivity and spark source mass spectrometry. 4) We have the capability to grow nearly all transition metals and many of their alloys and compounds.	Contact Us
V	"	11 " X 100		
Nb	"	12 " X 200		
Ta	"	9 " X 20		
Mo	"	3 " X 150		
Ni	"	10 " X 120		
Ir	"	5 " X 30		
Pt	"	6 " X 30		
Nb-Ta alloys (from 2.5 at.% to 33 at.% Ta) (2.5, 5, 7, 10, 15, 25, 33)	"	6 " X 200		
Nb + 5 at.% Zr	"	8 " X 70		
Nb-Mo alloys (5.4 wt.%, 10 at.% Mo)	"	3 " X 100		
Mo-Re alloys (15, 20, 25, 92 at.% Re)	"	4 " X 150		
Re + 8 at.% W	"	5 " X 20		
Ti ₃ Au	"	7 " X 12		

180 b				
Name(s): Yung-Kwang Chang				
Inst. & Address: Solid State Division, Oak Ridge National Laboratory, P. O. Box X, Oak Ridge, Tennessee 37830			Telephone: (615) 574-5496 FTS 624-5496	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ti ₃ Pt	Z	3 mm grain		Contact Us
V ₃ Si	"	5 mm grain		
Tb + 10 at.% Y	Grain	3 X 3 X 4		
Tb + 50 at.% Y	"	6 X 20 X 20		
Fe + 20 wt.% Ni + 15 wt.% Cr	"	3 X 3 X 20		
Fe + 3 wt.% Si	"	7 X 25 X 30		

181				
Name(s): G. W. Clark, C. E. Finch, and O. C. Kopp				
Inst. & Address: Metals and Ceramics Division, Oak Ridge National Laboratory P. O. Box X, Oak Ridge, TN 37830			Telephone: (615) 574-4814 FTS 624-4814	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Eu_2O_3 , Europia	Flu	0.2x0.2x5 mm	>99.99% pure single crystal for x-ray diffraction study [S. L. Bennett et al., <i>J. Cryst. Growth</i> 41, 309, (1977)]	*
Fe_2SiO_4 , Fayalite	Cz	10mmφ x 15mm	>99.99% pure single crystal for geodynamic research [C. E. Finch et al., <i>Proc. 4th Amer. Conf. Crystal Growth</i> , 1978]	*
SiO_2 , Quartz	Hyd	10 x 20 x 7mm	Typically >99.9% pure SiO_2 ; major impurities: Al, Fe, Alkalies; grown on AT-cut seeds; doped with various alkalies and alkaline earths, boron; infra-red spectroscopy, anelastic loss measurements; frequency control; O. C. Kopp and G. W. Clark, <i>J. Cryst. Growth</i> , V. 2, pp. 308-312, 1968; O. C. Kopp and P. A. Staats, <i>Jour. Phys. Chem. Solids</i> , V. 35, pp. 1029-1033, 1974; O. C. Kopp and P. A. Staats, <i>Jour. Phys. Chem. Solids</i> , V. 36, p. 356, 1975.	* *limited

182				
Name(s): John W. Cleland				
Inst. & Address: Oak Ridge National Laboratory, Solid State Division, P. O. Box X, Oak Ridge, TN 37830			Telephone: (615) 574-5493 FTS 624-5493	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Semiconducting Elements and Compounds - Ge, Si, InSb, InAs, GaAs, etc.		Ingot slices	Defect-impurity interactions, radiation effects, neutron transmutation doping, electrical and optical properties, test device fabrication, photovoltaics.	+

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Name(s): John D. Holder				
Inst. & Address: ORNL, P.O. Box X, 4500S, A254, Oak Ridge, TN 37830			Telephone: (615) 574-4816 FTS 624-4816	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cr_2O_3 - Mo Eutectic	Z	45x50x20	99.9pure, directionally solidified, Mo as fibers	+
LaCrO_3 - Mo Eutectic	Z	15x10x10	99.9pure, directionally solidified, Mo as fibers	+
$(\text{Cr}, \text{Al})_2\text{O}_3$ - Mo Eutectic	Z	15x10x10	99.9pure, directionally solidified, Mo as fibers	-
LaCrO_3 - W Eutectic	M	5x5x5	99.9pure, cold crucible, W as fibers	-
LaCrO_3 - Cr Eutectic	M	5x5x5	99.9pure, cold crucible, Cr as fibers	-
YCrO_3 - Mo Eutectic	M	5x5x5	99.9pure, cold crucible, Mo as fibers	-
YCrO_3 - W Eutectic	M	5x5x5	99.9pure, cold crucible, W as fibers	-
YCrO_3 - Cr Eutectic	M	5x5x5	99.9pure, cold crucible, Cr as fibers	-
$\text{ZrO}_2(\text{Y}_2\text{O}_3)$ - $\text{Al}_2\text{O}_3(\text{Cr}_2\text{O}_3)$ - Mo Ternary Eutectic	Z	10x10x10	99.9pure, directionally solidified, Mo as fibers	+
Pb-Sn-Cd Eutectic and alloys	Z	2x2x50	99.999pure, directionally solidified, both two and three phase composites	+

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Name(s): Edward Sonder				
Inst. & Address: Solid State Division, ORNL, P.O. Box X, Oak Ridge, TN 37830			Telephone: (615) 574-6292 FTS 624-6292	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
MgO		12 x 12 x 3	Transport properties studied as a function of species, state, and amount of impurity using the following technique: electrical conductivity, thermopower, EPR, and optical absorption	
MgAl ₂ O ₄		12 x 5 x 1		

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Name(s): Rosa T. Young				
Inst. & Address: Solid State Division, Oak Ridge National Laboratory P. O. Box X, Oak Ridge, TN 37830			Telephone: (615) 574-5500 FTS 624-5500	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Si, Silicon	FZ,Cz	diameter to 7 mm	n and p dopants for radiation damage, neutron transmutation doping and solar cell studies. Carrier concentration range from $\sim 10^{13}$ to 10^{17} cm ⁻³ .	+
GaAs, Gallium Arsenide	Br	.5 mm thick	Si doped. Carrier concentration $\sim 10^{18}$ cm ⁻³ for solar cell study.	+

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Name(s): W.E. Bulman; M.J. Arnoff				
Inst. & Address: Ohio Semitronics Inc., 1205 Chesapeake Ave., Columbus, Ohio 43212			Telephone: (614) 486-9561	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InAs - Indium Arsenide	Z	26x18x200	Polycrystalline - Hall coef. to 800 cm ³ /coul Electrical characterization includes resistivity and mobility. Used for Hall Effect devices and infrared windows. Standard material is undoped, but has been produced with dopants such as Ag, Cr, Cu and Mn. Applications of the Hall Effect, Bulman W. E. Solid State Electronics Vol. 9, No. 5, Pages 361-372.	+ -
InSb - Indium Antimonide	Z	32x25x380	Polycrystalline - Carrier Concentration from 1×10^{19} to 5×10^{13} /cm ³ mobility to 10^6 cm ² /volt sec. Electrically characterized as to resistivity and mobility at 77°K. Used as infrared windows, Hall Effect devices and magnetoresistors. Material is available undoped or doped with Be, Cd, Mg and Zn.	+ -

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Name(s): W. E. Bulman; M. J. Arnoff

Inst. & Address: Ohio Semitronics, Inc., 1205 Chesapeake Ave., Columbus, Ohio 43212

Telephone: (614) 486-9561

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InSb - Indium Antimonide	Cz	32 O.D. x 125 L.	Single Crystal - Carrier Concentration from 1×10^{19} to $2 \times 10^{14}/\text{cm}^3$. Electrical characterization and dopants same as InSb- Polycrystalline above. Standard orient- ations are $[111]$, $[211]$, $[110]$, others as re- quired.	+ -
Bi_2Te_3 - Bismuth Telluride	Br	4 to 25 O.D. 250 L. and 25 x 25 x 250 L.	Polycrystalline - Thermoelectric grade Electrical property characterization in- cludes resistivity, seeback and figure of merit.	+ -
PbTe - Lead Telluride	Br	25 O.D. x 200L.	Polycrystalline - High purity optical grade for evaporated coatings.	+ -
GaSb - Gallium Antimonide	Z	25 x 18 x 150	Polycrystalline - Undoped carrier concen- tration to $1.3 \times 10^{16}/\text{cm}^3$ and mobility to $3.0 \times 10^3 \text{ cm}^2/\text{volt sec.}$ at 77°K . Used in making infrared windows and other devices.	+

188

Name(s): Joel J. Martin

Inst. & Address: Department of Physics, Oklahoma State University, Stillwater, OK 74074

Telephone: 624-5799

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KMgF_3 , RbMgF_3	Cz & Br	5 x 5 x 15	Fluoride perovskites are grown undoped and doped with transition metals ore rare earths.	+
$\text{Mn}_{1-x}\text{Zn}_x\text{F}_2$ $\text{Bi}_4\text{Ge}_3\text{O}_{12}$	Cz		Properties studied include: optical, resonance, thermal, and radiation damage.	+

189

Name(s): W.J. Fredericks

Inst. & Address: Department of Chemistry, Oregon State University, Corvallis, OR

Telephone: (503)-754-2191

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NaCl Sodium Chloride	cz	25x25x85	All materials purified by ion exchange methods and treated with reactive gases prior to melting and while melted. Crystals usually pulled under Ar atmosphere. Various dopants as desired.	-
KCl Potassium Chloride	cz	25x25x85		-
KBr Potassium Bromide	cz	25x25x85		-
SrF_2 Strontium Fluoride	M	12 dia x 50	Diffusion conductivity dielectric relaxation, optical absorption and luminescence measurements made.	-
BaF_2 Barium Fluoride	M	12 dia x 50		-

190

Name(s): A. R. McGhie

Inst. & Address: L.R.S.M., Univ. of Pennsylvania, Philadelphia, PA 19104

Telephone: (215) 243-6461

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Organic Molecular Crystals and Charge Transfer Complexes	Br Cz Sol Sub VP Z	mm to cms on edge	Samples, pure or doped, oriented as required. Research use in solid state physics and optical spectroscopy	occa- sionally

191

Name(s): David P. Pope

Inst. & Address: Department of Metallurgy & Materials Science, University of
Pennsylvania, Philadelphia, PA 19104 U.S.A.

Telephone: (215) 243-7977

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cu_3Au	B	10x10x5	99.999% purity	Small quan- tities

192

Name(s): Donald L. Smith

Inst. & Address: M.S.# 283 - Perkin-Elmer Corp., Main Ave., Norwalk, Conn. 06856

Telephone: (203) 762-6916

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
(PbSn)Te and (PbSn)Se on BaF_2 (100) substrate	molec. beam epitaxy	8X10 X.01	Bi and Tl-doped for diode lasers	possibly
GaAs	"	"	Be/SnTe-doped junctions and MOS structures for solar cells	"
(GaAl)N	RF- activ. evap.	"		"

193 Name(s): M. Delfino Inst. & Address: Philips Laboratories, 345 Scarborough Rd., Briarcliff Manor, New York 10510 Telephone: (914) 762-0300				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
L(+) glutamic acid hydro- chloride $\text{HOOC}(\text{CH}_2)_2\text{CH}(\text{NH}_2)\text{COOH}\cdot\text{HCl}$ and $\text{DOOC}(\text{CH}_2)\text{CH}(\text{ND}_2)\text{COOD}\cdot\text{DC}$	sol.	70x30x20	nonlinear optics J. Crystal Growth <u>36</u> , 267-272 (1976). J. Solid State Chem. <u>23</u> , 289-296 (1978).	+
L(+) hydrazonium tartrate $\text{H}_2\text{NNH}_2\cdot\text{HOOC}(\text{CHOH})_2\text{COOH}$ and $\text{D}_2\text{NND}_2\cdot\text{DOOC}(\text{CHOD})_2\text{COOD}$	sol.	20x10x10	Nonlinear optics J. Solid State Chem. <u>21</u> , 243-251 (1977)	+

194 Name(s): Dr. Brian J. Fitzpatrick Inst. & Address: Philips Laboratories, 345 Scarborough Rd., Briarcliff Manor, N.Y. 10510 Telephone: (914) 762-0300				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnSe	LE, VP, Sol	15x15x0.05	(111), all acceptor dopants, conductivity, photoluminescence, TEM studies, for laser and display applications; Near Band Edge Luminescence in ZnSe Grown from Gallium Solution, submitted to J. Electrochem. Soc. Optically Pumped Stimulated Emission in ZnSe J. Quant. Electr., July 1978. Electron Beam Induced Dislocation Climb in ZnSe, J. Appl. Phys., 1978	+

195 Name(s): G. M. Loiacono Inst. & Address: Philips Laboratories, Exploratory Research Dept. 345 Scarborough Rd., Briarcliff Manor, N.Y. 10510 Telephone: (914) 762-0300 ext. 458				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$(\text{CH}_2\text{ND}_2\text{COOD})_3\cdot\text{D}_2\text{BeF}_4$ (DTGFB)	Sol.	60x60x60	Studies of thermal and electrical properties. High purity. % D ~75%.	+
$\text{M}_3\text{B}_7\text{O}_{13}\cdot\text{X}$ (Boracites)	Hyd.	1-5	Studies of pyroelectric properties	-

196

Name(s): DR. WALTER K. ZWICKER

Inst. & Address: PHILIPS LABORATORIES, BRIARCLIFF MANOR, NY 10510

Telephone: 914-762-0300

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnSe _x S _{1-x}	M,VP	10x10x15	DOPANT: Al	-
M*P ₅ O ₁₄ (*RARE EARTHS)	FLU	12x25x30	FOR SOLID STATE MINI-LASERS	-

197

Name(s): Andrew C. Lawson, Professor of Physics (Assistant)

Inst. & Address: Department of Physics, Pomona College, Claremont, Calif. 91711

Telephone: (714) 621-8000
ex. 2945

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
intermetallic compounds			usually polycrystalline	+

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Name(s): Quartz Products Corporation

Inst. & Address: P.O. Box 628
Plainfield, New Jersey 07061201-757-
Telephone: 4545

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NaI(Tl)	Kyrop*	Ø16"x2"		All The Crystals Listed in this form are for sale & avail. from stock.
CsI(Tl)	Kyrop	Ø16"x2"		
CsI(Na)	Kyrop	Ø12"x2"		
NaCl	Kyrop	Ø400mmx80mm		
KCl	Kyrop	Ø350mmx100mm		
LiF	Kyrop	Ø100mmx165mm		
LiNbO ₃	Kyrop	Ø50mmx50mm		
KBr	Kyrop	Ø350mmx100mm		
AgCl	Br	Ø100mmx100mm		
CsI	Br	Ø100mmx100mm		
KRS-5	Br	Ø100mmx100mm		
CsBr	Br	Ø100mmx100mm		
PET	Sol	6"x4"x10mm		
EDdT	Sol	5"x3"x10mm		
KAP	Sol	8"x5"x10mm		
RbAP	Sol	6"x4"x10mm		
NaAP	Sol	3"x2"x10mm		
NH ₄ AP	Sol	4"x3"x10mm		
TLAP	Sol	4"x2"x10mm		
NaNO ₃	Sol	1"x1"x1"		
KDA	Sol	15x15x15mm		
KDP	Sol	5"x5"x5"		
DKDP	Sol	2"x2"x2"		
TGS	Sol	4"x2"x1.5"		
DTGS	Sol	2"x2"x1.5"		
TGFB	Sol	2"x2"x1.5"		
DTGFB	Sol	1.5"x1.5"x1.5"		
TGS(Alanine)	Sol	4"x2"x1"		

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PAGE 2

Name(s): Quartz Products Corporation
P.O. Box 626
Inst. & Address: Plainfield, N.J. 07061

201-757-4545
Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Rochelle Salt	Sol	4"x4"x2"	The Last Dimension is along the Polar Axis OY for all the crystals listed on this page.	
Alpha Iodic Acid	Sol	2"x2"x2"		
LiIO ₃	Sol	1"x1"x1"		
LiFormate	Sol	1/2"x1/2"x1/2"		
KPB	Sol	1/2"x1/2"x1/2"		

200

Name(s): D. S. Rimal

Inst. & Address: Physics Dept., Purdue University, W. Lafayette, IN 47907

Telephone: (317) 493-9247

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NbO ₂ Niobium dioxide	Cz	5 x 5 x 5	Oriented along [001], [110]	-
ZnF ₂ Zinc fluoride	M	8 x 8 x 8	Oriented same as above - Phys. Rev. B1,9, 4069(1977)	-
MgF ₂ Magnesium fluoride	M	13 x 13 x 13	Oriented for elastic constant determinations	-
NiF ₂ Nickel fluoride	M	6 x 6 x 6	Grown using self-sealing crucible	-
Ti ₂ O ₃ Titanium sesquioxide	Cz	25 x 25 x 25	Also doped with vanadium	-
a-As ₂ S ₃ amorphous arsenic trisulfide	--	10 x 10 x 5	Used in ultrasonic studies	-
a-As ₂ Se ₃ amorphous arsenic triselenide	--	10 x 6 x 4	Used in ultrasonic studies	-
TTF-TCNQ tetrathiofulvalene-tetracyanoquinodimethane	Sol.	3 x 1 x 0.1	Electrical conductivity studies	-

201

Name(s): H. Sato and W. M. Becker

Inst. & Address: Central Materials Preparation Facility, Physics Building
Purdue University, West Lafayette, Indiana 47907

Telephone: (317) 493-2197

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. V_2O_3	C_z (tri-arc)	4 dia x 20	Pure & doped	+
2. Ti_2O_3	C_z	15 dia x 20	Pure & doped	-
3. NiO , CoO	arc transfer	4 dia x 30	Pure	-
4. Alkali halides	Br	380 dia x 1500	Pure & doped	+
5. Fe_3O_4 , magnetite	skull melter	20 x 20 x 20	Pure	+
6. $(Fe_3O_4)_x \cdot (Fe_2TiO_4)_{1-x}$ ($0 < x < .95$) Magnetite-ulvöspinel solid solutions	skull melter	10 x 10 x 10		-
7. HgS , cinnabar	VP, Sol	4 x 5 x 7	Pure	+
8. GaP	VP	4 x 10 x 7	Pure	+
9. UO_2	VP	3 x 5 x 17	Pure	+
10. V_nO_{2n-1}	VP	2.5x2.5x20	Pure	-
11. TiS_2 , HfS_2	VP	.5 x 20 x 20	Pure	+

202

Name(s): Radiation Monitoring Devices, Inc.

Inst. & Address: 44 Hunt Street, Watertown, MA 02172

Telephone: (617)926-1167

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cadmium Telluride	Sol	20 X 20	Chlorine doped CdTe for nuclear detectors; $10^9 \Omega$ cm, good lifetimes and mobilities	+

203

Name(s): James Pappis

Inst. & Address: Raytheon Research Division, 28 Seyon St., Waltham, MA 02154 USA

Telephone: (617)899-8400
ext. 3941

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Zinc Selenide, ZnSe, Raytran ZnSe	CVD	800 x 800 x 25	Polycrystalline, high purity, well characterized. Can also order as shaped component.	com- mercial
Zinc Sulfide, ZnS, Raytran ZnS	CVD	700 x 700 x 15	Polycrystalline, high purity, well characterized. Can also order as shaped component.	com- mercial
Calcium Fluoride, CaF ₂	M	200 dia x 25	Polycrystalline, high purity, well characterized.	special order
Boron Nitride, BN, H ⁺ BN	CVD	150 x 80 x 8	Polycrystalline, isotropic, high purity, well characterized.	com- mercial
Magnesium Aluminate, MgAl ₂ O ₄ , Spinel	M	200 dia x 25	Polycrystalline, optical and mechanical properties being studied.	

204

Name(s): A. W. Swanson, S. R. Steele, K. K. Johnson

Inst. & Address: Raytheon Company, Research Division
28 Seyon Street, Waltham, MA 02154Telephone: (617)899-8400
ext. 3765

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gallium Arsenide (GaAs)	VPE	4 x 4 cm	High-purity single-crystal layers grown on <100> GaAs substrates. Wafers are character- ized using normal C-V techniques plus Hall measurements, etc. The VPE material is grown to specifications for the fabrication of the following devices: FET's Mixers Guns IMPATT's Single-drift Reads Double-drift Reads Varactors	+

205 S.Berkman R.Metzl Name(s): A.B.Dreeben R.E.Novak Inst. & Address: RCA Corp., David Sarnoff Research Center, Princeton, New Jersey Telephone: (609)452-2700				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Al_2O_3 - Sapphire	M(EFG)	1x75x1500	Single crystal substrates for integrated circuits fabrication.	-

206 Name(s): Dr. David Emil Carlson Inst. & Address: RCA Laboratories, Princeton, NJ 08540 Telephone: (609)452-2700 Extension 3205				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
a-Si:H, Hydrogenated Amorphous Silicon	glow-discharge deposition	300x300x0.01	Deposited from semiconductor grade silane (SiH_4); PH_3 and B_2H_6 are common dopant sources; potential application for low cost, thin-film solar cells; ref: D.E. Carlson, IEEE Trans. Electron Devices, ED-24, 449 (1977).	No

207 Name(s): G. H. Olsen and A. V. Cafiero Inst. & Address: RCA Laboratories, P. O. Box 432, Princeton, New Jersey 08540 Telephone: 609-452-2700				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LaB ₆ EuB ₆ BaB ₆ CeB ₆ (La,Eu)B ₆ (La,Y)B ₆ (La,Ba)B ₆ (La,Cs)B ₆ (Eu,Y)B ₆ (Eu,Ba)B ₆	SoI	~1	Single crystal rods and cubes "	
J. Cryst. Growth 44 (1978)				

208 Name(s): G. H. Olsen and T. J. Zamerowski Inst. & Address: RCA Laboratories, P. O. Box 432, Princeton, New Jersey 08540 Telephone: 609-452-2700				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
(In,Ga)(As,P)	VP	~.02 mm	Zinc and sulfur doped CW lasers Lattice matched to InP EMR Abstracts Santa Barbara, CA (1978)	

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Name(s): W. M. Yim

Inst. & Address: RCA Laboratories, David Sarnoff Research Center, Princeton, N.J. 08540, U.S.A. Telephone: (609)452-2700

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
AlN	VP	20x10x0.2	Optical bandgap, thermal expansion J. Appl. Phys. <u>44</u> , 292 (1973); <u>45</u> , 1456 (1974).	-
$\text{Bi}_{1-x}\text{Sb}_x$	Z	20x10x100	Thermoelectric, thermomagnetic "Crystal Growth", 187 (1967); Solid State Elect. <u>15</u> , 1141 (1972).	-
ScAs, ScP	VP	20x10x0.1	Optical bandgap, electrical J. Appl. Phys. <u>43</u> , 254 (1972).	-
BeTe, BeSe, BeS	M, VP	5x3x0.3	Optical bandgap, electrical J. Phys. Chem. Solids <u>33</u> , 501 (1972).	-
AlAs	VP	2x10x0.1	Optical bandgap, luminescence J. Appl. Phys. <u>42</u> , 2854 (1971); Solid State Commu. <u>8</u> , 1407 (1970).	-
ZnSe, ZnS, CdSe, CdS	VP	20x10x0.2	Optical, electrical, doping J. Electrochem. Soc. <u>119</u> , 381 (1972); <u>121</u> , 965 (1974).	-
ZnSc_2S_4 , CdSc_2S_4	VP	5x10x0.2	Optical, electrical, doping J. Electrochem. Soc. <u>120</u> , 441 (1973).	-
ScN, LuN	VP	20x10x0.2	Optical, electrical, doping RCA Rev. <u>31</u> , 680 (1970); J. Cryst. Growth <u>13/14</u> , 365 (1972).	-

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Name(s): W. M. Yim

Inst. & Address: RCA Laboratories, David Sarnoff Research Center, Princeton, N.J. 08540, U.S.A. Telephone: (609)452-2700

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$(\text{ZnSe})_{1-x}(\text{GaP})_x$	VP, Sol	20x10x0.2	Optical, electrical, doping J. Appl. Phys. <u>40</u> , 2617 (1969); Appl. Phys. Lett. <u>16</u> , 366 (1970); RCA Rev. <u>31</u> , 662 (1970).	-
$\text{Bi}_2\text{Te}_3\text{-Sb}_2\text{Te}_3\text{-Sb}_2\text{Se}_3$	Br	10x5x50	Thermoelectric J. Mater. Sci. <u>1</u> , 52 (1966); J. Electrochem. Soc. <u>115</u> , 556 (1968).	-
SrTiO_3 , CaTiO_3	Flu	5x5x5	Optical, electrical, luminescence ECOM 0155-3, 0155-F Technical Reports (1971, 1972).	-
Bi-MnBi	Z	10x10x70	Electrical J. Appl. Phys. <u>38</u> , 5211 (1967).	-

211

Name(s): Dady B. Dadyburjor

Inst. & Address: Rensselaer Polytechnic Institute, Troy, New York 12181

Telephone: (518) 270-6381

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Direct, or crowded, ripening			A possible effect in unsaturated solutions. If two particles are close enough, single atoms may transfer "directly" from the smaller to the larger particle. See Dadyburjor and Ruckenstein, J. Crystal Growth <u>38</u> , 285 (1977); Ruckenstein and Dadyburjor, Thin Solid Films (1978) to be published.	

212

Name(s): Sorab K. Ghandhi

Inst. & Address: Rensselaer Polytechnic Institute, Troy, New York 12181

Telephone: [518] 270-6333

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gallium Arsenide	VI	1 cm x 1 cm x 15 μ m	Grown from trimethylgallium and arsine. $2 \times 10^{15}/\text{cm}^3$ (100), (111) sulfur, zinc, mobility carrier concentration, diffusion length, Schottky diodes, photo-voltaics, FET's. J. Cryst. Growth, 26, 314 (1974); J. Electrochem. Soc., 122, 1378 (1975); J. Electrochem. Soc., 124, 1447 (1977); J. Electrochem. Soc., 125, 771 (1978).	-
Indium Arsenide	VP	1 cm x 1 cm x 5 μ m	Grown from triethylindium and arsine on gallium arsenide substrates. $2 \times 10^{16}/\text{cm}^3$, (100) mobility, carrier concentration. J. Electrochem. Soc., 121, 1642 (1974); J. Electrochem Soc., 121, 1646 (1974).	-
Gallium _x Indium _{1-x} Arsenide $0 \leq x \leq 1$	VP	1 cm x 1 cm x 6 μ m	Grown from trimethylgallium, triethylindium and arsine on gallium arsenide substrates (100), mobility, carrier concentration, energy gap, composition, carrier velocity, microwave FET's. JECS, 122, 683 (1975); J. Appl. Phys., 46, 3941 (1975); J. Appl. Phys., 46, 4608 (1975).	-
Tin Oxide	VP	2 cm x 2 cm x 1 μ m	Grown from tetramethyltin and oxygen, both undoped and phosphorus doped (from phosphine) glass and silica substrates, annealed and unannealed, composition, conductivity, transmittance, solar cells. CONTINUED ON NEXT PAGE	-

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Name(s): Sorab K. Ghandhi

Inst. & Address: Rensselaer Polytechnic Institute, Troy, New York 12128

Telephone: [518] 270-6333

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tin Oxide (Cont.)			J. ECS, 123, 941 (1976); J. ECS, 124, 1060 (1977); Sol. State Electronics, 20, 773 (1977).	
Silicon Dioxide	VP	2 cm x 2 cm x 1 μ m	From silane and oxygen, both SiO_2 and phosphosilicate glass (from phosphine). Kinetics of growth, composition, diffusion masks for gallium arsenide. J. Appl. Phys., 44, 990 (1973); IEEE Trans. Elec. Dev., ED-19, 761 (1972); IEEE Trans. Elec. Dev., ED-21, 410 (1974).	-

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Name(s): M. E. Glicksman

Inst. & Address: Materials Engineering Department, Rensselaer Polytechnic Institute,
Troy, New York 12181

Telephone: (518) 270-6372

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Succinonitrile $[\text{CN}(\text{CH}_2)_2\text{CN}]$	SUB Z	25 dia. unlimited length	99.9999+ % Random large crystals, B.C.C. plastic crystals at room temperature.	+

215

Name(s): S. B. Austerman

Inst. & Address: Rockwell International, 3370 Miraloma Ave., Anaheim CA 92803

Telephone: 714/632-1562
632-1931

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Beryllia, BeO, Beryllium Oxide	Flux	Up to 20 x 20 x 40	<200 ppm total impurities Faceted bulk crystals with Wurtzite structure. Properties described in many open publications.	By Arrangement

216

Name(s): H. DAVID LAW

Inst. & Address: Rockwell International Science Center, 1049 Camino dos Rios,
Thousand Oaks, CA 91360Telephone: (805) 498-4545
x. 222

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. InGaAsP, Indium Gallium Arsenide Phosphide	LE	1cmx1.5cmx 40µm	Grown on InP substrate; <111>B and <100>; Te n-type dopant; Zn p-type dopant; Double Heterojunction Lasers; Ion Implanted Avalanche Photodiodes; LED	-
2. GaAlSb, Gallium Aluminum Antimonide	LE	1cmx1.5cmx 400µm	Grown on GaSb substrate; <111>B and <100>; Te-n-type dopant; Ge-p-type dopant; Avalanche Photodiodes, Double Heterojunction Lasers	-
3. GaAlAsSb, Gallium Aluminum Arsenide Antimonide	LE	1cmx1.5cmx 400µm	Grown on GaSb substrate; <100>; used to lattice match GaAlSb; maximum As in solid = 3.5%; Te-n-type dopant, Ge-p-type dopant; Avalanche Photodiodes; Double Heterojunction Lasers.	-
4. GaAlPSb, Gallium Aluminum Phosphide Antimonide	LE	1cmx1.5cmx 400µm	Grown on GaSb substrate; <100>; very low solubility in Ga; Not recommended.	-
5. GaAsSb, Gallium Arsenide Antimonide	LE	1cmx1.5cmx 400µm	Grown on GaAs substrate; <100>; Sn, Te-n-type dopants; Ge-p-type dopant; Multi-lattice matching layers required to reduce defect density; Avalanche Photodiodes; LED; Double Heterojunction Laser.	-
6. GaAlAsSb, Gallium Aluminum Arsenide Antimonide	LE	1cmx1.5cmx 600µm	Grown on GaAs substrate; <100>; Sn, Te n-type dopants; Ge-p-type Dopant; used as confining layers in GaAsSb alloys; Double Heterojunction Laser; LED	-
7. GaAlAs, Gallium Aluminum Arsenide	LE	1cmx1.5cmx 400µm	Grown on GaAs substrate; <100>, <111>B, Sn, Te-n-type dopants; Ge-p-type dopants; Double Heterojunction Laser; LED; Avalanche Photodiodes.	-
8. GaAs, Gallium Arsenide	LE	1cmx1.5cmx 400µm	Same as 7.	-

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Name(s): Joel S. Miller

Inst. & Address: Rockwell International Science Center
P.O. Box 1085, Thousand Oaks, California 91360

Telephone: (805) 498-4545

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
K ₂ Pt(CN) ₄ Br _{0.3} ·3H ₂ O* Potassium tetracyanoplatinate bromide (2:1:0.3) trihydrate	EC	~ 10x1x1 mm	High purity, highly anisotropic, highly conducting (σ > 10 ³ ohm ⁻¹ cm ⁻¹) one-dimensional metal-like complexes. For a review article see J.S. Miller and A.J. Epstein, <i>Prog. Inorg. Chem.</i> , 20, 1 (1976). For preparative details see J.S. Miller, <i>Science</i> , 194, 189 (1976) and J.S. Miller, <i>Inorg. Synthesis</i> , 19, 000 (1978).	+
K ₂ Pt(CN) ₄ C ₂ O ₄ ·3H ₂ O* Potassium tetracyanoplatinate chloride (2:1:0.3) trihydrate	EC	~ 10x1x1 mm		+
K _{1.75} Pt(CN) ₄ ·1.5H ₂ O* Potassium tetracyanoplatinate (1.75:1) sesquihydrate	EC	~ 5x1x1 mm		+
K _{1.64} Pt(O ₂ C ₂ O ₂)·2H ₂ O* Potassium bis(oxalato)platinate (1.64:1) dihydrate	EC	~ 1x0.05 x0.05 mm		+
*The deuterio analogues can also be prepared.				

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Name(s): Wayne Anderson

Inst. & Address: Rutgers University, Electrical Engineering, P.O. Box 909, Piscataway, NJ 08854 Telephone: 201/932-3466
*SUNY at Buffalo, Electrical Engineering, 4232 Ridge Lea Road, Amherst, NY 14226 * 716/831-1835

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
polycrystalline silicon	electron beam deposi- tion		<110> orientation p-type, 1 Ω -cm crystallinity, columnar structure, x-ray diffraction, SEM, TEM applied to MIS solar cells	limited

*August 1978-August 1979

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Name(s): Dr. Martha Greenblatt

Inst. & Address: Department of Chemistry, Rutgers University, New Brunswick, N.J. 08903 Telephone: 201-932-3277

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Sr}_5(\text{PO}_4)_3\text{Cl}$, $\text{Ba}_5(\text{PO}_4)_3\text{Cl}$ $\text{Cd}_5(\text{PO}_4)_3\text{Cl}$	Flu	2x1x0.5	doped with Cr(V) for ESR study to detect structural changes in the apatite struc- ture.	+
Li_3PO_4 , Li_3AsO_4	Flu	3x4x1		
Li_3VO_4	M	2x1x0.1	All samples were doped with small concen- trations of Cr(V) to study electronic structure by ESR.	
YPO_4 , YVO_4	Flu			
K_2SO_4	Sol	3x2x0.5	doped with Mn(VI) for ESR study of the electronic structure of MnO_4^{2-}	

220

Name(s): Dr. Lisa C. Klein

Inst. & Address: Rutgers University, Ceramics Dept.
P.O. Box 909 Piscataway, NJ 08854

Telephone: (201)932-2096

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
K feldspar- SiO_2	M	1-5	crystallization from eutectic melts	

221

Name(s): R.C. Folweiler

Inst. & Address: Sanders Associates, Inc., 95 Canal Street, Nashua, N.H. 03061 Telephone: (603) 885-2828

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiYF ₄ (YLF, Nd:YLF, α -YLF)	Sol/Cz	35 dia. 150 long	4 to 5 nines purity; doped with Nd, Tb, Ho, Er, Tm, Yb and other rare earths; primarily "a" axis; laser host and UV optical material; some compositions stock.	+
LiHoF ₄	Sol/Cz	30 dia. 100 long		
HoF ₃	M/Cz	30 dia. 100 long		
ErF ₃	Sol/Cz	30 dia. 100 long		
K Tb ₃ F ₁₀	Sol/Cz	30 dia. 50 long	Cubic; Faraday rotator material with large verdet constant; low n and n ₂ .	+
BaTiO ₃	Sol/Cz	10x10x10	No flux contamination.	-
SrTiO ₃	Sol/Cz	10x10x10	Low strain	-
CaY ₂ Mg ₂ Ge ₃ O ₁₂ (CAMGAR)	Sol/Cz	30x30x20	Cubic; Nd doped (medium gain)	-

222

Name(s): Arthur H. Lockwood

Inst. & Address: Santa Barbara Research Center, 75 Coromar Drive, Goleta, CA 93017

Telephone: 805-968-3511

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InGaSb	LE	40x32x.1	<111> B, electrical and optical properties, IR detectors	- -
InAsSb	LE	40 x 32 x .1	<111> B, electrical and optical properties, IR detectors	- -
HgCdTe	LE	25 x 25 x .1	<111>B, electrical and optical properties, IR detectors	- -
InSb	LE	40 x 32 x.1	n-type, $2 \times 10^{14} \text{ cm}^{-3}$, IR Detectors <111> B, electrical and optical properties	- -

223

Name(s): D. R. Kinloch, Research Manager

Inst. & Address: Sawyer Research Products, a subsidiary of Brush Wellman Inc.
35400 Lakeland Boulevard, Eastlake, Ohio 44094

Telephone: (216) 951-8770

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SiO ₂ , Quartz	Hyd	300/160/100	Crystal produced commercially for Piezoelectric and optical uses.	Yes

224 Name(s): T.F. Ciszek Inst. & Address: Solar Energy Research Institute (SERI) 1536 Cole Blvd. Golden, Colorado 80401 Telephone: (303) 234-7418				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability: + -
Si Silicon	M	50x50x150	Crystallization techniques applicable to photo-voltaics are studied.	--

225 Name(s): Francis M. Uno Inst. & Address: Spectrolab, Inc.; 12500 Gladstone Avenue, Sylmar, CA 91335 Telephone: 213 365-4611				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Analysis of Effects of Impurities Intentionally Incorporated into Silicon	CZ, FZ	20x20x0.20	Electrical Properties of Silicon Photovoltaic Cells Deliberately Incorporated with 26 single, binary and Ternary Compounds. Final Report: JPL Contract 954694, 15 December 1977	-

226 Name(s): Dr. Michael Kestigian Inst. & Address: Sperry Research Center, Sudbury, MA 01776 Telephone: (617)369-4000				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gadolinium gallium garnet ($Gd_3Ga_5O_{12}$)	Cz	30mm dia. 150mm l	6-nines purity all major crystallographic orientations, substrate material	
Samarium gallium garnet $Sm_3Ga_5O_{12}$	Cz	30mm dia. 150mm l	Same as above	
Gadolinium scandium gallium garnet $Gd_3Sc_2Ga_3O_{12}$	Cz	30mm dia 150mm l	Same as above	
Gadolinium calcium zirconium gallium garnet $Gd_{(3-x)}Ca_{(x)}Ga_{(5-x)}Zr_{(x)}O_{12}$	Cz	30mm dia 250mm l	Same as above	
Gadolinium magnesium zirconium gallium garnet $Gd_3Mg_{(x)}Zr_{(x)}Ga_{(5-2x)}O_{12}$	Cz	30mm dia 150mm l	Same as above	
Samarium thulium gallium iron garnet	IE	204mm x 30mm dia	Small-bubble-diameter cylindrical-domain magnetic film wafers	
Yttrium samarium thulium gallium iron garnet	IE	204mm x 30mm dia	Same as above	

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Name(s): M. Kestigian

Inst. & Address: Sperry Research Center, Sudbury MA 01776

Telephone: (617)369-4000

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Yttrium samarium lutetium calcium germanium iron garnet	LE	20mm x 30mm dia	Same as above	
Combination of any rare earth cations with either gallium, aluminum, or germanium substitutions to control μ_{eff}				
Bismuth lutetium iron garnet	LE	40mm x 30mm dia	Magneto-optic thin films	

228

Name(s): Drs. V. K. Kapur, L. Nanis and A. Sanjurjo

Inst. & Address: SRI International, 333 Ravenswood Ave., Menlo Park, CA 94025

Telephone: (415) 326-6200

Ext. 2541

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Si Silicon	M, Sol, VP, Cz	100 μm^3	> 99.99% polycrystalline, p type, resistivity 0.2-0.5 ohm cm, low boron and phosphorus content (100-200 ppb), solar cells; Quarterly reports 1 through 9 for the project, "Novel Duplex Vapor- Electro-chemical Method for Silicon Solar Cells" Contract No. DOE/JPL-954471	+
SiF ₄ Silicon Tetrafluoride Gas	Sol, VP	---	> 99.99% pure gas, inexpensive and abundantly available source for silicon according to $\text{SiF}_4 + 4\text{Na} \rightarrow \text{Si} + 4\text{NaF}$ $\Delta H_{298}^\circ = -166 \text{ Kcal/mole}$	+
Na ₂ SiF ₆ Sodium fluosilicate	Sol	---	Convenient source for obtaining SiF ₄ gas according to $\text{Na}_2\text{SiF}_6 \xrightarrow{\Delta} 2\text{NaF} + \text{SiF}_4$	+

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Name(s): A. L. Fahrenbruch, J. Aranovich

Inst. & Address: Department of Materials Science and Engineering, Stanford University,
Stanford, CA 94305

Telephone: (415) 497-2951

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Zinc Oxide	SP*	10 x 10 x 10^{-3} mm^3	Films produced by * spray pyrolysis, 10^5 ohm-cm as sprayed without doping, resistivity reduced to 10^{-2} ohm-cm by heat-treatment in H_2 . Optical transmission 80 to 90% corrected for reflection. (111) axis perpendicular to substrate. Produced solar cells on CdTe with about 4% solar efficiency. See 13th IEEE Photovoltaic Specialists Conference Volume, Washington, D.C., June 5, 1978. (Author, A. Fahrenbruch).	

230 Name(s): Dr. Richard W. Vook Inst. & Address: Syracuse University 409 Link Hall Syracuse, N.Y. 13210 Telephone: 315-423-3466				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cu, Ag, Au	VP	10 5 2x10 ⁻⁴	(001) and (111) orientation (111) orientation is exceptionally smooth RHEED, TEM	To order

231 Name(s): Dr. Michael P. Rosynek Inst. & Address: Dept. of Chemistry, Texas A & M University College Station, Texas 77843 Telephone: (713)845-3552				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KC ₈ , Potassium-Graphite Intercalate	M/SUB	Powder	Investigated behavior as a heterogeneous catalyst for gas-phase hydrogenation of CO to hydrocarbon products at one atmosphere pressure.	+
Iron-Graphite Intercalate (2.2 wt% Fe)	SUB	Powder	"	+
Cobalt-Graphite Intercalate (3.4 wt% Co)	SUB	Powder	"	+
La ₂ O ₃ , Lanthanum Sesquioxide	SOL	Powder	Investigated surface properties by infrared spec- troscopy and catalytic behavior for double-bond isomerization of alkenes and dehydration/dehydrog- enation of alcohols.	

232 Name(s): D. R. Chopra Inst. & Address: East Texas State University, Physics Dept, Commerce, Texas 75428 U.S.A. Telephone: (214) 886-5488				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. Nickel		15x15x0.5	99.9% electropolished grade A polycrystalline foil x-ray L-absorption & emission, Phys. Rev. A, 1, 230 (1970)	+
2. Potassium Acid phthalate KAP		50x50x10	99.9% single crystals x-ray spectrometric characteristics Rev. Sci. Instr. 41, 1004, (1970).	-
3. Vanadium metal & oxides V ₂ O ₃ , V ₂ O ₄ , V ₂ O ₅		10x5x0.002 powders	99.9% polycrystalline foil powder x-ray K-spectral Absorption study of semiconductor- to-metal transition, proceedings of 10th Symposium on Nondestructive Evaluation, 230-242 (American Society of Nondestructive Testing and Southwest Research Institute, 1975).	-
4. Iron		10x10x0.25	t4N polycrystalline foil (Ventron) Electronic band structure	+
5. Stainless steel 316		10x10x2	polished and unpolished samples corrosion studies proceedings of 11th Symposium on Nondestructive Evaluation, 31-37 (American Society of Nondestructive Testing and Southwest Research Institute, 1977)	+
6. Terbium		10x10x0.62	m3N polycrystalline foil (Ventron) Electronic band structure	+

233 Name(s): D. R. Chopra Inst. & Address: East Texas State University, Physics Dept., Commerce, Texas 75428 U.S.A. Telephone: (214) 886-5488				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
7. Dysprosium		10x10x0.62	m3N polycrystalline foil (Ventron) Electronic band structure Rare Earth in Modern Science and Technology, 347-54 (Plenum Publishing Corporation, New York, 1978)	+
8. Holmium		10x10x0.62	m3N polycrystalline foil (Ventron) Electronic band structure Phys. Rev. B, 14, 5231 (1976).	+
9. Erbium		10x10x0.75	99.9% polycrystalline foil (Research Chemicals) Electronic band structure	+
10. Ytterbium		10x10x0.64	m3N polycrystalline foil (Ventron) Electronic band structure	+

234 Name(s): Glenn H. Sherman Inst. & Address: II-VI Incorporated, Saxonburg Boulevard, Saxonburg, PA 16056 Telephone: 412-352-1504				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe, Cadmium Telluride	Br	poly, 50mm dia. 60mm thick single 15x15x45 25x25x25	Optical, electrooptical, and nuclear detector grades	Good
ZnSe, Zinc Selenide	VP (CVD)	100x100x10	Polycrystalline, Optical grade	Good

235 Name(s): Tyco, Saphikon Division Inst. & Address: 51 Powers Street, Milford New Hampshire 03055, U.S.A. Telephone: 603-673-5831				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SAPPHIRE	M (FFG)	Filament: 0.25 MM Dia. x 100 Meters	1. Undoped. 2. Several orientations are routinely produced. 3. Applications: Lids for FEPROM Substrates (SOS any Hybird) Medical Parts Wear Parts Other Custom Work	
		Tube: 25 MM Dia. x 50 CM		
		Ribbon: 75 MM x 3 MM x 150 CM		
		Special Shapes		

236 Name(s): T. H. (Tom) Barker Inst. & Address: Union Carbide Corp. Box 180 Sistersville, WV 26175 Telephone: (304) 652-3211				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Trichlorosilane	VP	Bulk, 18 ton/load	Boron < 0.3 ppbw parts per billion, weight Phosphorus < 0.3 ppbw Arsenic < 0.5 ppbw	+
Dichlorosilane	VP	Bulk 5 ton/load	Boron < 0.3 ppbw Phosphorus < 0.3 ppbw Arsenic < 0.5 ppbw	

237 Name(s): W. C. (Bill) Breneman, Dr. T.E. Childress Inst. & Address: Union Carbide Corp. Box 180 Sistersville, WV 26175 Telephone: (304) 652-3211				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silane, SiH ₄	VP	5 Kg	Purity: 98 + % Resistivity by silicon epitaxy: > 50 cm Process Development research funded by DOE under JPL Contract 954334	-

238 Name(s): Dr. Hiroshi Morihara Inst. & Address: Union Carbide Corp. 61 E. Park Dr. Tonawanda, N.Y. 14150 Telephone: (716) 877-1600				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon	VLS	Variable	Process design for large scale (1000 ton/yr.) plant based on prep. of silane and pyrolysis to silicon. Research funded by DOE under JPL Contract 954334	-

239 Name(s): Lawrence C. Greene Inst. & Address: AFML/LPO, W-PAFB, Ohio 45433 Telephone: (513) 255-2334				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cadmium Sulfide	VP	platelets	undoped, doped with Na, Li, K, Rb, Ag	+
Cadmium Sulfide	M	boules 10x15 several grains	undoped, doped with Na, Li, K, Ag	+
Cadmium Selenide	VP	platelets	undoped	+
Cadmium Selenide	M	boules 10x15 several grains	undoped	+
Zinc Selenide	M	boules 10x15 several grains	undoped, doped with Na, Li, K, Ag, Ag, Fe 107 109	+
Zinc Telluride	M	boules 10x15 several grains	undoped	+

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Name(s): Elizabeth H. Tarrants

Inst. & Address: Air Force Materials Laboratory, AFML/LTE, Wright-Patterson AFB, OH 45433

Telephone: (513)255-2280/
6744

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
High Purity Silicon (Si)	VAC Z	25 mm dia. 300 mm length	Purity: $1-2 \times 10^{12}$ N _B Orientation: (111) Resistivity: 15,000 -30,000 ohm-cm Type: P Dislocations: Non-detectable Applications: 1.06 micrometer detectors Power Si devices * Hughes Aircraft Industrial Products Division ATTN: W. J. Heinecke, Marketing Manager 6155 El Camino Real Carlsbad, CA 92008 (714) 438-9191	+ *

241

Name(s): Dr. Scott Morrow, Dr. Jagadish Sharma

Inst. & Address: ARRADCOM, Dover, N. J. 07801 ATTN: DRDAR-LCE

Telephone: 201-328-5490

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Single Crystals of 2,4,6 Trinitrotoluene (TNT) $C_7H_5N_3O_6$	Sol	20x10x5	Polarizing microscope studies and x-ray diffraction confirm that the crystals are single (no twins). Explosive properties are being investigated.	20 Crystals

242

Name(s): Dr. T. A. Richter

Inst. & Address: USARRADCOM
DRDAR-LCE Dover, NJ 07801

Telephone: (201) 328-4914

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Inorganic Azides	Sol	Best effort	Many inorganic azides have the property of explosion. See: T. A. Richter, in: Energetic Materials, Vol. 1, Physics and Chemistry of the Inorganic Azides. (H.D. Fair and R.F. Walker, ed.) New York, 1977	+

243

Name(s): Dr. J. W. McCauley

Inst. & Address: Army Materials and Mechanics Research Center
Watertown, MA 02172

Telephone: (617) 923-3463

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
MICA/ALUMINA	Hot Press- ing	1" x 4"	Alumina based ceramic with 5-50 Vol % Ba-mica (BaMg ₃ Al ₂ SiO ₁₀ F ₂) All properties; publications available.	+
Al ₂₃ O ₂₇ N ₅ (ALON)	Sinter- ing Vapor	1/4" x 1/2"	Polycrystalline and single crystals of cubic aluminum oxynitride spinel; some properties; publications available.	+

244 Name(s): Dennis J. Viechnicki Jaroslav L. Caslavsky Inst. & Address: U.S. Army Materials & Mechanics Research Center Watertown, MA 02172 Telephone: (617) 923-3463 -3352				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$Y_3Al_5O_{12}$ (YAG)	M,HEM**	d = 76 h = 102	Impurities - 100 ppm Orientation - [111], [100] Dopant - Undoped or doped with Nd Properties - Laser related Characterization - XRD, optical, SEM Applications - Lasers	+
$YAlO_3$	M,HEM**	d = 50 h = 30	Same as above except for orientation which is [001]	+
** HEAT EXCHANGER METHOD				

245 Name(s): CHARLES S. SAHAGIAN et al Inst. & Address: DEPUTY FOR ELECTRONIC TECHNOLOGY RADC/ESM, HANSCOM AFB, MA 01731 Telephone: (617) 861-4034				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$KPb_2Nb_5O_{15}$ (PKN)	Cz	2x2x2	Surface acoustic wave applications	-
InP (indium phosphide)	Cz	100x20x20	- undoped and doped with Fe or Cr - optoelectronic applications - RADC technical report TR-77-398	+
ZrO_2 (zirconium dioxide)	Cz	25x25x25	- cubic, stabilized with yttria. - undoped and doped with Nd, Ni, etc	+

246 Name(s): Hubert H. Caspers, D. Inst. & Address: Naval Ocean Systems Center, Code 9222, San Diego, CA 92152 Telephone: (714) 225-6591				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InP	LE	10x10x1	Measurements to determine the optical properties of bulk and layered structures.	-
$In_{1-y}Ga_yAs_xP_{1-x}$ /InP	LE	thin films on InP sub- strate	Measurements to determine the optical properties of bulk and layered structures.	-

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Name(s): Arthur R. Clawson

Inst. & Address: Naval Ocean Systems Center, Code 9222, San Diego, CA 92152

Telephone: (714) 225-6591

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InP	LE	$10^{-3} \times 7 \times 9$	Preparation of submicron layers on {100} and {111}B InP:Fe substrates; for microwave MISFETs; Sn doping (10^{17}cm^{-3}); thermal degradation studies of Fe-doped semi-insulating InP polished surfaces and its control with phosphorus overpressure using decomposition of PH_3 ; photoluminescence, transport properties.	-
$\text{In}_{1-y}\text{Ga}_y\text{As}_x\text{P}_{1-x}$	LE	$10^{-2} \times 7 \times 9$	Preparation of layers lattice-matched to {100} and {111}B InP for IR detectors ($\lambda \approx 1.0$ to $1.3 \mu\text{m}$); Sn doping (10^{17} to 10^{18}cm^{-3}); Zn doping (10^{16} to 10^{18}).	-
GaAs	LE	$10^{-3} \times 7 \times 9$	Preparation of submicron layers on {100} GaAs:Cr substrates for microwave FETs; Sn-doping (10^{17}cm^{-3}); Ge doping (10^{16} to 10^{18}cm^{-3}).	-

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Name(s): H. Lessoff, P. Klein, J. Milstein and E. Swiggard

Inst. & Address: Naval Research Laboratory, Code 5220, Washington, DC 20375 Telephone: 202-767-3096

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KCl Potassium Chloride	Br	35 dia. x 200	IR abs. coef. as low as 10^{-5} high purity	Direct Inquiry (Research)
KBr Potassium Bromide	Br	35 dia. x 200	" " " " " " " " " "	
NaCl Sodium Chloride	Br	25 dia. x 80	CCl_4 vapors used in growth	
NaF Sodium Fluoride	Cz	20 dia. x 120	OH^- conc. as low as 0.01 ppm	
LiF Lithium Fluoride	Cz	20 dia. x 120	" " " " " " " " " "	
$\text{Bi}_{12}\text{SiO}_{20}$ Bismuth Silicon Oxide	Cz	12 dia. x 50		Inquiry (Research)
GaAs Gallium Arsenide	Cz	25 dia slices	Cr doped and undoped (approx. 10^{15}cm^{-3}) semi-insulating, high purity bulk	
InP Indium Phosphide	Cz	25 dia slices	Fe-doped and undoped, semi-insulating high purity bulk	
RPO_4 Rare Earth Orthophosphate	Flu	3x3x10		
Rare Earth Transition Metal Intermetallics	Cz	10x10x50		
NbO_2 Niobium Oxide	Cz	10x10x20		
ZnGa_2O_4 Zinc and other gallates	Flu	5x5x5		

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Name(s): K. Peter Scharnhorst

Inst. & Address: Naval Surface Weapons Center, White Oak, Silver Spring, MD 20910

Telephone: 394-2774

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GeTe (Germanium Telluride)	M/Sub	Films	<u>Purity:</u> Starting Elements—99.9999% <u>Properties Studied:</u> 1/f-Noise, Photoconductivity, Electrical Resistivity of the amorphous state. <u>Application:</u> Infrared Detectors. <u>Publications:</u> 1) The Specific Current Noise Power of Amorphous GeTe, J. of Non-Crystalline Solids, <u>23</u> , 435 (1977), K. P. Scharnhorst. 2) 1/f-Noise in Amorphous GeTe, NSWC TR 76-97 (1976), K. P. Scharnhorst. 3) The Noise Equivalent Power of the Amorphous GeTe/ SiO Two Color Detector, NSWC TR 76-98(1976), K. P. Scharnhorst, D. Domske, G. Black. 4) Exponential Photoconductive Edge in Amorphous GeTe, Infrared Physics, <u>14</u> , 139 (1974), H. R. Riedl, K. P. Scharnhorst, D. G. Simons. 5) A 7-12 Micron Infrared Bolometer, Infrared Physics <u>13</u> , 77 (1973), H. R. Riedl, K. P. Scharnhorst. 6) Effects of Annealing on the Electro-Optical Properties of Amorphous GeTe Films, NOLTR 72-196 (1972), K. P. Scharnhorst, H. R. Riedl.	

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Name(s): K. P. Scharnhorst

Inst. & Address: Naval Surface Weapons Center, White Oak, Silver Spring, Maryland 20910

Telephone: 394-2774

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ge _x Te _{1-x}	M/Sub	Films	<u>Purity:</u> Starting Elements—99.9999% <u>Properties Studied:</u> Photoconductivity, Electrical Resistivity of the Amorphous State. <u>Applications:</u> Infrared Detectors. <u>Publications:</u> 1) Photoconductivity in the Amorphous Ge-rich Ge _x Te _{1-x} System, J. Appl. Phys., <u>45</u> , 2971 (1974), K. P. Scharnhorst, H. R. Riedl.	
[(Pb _{0.8} Sn _{0.2}) _x Ge _{1-x}]Te (Germanium Lead Tin Telluride)	M/Sub	Films	<u>Purity:</u> Starting Elements—99.9999% <u>Properties:</u> Photoconductivity, Electrical Resist- ivity of the Amorphous State. <u>Applications:</u> Infrared Detectors. <u>Publications:</u> 1) Photoconductivity in Amorphous Ge-Pb-Sn-Te Alloys, Proceedings of the Special Meeting on Unconventional Detectors, IRIS Specialty Group, p. 113, University of Michigan, K. P. Scharnhorst, H. R. Riedl. 2) Temperature and Intensity Dependence of Photocon- ductivity in Amorphous (GePbSn)Te Films, J. Appl. Phys. <u>43</u> , 5142 (1972), K. P. Scharnhorst, H. R. Riedl.	

251 Name(s): K. P. Scharnhorst Inst. & Address: Naval Surface Weapons Center, White Oak, Silver Spring, Maryland 20910 Telephone: 394-2774				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
(Pb _{1-x} Sn _x)Se (Lead Tin Selenide)	M/Sub	Films	Purity: Starting Elements—99.9999% Properties Studied: Photoconductivity. Application: Infrared Detectors. Publications: 1) Epitaxial (Pb _{1-x} Sn _x)Se Films, NOLTR 70-233 (1970), H. R. Riedl, K. P. Scharnhorst.	

252 Name(s): Richard Schoolar, James Jensen Inst. & Address: Naval Surface Weapons Center, Silver Spring, MD 20910 Telephone: 202-394-2783				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Pb _x Cd _{1-x} S 1 ≥ x ≥ .92	Sub	15x15x.01	<111> n- and p-type, electrical, optical, photovoltaic, ir detectors	+
PbS _y Se _{1-y} 1 ≥ y ≥ 0	Sub	15x15x.01	<111> n- and p-type, electrical, optical, photovoltaic, ir detectors	+
Pb ₂ Sn _{1-z} 1 ≥ z ≥ .90	Sub	15x15x.01	<111> n- and p-type, electrical, optical, photovoltaic, ir detectors	+

253 Name(s): F. Rosenberger, with M.C. DeLong and G.F. Robertson Inst. & Address: Crystal Growth Laboratory, 329 N-Physics University of Utah, Salt Lake City, Utah 84112 Telephone: (801) 581-8372				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Bi ₄ Ge ₃ O ₁₂	Cz	20ø x 30	Purity: Crystals are grown from reagent grade and ultra-pure starting materials. Synthesis from the elements or further purification by zone refining, vacuum sublimation and/or high temperature chemical treatments, if required (Refs. 1-4) Orientation: Any desired orientation can be grown and/or cut. Routinely cylinder axis (100) Dopants: Alkali metals (e.g. 800 ppm Li ⁺ in KCl) alkaline earths, heavy metals, rare earths, OH ⁻ , OD ⁻ , CN ⁻ , etc. inquire. Mixed crystals: Alkali halides-alkali halides Alkali cyanides-alkali halides Dimensions: Unless stated otherwise minimum: 25ø x 25 mm maximum: 70ø x 50-60 mm Characterization: Laue diffraction Chemical analysis by atomic absorption/emission	All crystals listed are available. Inquire for terms.
CdWO ₄	Cz	15ø x 25		
CsBr	Cz, Br			
CsCl	Cz, Br			
CsF	Cz, Br			
CsI	Cz, Br			
HgI ₂	VP	?		
KBr	Cz, Br			
KCl	Cz, Br			
KCN	Z, Cz			
KF	Cz, Br			
KMnF ₃	Cz			
KI	Cz, Br			
LiBr	Cz, Br			
LiCl	Cz, Br			
LiD	Cz			
LiF	Cz, Br			
LiH	Cz			
			Contd.	

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F. Rosenberger, with M.C. DeLong and G.F. Robertson

Name(s):

Crystal Growth Laboratory, 239 N-Physics

Inst. & Address:

University of Utah, Salt Lake City, Utah 84112

Telephone: (801) 581-8372

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiI	Cz,Br	15ø x 25	See also previous page	All crystals listed are available. Inquire for terms.
Mg	VP,Br		The melt grown alkali halides have dislocation densities (D) between 10^4 - 10^7 cm ⁻² . Small samples (1 cm ³) with $D < 10^6$ cm ⁻² are grown from the vapor. (Ref. 5)	
NaBr	Cz,Br			
NaCN	Z,Cz			
NaF	Cz,Br			
NaI	Cz,Br			
RbAg ₄ I ₅	Cz			
RbBr	Cz,Br			
RbCl	Cz,Br			
RbF	Cz,Br			
RbI	Cz,Br			
TiO ₂	VP	10 x 10 x 3 ?	References:	
	M		1). Mat. Res. Bull <u>1</u> ,55 (1966). 2). Mat. Res. Bull <u>1</u> , 123 (1966). 3). <u>Crystal Growth</u> , H.S. Peiser ed. (Pergamon 1967) pp. 141-145. 4). <u>Ultrapurity</u> , M. Zief and R. Speights, eds. pp. 3-69. 5). J. Crystal Growth <u>43</u> , 148 (1978).	

255 Name(s): Roger A. Castonguay Inst. & Address: Varian/Lexington Vacuum Division 121 Hartwell Avenue, Lexington, MA 02173 Telephone: (617) 861-7200				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gallium Arsenide	CZ	37 mm Dia. 150 mm Long	<111> Photocathodes, LED'S FET'S	+
Lanthanum Hexaboride	Z	8 mm Dia. 100 mm Long	<111> Cathodes	+
Gallium Phosphide	CZ	37 mm Dia. 75 mm Long	<111> <100> LED'S	+
Gadolinium Gallium Garnet GGG	CZ	37 mm Dia. 150 mm Long	<100> Bubble Domain Sustrates	+
Calcium Tungstate	CZ	37 mm Dia. 150 mm Long	Neodymium Doped	
Calcium Fluoride	CZ	37 mm Dia. 150 mm Long	Rare Earth Dopants	
Refractory Oxides	Skull Melting			

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Name(s): Dr. Kumar S. Majumder (Stanford University, California)

Inst. & Address: Varian Associates, 611 Hansen Way Palo Alto, Ca. 94303

Telephone: (415) 493-4000
X 3519

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Copper with Alumina Dispersion (Cu + Al ₂ O ₃)	PVD	100x100x2	OFBC copper and ultrapure sapphire were used. Equiaxed matrix with very fine (~40 Å) amorphous dispersion. Max. amount ~2 vol. %. Deposition parameters (substrate temp., deposition rate, surface roughness, amount of dispersoid, etc.) were investigated. Microhardness and room temperature tensile properties, and creep at elevated temperatures were studied. Fractography was done. Various characterization techniques employed. Excellent TEM photos show the fine dispersions. Applications: High strength conductors at elevated temperatures. Publications: 1) Thin Solid Films, <u>42</u> (1977) 327. 2) Thin Solid Films, <u>42</u> (1977) 343. 3) Thin Solid Films, <u>45</u> (1977) 527.	+ only for research purposes

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Name(s): G. E. Duvall, J. R. Asay, Y. M. Gupta, G. Rosenberg, G. R. Fowles

Inst. & Address: Washington State University, Pullman, WA 99164

Telephone: 509-335-3310

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Lithium Fluoride, LiF		9mm x 25 mm diameter	Doped with Mg, <10 to 600 ppm, shock experiments, propagation directions <100>, <110>, <111>; dynamic yield strength measurements.	

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Name(s): M. G. Fey, T. N. Meyer, W. H. Reed

Inst. & Address: Westinghouse Electric Corporation, Power Circuit Breaker Division
Forbes Road, Trafford PA 15085Telephone: (412) 256-5210
(412) 256-5214

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Polycrystalline, Solar Grade Silicon Development of a Process for High Capacity-Arc Heater Production of Solar Grade Silicon via a high temperature, sodium reduction of silicon tetrachloride (4Na + SiCl ₄ → Si _(l) + 4NaCl _(v))			Expected silicon purity to be solar grade, silicon product will be collected in liquid phase, project sponsored by the Jet Propulsion Laboratory (JPL) for the U.S. Department of Energy. Publications: Project Quarterly Reports distributed by JPL W. H. Reed, et al, "Development of a Process for High Capacity-Arc Heater Production of Silicon for Solar Arrays", <u>Proc. of IEEE Photovoltaic Specialists Conf.</u> , to be published late 1978.	Experi- mental

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Name(s): D. L. Barrett				
Inst. & Address: Westinghouse Research & Development Center 1310 Beulah Road, Pittsburgh, PA 15235			Telephone: (412)256-3360	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs - Gallium Arsenide	Cz	60 diam.	Semi-insulating for microwave semiconductor application.	-
GaAs - Gallium Arsenide	VP	25x.2x40	Epitaxial GaAs layers for microwave semiconductor application - undoped buffer layers, doped active layers, n ⁺ layers. Hall, CV and profile analysis.	-
Si - Silicon Web	M	30x2x1200	Silicon web for large area semiconductor applications, optical and solar cell.	-
SiC - Silicon Carbide	Sub	10x.5x10	SiC single crystal for semiconductor applications - high purity, polytype mobility, grown junction.	-

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Name(s): H. M. Hobgood, J. C. Swartz, R. N. Thomas				
Inst. & Address: Westinghouse Research & Development Center 1310 Beulah Road, Pittsburgh, PA 15235			Telephone: (412)256-7588	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Si, silicon	Cz, Z	50mm-75mm dia.x40cm	20-10,000 Ω -cm/<100>, <111>/boron, phosphorus (conventional, NTD), sulfur, selenium/photoconductivity/resistivity, i.r. absorption, Hall-effect/photoconductive devices.	-
Si:In, indium-doped silicon	Cz, Z	30-75mm dia.x40cm	0.4-1 Ω -cm/<100>, <111>/indium, phosphorus (conventional and NTD)/photoconductivity/resistivity, i.r. absorption, variable temperature Hall effect/photoconductive devices.	-
Si:Ga, gallium-doped silicon	Cz, Z	30-75mm dia.x40cm	0.1-1.5 Ω -cm/<100>, <111>/gallium, phosphorus/photoconductivity/resistivity, i.r. absorption, variable temperature Hall-effect/photoconductive devices.	-

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Name(s): F. A. Shirland				
Inst. & Address: Westinghouse R&D Center, Pittsburgh, PA 15235			Telephone: (412) 256-3222	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cadmium Sulfide	VP	.03 x 150 x 150	+C oriented polycrystalline thin films for solar cells. Stoichiometrically doped, 1 to 10 ohm-cm, on various substrates	+ for research

262				
Name(s): Arthur B. Ellis				
Inst. & Address: Chemistry Department, University of Wisconsin-Madison, 1101 University Avenue, Madison, WI 53706			Telephone: 608-262-0421	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdS, CdSe, CdTe, GaP, GaAs, InP, Si		10x10x1 typically	Use of commercially available materials in photoelectrochemical cells for the conversion of optical energy to electricity. See J. Amer. Chem. Soc., <u>99</u> , 2839, 2848 (1977) and references therein.	

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Name(s): Tu Chen

Inst. & Address: Xerox Palo Alto Research Center, 3333 Coyote Hill Road
Palo Alto, CA 94304

Telephone: (415) 494-4175

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
MnSb Single Crystal	Cz	5x5x10	Mn _{0.55} Sb _{0.4} to ~ MnSb (99.9% pure material), nonstoichiometric	+
Mn ₂ Sb " "	"	10x10x20	" "	+
NiSb " "	"	10x10x10	" "	+
CoSb " "	"	10x10x10	" "	+
AuSn " "	Br	10x10x10	Good stoichiometry (>99.99% pure)	+
CuSn " "	"	2x5x10	" "	+
AuGa " "	"	5x10x10	" "	+
MnSb, NiSb and CoSb published on growth, structures and physical properties.				

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Name(s): James C. Mikkelsen, Jr.

Inst. & Address: Xerox Palo Alto Research Center, 3333 Coyote Hill Road, Palo Alto, CA
94304

Telephone: (415) 494-4171

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
layered dihalides: PbI ₂ , CdI ₂ , CdCl ₂ , CdBr ₂ , MgBr ₂	Br	5 x 5 x 5	CdBr ₂ , CdCl ₂ , and MgBr ₂ are hygroscopic and are encapsulated in silica ampoules	+
monohalides: InBr, InI	Br	5 x 5 x 5		+
layered dichalcogenides: SnS ₂ , SnSe ₂	Br	8 x 5 x 5	several polytypes of SnS ₂ ; phonon properties studied	+
layered monochalcogenides: GaS, GaSe	Br	8 x 5 x 3		+
NiAs-type: NiAs, NiS _{1+x} , NiTe _{1+x} , MnTe _{1+x} , CoTe _{1+x} , Ni _{1+x} Sb, Co _{1+x} Sb, Cr _{1+x} Sb	Br	5 x 5 x 5	composition, lattice parameter, and reflectivity measured	+
chalcopyrite-type: AgGaSe ₂ , AgGaS ₂ , CuGaS ₂ , CuGaSe ₂ , AgGa(S _x Se _{1-x}) ₂ for x = 0.3, 0.5, and 0.7	Br	5 x 5 x 5 3 x 2 x 2	optical and nonlinear optical properties	+
SiAs, GeAs	Br	3 x 2 x 2	phonon properties studied	+
Li ₂ Ti ₃ O ₇	Br	5 x 3 x 3	Raman, optical properties, NMR, conductivity (AC)	

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Name(s): Stanley Mroczkowski

Inst. & Address: Yale University, 427 Becton Center, New Haven, CT. 06520
(Department of Engineering & Applied Science)

Telephone: (203)436 3630

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Magnetic & non-magnetic garnet	Flu.	3-5 m/m ²	J. of Crystal Growth <u>24/25</u> , 1974. Random. Magnetic studies and IR tunable lasers.	+ some
Rare earth-halides anhydrous RECl ₃	Br.	1-2 cm ²	J. of Crystal Growth <u>6</u> , 1970 Magnetic studies.	+ some
Rare earth-hydroxides RE(OH) ₃	Hyd.	1-5 m/m long	J. of Crystal Growth <u>13/14</u> , 549-551, 1972. C-axis along the crystal.	+ some
Rare earth-oxycarbonates	Hyd.	1-2 m/m ²	Optical and structural studies.	-
Bi ₁₂ SiO ₂₀	Cz.	1/2 cm x 10 cm.	J. of Crystal Growth <u>42</u> , 1977. 100 111 Electro-optical application.	+
K ₂ PtCl ₆	Sol.	3-5 m/m ²	Low temperature phase transition studies.	+
K ₂ OsCl ₆	Sol.	3-5 m/m ²		+
BaClF:Eu ⁺²	Cz.	5 m/m x 50m/m	For tunable solid state lasers.	-
BaCl ₂ :Eu ⁺²	Br.	1 cm x 3 cm		-
CDP:Na ₃ [Ce(C ₇ H ₃ NO ₄) ₃].15H ₂ O	Sol.	5 m/m ²	J. of Crystal Growth <u>16</u> , 1972. Material for millikelvin thermometry	+
Ni and Ni Alloy: single crystal Platelets	VP	3-4 m	110 Magnetic studies	-
TaS ₂ , TaSe ₂ , NbS ₂ , NbSe ₂	VP	0.1 m/m max.	Hexagonal platelets Superconductivity measurements	+

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Name(s): H. Catherine W. Skinner

Inst. & Address: Yale University, New Haven, Connecticut

Telephone: 436-1451

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ca ₅ (PO ₄) ₃ (OH) Hydroxyapatite	Hyd	0.2x0.2x1.0	High purity, Studied: optical properties x-ray dif- fraction characteristics phase diagram determination	0
Ca ₂ P ₂ O ₇	Hyd	1.0x0.5x5.0		+
CaHPO ₄ Monetite	Hyd	1.0x1.0x5.0	X-Ray Diffraction Analysis Techniques to Monitor Com- position Fluctuations Within the Mineral Group:	+
CaHPO ₄ .2H ₂ O Brushite	Sol	.01x3x5	Apatite. Applied Spectroscopy, 22:412-414, #5, Part I, 1968. Transition of Calcium Orthophosphate to Pyrophosphate at Elevated Pressures. Mat. Res. Bull., 5:437-448, 1970. Studies in the Basic Mineralizing System, CaO-P ₂ O ₅ - H ₂ O. Calc. Tiss. Res., 14:3-14, 1973. Phase Relations in the CaO-P ₂ O ₅ -H ₂ O System from 300- 600°C at 2 kb H ₂ O Pressure. ²⁵ Amer. J. Sci., 273:545-560, Summer 1973. Thermal Instability in Synthetic Hydroxyapatites. J. of Physical Chemistry, 79:2017-2019, 1975.	+

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Name(s): ENRIQUE GRUNHUT

Inst. & Address: Avda. Gral. Paz y Albarelos, C.C. 157, 1650 San Martín, Bs.As., ARGENTINA Telephone: 752-0196

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe-Cadmium telluride	Sol. + <u>Z</u>	50x10x10	99.999 ⁺ , undoped. For nuclear radiation detectors	+
CdTe-Cadmium telluride	<u>Z</u>	30x10x10	99.999 ⁺ , undoped. For solar cells	-
HgI ₂ -Mercury Iodide	<u>Sub</u>	10 x 5 x 3	Undoped. For nuclear radiation detectors	-
Si-Silicon epitaxial	<u>GE</u>	30x30x10 ⁻³	Semiconductor grade. n and p doped. For semiconductor devices	+

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Name(s): M.A.R. de Benyacar

Inst. & Address: Comisión Nacional de Energía Atómica - ARGENTINA

Telephone:

Investigaciones - Avda. del Libertador 8250 - 1429 Bs.As.

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
(AsO ₄) ₂ H ₂ (UO ₂) ₂ · 8H ₂ O Hydrogen uranospinite	Sol.	2x2x0.5	<u>Publications:</u> - American Mineralogist <u>59</u> , 763, (1974) - Ferroelectrics <u>9</u> , 241, (1975) - " <u>17</u> , 469, (1978) (Studies on dielectric, optical and thermal behaviour).	Only in small quantities

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Name(s): Juana Bebczuk de Cusminsky

Inst. & Address: Facultad de Ingeniería, Universidad Nacional de La Plata,
calle 47 esq 1, La Plata 1900, Argentina

Telephone: 34871

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cu (copper)	EC	8x8x10	<u>Purity:</u> 99.98% and 99.999% <u>Orientation of single crystals:</u> (100), (110), (111)	
	Br	10x10x40	<u>Properties studied:</u> 1. Habit and Structure of Electrodeposits 2. Epitaxial Growth by Electrodeposition 4. Polygonization <u>Publications:</u> 1) J. Bebczuk de Cusminsky; Electrochim. Acta <u>15</u> , 73 (1970). 2) J. Bebczuk de Cusminsky and H. Wilman; Electrochim. Acta, <u>17</u> , 237 (1972). 3) J. Bebczuk de Cusminsky; Scripta Metall. <u>10</u> , 1071 (1976). 4) J. Bebczuk de Cusminsky; J. Cryst. Growth <u>41</u> , 330 (1977). 5) J. Bebczuk de Cusminsky and E. T. Niemela; J. Mat. Sci. <u>12</u> , 1492 (1977).	
Ag (silver)	Br	10x10x40	<u>Purity:</u> 99.98% and 99.999% <u>Orientation:</u> (111) <u>Properties studied:</u> Polygonization <u>Publications:</u> J. Bebczuk de Cusminsky; Surface Technology <u>7</u> , 129 (1978).	

270

Name(s): John S. Cook

Inst. & Address: National Measurement Laboratory, C.S.I.R.O., P.O. Box 218,
Lindfield, N.S.W. 2070, AUSTRALIA

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Alkali halides	Cz	25x25x100	High purity, cation and anion doped, dielectric absorption, electrical conductivity, hardness, UV and IR absorption and emission. Series of publications with Dryden, Meakins, Harvey and Heydon.	+

271

Name(s): S. MYHRA

Inst. & Address: School of Science, Griffith University, Nathan 4111, Qld., Australia

Brisbane

Telephone: 275-7271

MATERIALS (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InSb (n- and p-type) Also high-purity.			Electron radiation damage and the effect on transport properties at low temperatures (4K and 80K) including galvanomagnetic, thermo-electric and thermomagnetic effects.	-
High-purity Zn (6 9's)			Electron radiation damage with electrical resistivity as monitor	+

272

Name(s): Prof. R.L. Segall, Dr R.St.C. Smart and Dr P.S. Turner.

Inst. & Address: School of Science, Griffith University, Nathan, Queensland 4111, Australia Telephone: Brisbane 275 7111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Nickel oxide Cobalt oxide	VP	Powders 0.1 - 1 μ m	Characterisation of surfaces using infrared spectroscopy, scanning and transmission electron microscopy - dissolution kinetics (in acid solution) of semi-conducting oxides. 1. Semiconducting Oxides. Effect of Prior Annealing Temperature on Dissolution Kinetics of Nickel Oxide, J. Chem. Soc. Faraday I, 1977, 73, 1710. 2. Semiconducting Oxides. Effect of Electronic and Surface Structure on Dissolution Kinetics of Nickel Oxide, J. Chem. Soc. Faraday I, in press. 3. Semiconducting Oxides. Rate and Infrared Studies of Surface Blocking by Surfactants in Dissolution of Nickel Oxide, J. Chem. Soc. Faraday I, in press. 4. Semiconducting Oxides. Dissolution Kinetics of Cobaltous Oxide, to be submitted to J. Chem. Soc. Faraday I.	-
Magnesium oxide	VP	Powders 0.05-0.5 μ m	Similar Methods - dissolution kinetics of ionic oxides. 1. Ionic Oxides. The Dissolution of Magnesium Oxide, submitted to J. Chem. Soc. Faraday I. 2. Growth Twinning in Magnesium Oxide Smoke Crystals, submitted to Phil. Mag. A.	-

273

Name(s): A. R. BILLINGS, Y. C. MAK, N. PAZOLLI, A. BENN

Inst. & Address: Electrical & Electronic Engineering Department, University of Western Australia, Nedlands, Western Australia 6009

Telephone: 380-3105
380-3048

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gallium Arsenide (and related ternary compounds)	MBE*	25x25x0.005	Epitaxial thin films being investigated with respect to infrared waveguides and integrated optics applications *Molecular Beam Epitaxy	
Titanium-diffused Lithium Niobate channel waveguides	diffusion	25x10x1	Construction of Kogelnik-type thin film optical directional couplers for integrated optics applications Orientation: y-cut & z-cut LiNbO ₃ crystals Dopant: Titanium	

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Name(s): A. R. BILLINGS, Y. C. MAK, N. PAZOLLI, A. BENN

Inst. & Address: Electrical and Electronic Engineering Department
University of Western Australia, Nedlands, Western Australia 6009Telephone: 380-3105
380-3048

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Lithium Niobate (LiNbO ₃) thin films deposited on Lithium Tantalate (LiTaO ₃) substrate	Diode R.F. sputtering	25x10x10 ⁻⁵	Investigation of the effect of substrate temperature and sputtering gas on epitaxial growth of sputtered LiNbO ₃ films on LiTaO ₃ substrate has shown that transparent and chemically stable films were obtained when the substrate temperature was approximately 400°C and the Ar:O ₂ ratio was 2:1. Higher substrate temperature and/or lower oxygen content produced brown films and decolourized the transparent LiTaO ₃ substrate. Optical and waveguide properties of the LiNbO ₃ -LiTaO ₃ film structure are being studied.	

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Name(s): A. R. BILLINGS, Y. C. MAK, N. PAZOLLI, A. BENN

Inst. & Address: Electrical and Electronic Engineering Department,
University of Western Australia, Nedlands, Western Australia 6009Telephone: 380-3105
380-3048

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Lithium Niobate (LiNbO ₃) thin films deposited on optically polished vitreosil glass substrate	Diode	25x10x10 ⁻⁴	All freshly deposited LiNbO ₃ films were transparent and extremely adherent. After exposure to air, white haze formed on all film surfaces. Results obtained from electron microprobe measurement showed definite Nb and O ₂ elements. Li was too high to be detected by the probe. Film refractive index obtained from ellipsometer measurement was found to be below 2, less than those of bulk material. A rutile prism coupler and helium-neon laser of wavelength 6328 Å were used to study the waveguide properties of these films. No waveguide modes were coupled into the LiNbO ₃ films. A bright short strip of light was observed terminating just outside the edge of the prism. It is considered that the poor waveguide performance of the LiNbO ₃ films was due to undesirable crystalline structure of the film and mismatched boundary conditions between film and substrate surface.	

276

Name(s): A. G. NASSIBIAN

Inst. & Address: Electrical & Electronic Engineering Department
University of Western Australia, Nedlands, Western Australia 6009

Telephone: 380 3047

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon	MOS	N.A.	Material studies, minority carrier lifetime, capture cross section of generation impurities, and gettering of generation impurities by Ar ⁺ implant damage method. Properties of silicon are studied by using non- equilibrium linear voltage ramp method. Materials are doped intentionally with gold, the lifetime of material and capture cross section of gold is measured, then gettering is carried out and various material parameters are related to the presence and absence of gold.	N.A.

276a

Name(s): Robert S. SEYMOUR

Inst. & Address: Laser Group, Defence Research Centre Salisbury.
Box 2151 G.P.O. ADELAIDE, South Australia 5001

Telephone: (08)259-807

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiNbO ₃ , lithium niobate	Cz	50 x 20 dia	c or b axis orientation	+
KH ₂ PO ₄ , potassium dihydrogen phosphate	Sol.	30 x 30 x 60	c axis orientation, laser frequency doubling	+
NaNO ₃ , sodium nitrate	Cz	40 x 20 dia	c axis orientation, polarising prism studies	+

277

Name(s): Dr. Z. Boncheva-Mladenova

Inst. & Address: Higher School of Chemical Technology and Engineering,
Sofia-56, Darvenitza, Bulgaria

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ag ₂ Te	VP (chem. trans- port)	4.5x1x1	The Laue photographs indicate that the material is in single crystal form. Publications: Compt. Rend. l'Acad. Bulg. Sci. 22, N2 (1969); ibid. 25, N2 (1972). Phys. Stat. Sol. (a) 47, K59 (1978). O. Pavlov, thesis, Higher School of Chemical Technology and Engineering, 1977.	-
Ag ₂ Se	VP (chem. trans- port)	1x1x0.5	X-ray rotating photographs indicate that the material is in single crystal form. Emission spectroscopy indicates that the material is contaminated with Cu, Ti, V. N. Aramov, thesis, Higher School of Chemical Technology and Engineering, 1977.	-
Bi ₂ (SeO ₄) ₃	Gel growth	5x3x2	Electron diffraction indicates that the material is in single crystal form, rhombic system, d ₂₀ = 3.92g/cc, parametric ratio a:b:c = 0.64:1.00:1.08. Emission spectroscopy of the Bi ₂ (SeO ₄) ₃ and start- ing materials shows that the growth process is effective in improving the purity of the crystals.	-

278 Name(s): Dr. Z. Bancheva-Mladenova Inst. & Address: Higher School of Chemical Technology and Engineering Sofia-56, Darvenitza, Bulgaria Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ag ₂ SeO ₄	Gel growth	50x3x2	X-ray rotating photographs and electron diffraction indicate that the material is in single crystal form. Ag ₂ SeO ₄ crystals are successfully doped with Cu and Ni. The material is characterized by X-ray diffraction powder pattern and infrared spectrum.	-

279 Name(s): G.J.C. Carpenter and J.F. Watters Inst. & Address: Atomic Energy of Canada Limited Chalk River Nuclear Labs. Telephone: (613) 584-3311 Chalk River, Ontario Canada KOJ 1J0 local 743/672				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Zirconium	Z	100 5 5	Iodide purity Random orientation Used mainly for studies of radiation-induced growth and creep, and of hydride precipitation.	On a collabo- rative basis

280 Name(s): M.L. Swanson, A.F. Quenneville Inst. & Address: Atomic Energy of Canada Limited, Chalk River Nuclear Laboratories Telephone: (613) 584-3311 Chalk River, Ontario KOJ 1J0 Canada				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Al	Br	13x12x70	99.999% and dilute alloys containing 0.01-0.2 at% Mn, Cu, Zn, Ga, Ge, Ag, Sn. Random orientation for channeling studies	+
Cu	Br	13x13x70	99.999% and dilute alloys containing 0.1% Be, Ag, Sb, Au Random orientation for channeling studies.	+
Zn	Br	13x13x70	99.999%	+
Au	Br	10x10x40	99.999%	+

281 Name(s): W.K. Unger, J.C. Irwin Inst. & Address: Dept. of Physics, SFU, Burnaby, B.C., Canada Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdIn ₂ S ₄	VP	2x2x2	Phys. Stat. Sol. (a) 46, 81 (1978)	-
Zn _x In _{2-2x} S ₄	VP		Solid State Commun. 25, 913 (1978)	

282

Name(s): Martin Melvin Perlman

Inst. & Address: Department of Physics, College militaire royal
Saint-Jean, Quebec, CANADA

Telephone: 1-347-5381

Ext. 587

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KCl (Potassium Chloride)	<u>C_z</u>	20 × 20 × 20	Dopants - Divalent Elements e.g. Ca ²⁺ , Eu ²⁺ , Sr ²⁺ .	+
NaCl (Sodium Chloride)	<u>C_z</u>	20 × 20 × 20	Electrical Properties Dielectric Measurements Ionic Thermal Currents Publications: 1) Phys. Rev. <u>B 6</u> , 3973 (1972) 2) Phys. Rev. <u>B 10</u> , 3692 (1974) 3) J. Electrostatics (England) 1, 231 (1975) 4) Phys. Rev. <u>B 12</u> , 809 (1975) 5) Phys. Rev. <u>B 12</u> , 5997 (1975) 6) Phys. Rev. <u>B 15</u> , 4105 (1977)	

283

Name(s): A. G. White, R. F. Redden or H. E. Hirsch

Inst. & Address: Cominco Ltd. Trail B. C., Canada V1R 4L8

Telephone: (604) 364-4758

(604) 364-4751

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)		METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.		Material Avail- ability + -
Al	Aluminum	Z	Not Restricted	99.9999+%	Resistivity Ratio >30,000,Single Crystal <	

284

Name(s): Dr. B.A. LOMBOS

Inst. & Address: Concordia University, 1455 de Maisonneuve Blvd. W, Montreal H3G 1M8, Quebec

Telephone: (514) 879-4078

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Hg Te	Br	15x5x5	Transport properties	n
Hg Se	Br	15x5x5	I.R. detectors	n
Hg Te _x Se _{1-x}	Br	15x5x5		n
Hg S	vP	1	Optical properties	-
Ga As	Z	48x22x300	Semi-insulating, Optoelectronics	p,n
Si	C _z , vP	50 O.D x 200	Solar cells	p,n
Ge	C _z	500(O.D.)x200	Discrete impurity band conduction (under uniaxial stress)	p,n

285

Name(s): Wm. E. Pinson Infrared Photo Ltd., Ottawa, K2G 2W3. and

Inst. & Address: c/o Physics Dept, Univ. of Ottawa, Ottawa, Ont. K1N 6N5.

Telephone: 613 224 2443

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdS _x Se _{1-x} , prep & stud.	Sub, GE	10x10x10 ⁻³	Films prepared under near-equilibrium conditions as part of heterojunction anodes for semiconductor-liquid-junction solar cells.	+
GaP, GaAs _{0.6} P _{0.4} , GaAs, Cu ₂ O			Studied as electrodes in semiconductor liquid-junction solar cells.	+

286

Name(s): Dr. S.J. Jones

Inst. & Address: Inland Waters Directorate, Dept. of Environment, 562 Booth St., Ottawa, K1A 0E7

Telephone: (613) 995-4058

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ice (H ₂ O)	Z	100 Diam 1000 Long	Very high purity as shown by dielectric measurements. Orientation as desired. Doped with HF, NH ₃ . Also one D ₂ O ice sample. Electrical and mechanical studies carried out.	On request.

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Name(s): C. H. CHAMPNESS

Inst. & Address: McGill University, Electrical Engineering Dept., 3430 University Street,
Montreal, P.Q., Canada H3A 2A7

Telephone: 514-392-4208

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tellurium	Cz	20 x 20 x100	Pulled from high purity melt Sb - doped	+ -
Selenium hexagonal films	GE	10 μ thick	undoped or Cl-doped (1010) or (0001) orientations	-
Se-Te alloys	Cz	-	under study	-

288

Name(s): DR. J. E. GREEDAN AND MR. J. D. GARRETT

Inst. & Address: Institute for Materials Research, McMaster University, Hamilton, Ontario, Telephone: (416) 525-9140
Canada. L8S 4M1 Ext. 4563 & 2005

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiF (Mg)	Cz	25x25x50	Mg-DOPED <100> .225 mole %	-
LiBO ₂	Cz	5x10x15	(010) Raman & IR various isotopic substitution	+
Li ₂ B ₄ O ₇	Cz	10x10x10	(102) Same as LiBO ₂	+
LnTiO ₃ Ln = Lanthanide	Cz, Br	5x3x2	Magnetic, optical, electrical properties	-
Heusler alloys NiMnSn ₂ NiMn _{.8} V _{1.2} Sn ₂	Cz, Br	25x25x50	Neutron inelastic scattering	-
CsMnBr ₃ RbMnBr ₃	Cz, Br	10x20x25	Neutron elastic and inelastic scattering	+
CuO	V.P.	1x1x0.1	Neutron scattering	-
ReO ₃	V.P.	2x2x0.5	DIVA (de Haas - van Alphen Effect)	-
CdRbCl ₃ KCdCl ₃ KCdBr ₃	Cz, Br	5x5x5. 15 dia.x10	Raman scattering	+
NaNO ₂	Cz		NMR studies	-
LiNaV ₂ O ₆		1x1x5	Structure & phase changes	-

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Name(s): Brooker, Murray H.

Inst. & Address: Chemistry Department, Memorial University of Nfld. St. John's Nfld Canada Telephone: 709-753-1200
A1B 3X7

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NaNO ₃ sodium nitrate	sol	3 x 3 x 3	Infrared and Raman studies have been performed on the single crystals. Normal modes of vibration have been assigned and phase transitions studied. Recent publications J. Chem. Phys. <u>68</u> , 67 (1978)	+
KNO ₃ potassium nitrate	sol	6 x 4 x 2		+
RbNO ₃ rubidium nitrate	sol	4 x 2 x 2		+
CsNO ₃ cesium nitrate	sol	4 x 2 x 2		+
TlNO ₃ thallium nitrate	sol	2 x 2 x .5		+
Sr(NO ₃) ₂ strontium nitrate	sol	4 x 4 x 2	J. Chem. Phys. <u>63</u> , 3054 (1975)	+
Pb(NO ₃) ₂ lead nitrate	sol	4 x 4 x 4		+
Ba(NO ₃) ₂ barium nitrate	sol	2 x 2 x 2		+
UO ₂ (NO ₃) ₂ ·6H ₂ O Uranyl nitrate	sol	4 x 4 x 2		+
Tl ₂ CO ₃ thallium carbonate	sol	6 x 2 x .5		+
K ₂ MgCl ₄ potassium tetrachloromagnesate	M	2 x 1 x 1		+
Cs ₂ MgCl ₄ cesium " "	M	2 x 1 x 1		+
Cs ₂ MnCl ₄ " tetra	M	2 x 1 x 1		+
Cs ₂ ZnCl ₄ " tetrachlorozincate	M	2 x 1 x 1		+
CsMnCl ₃ " trichloromanganese	M	2 x 1 x 1		+

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Name(s): M.J. Graham, R.J. Hussey, D.F. Mitchell, B.R. MacDougall

Inst. & Address: Division of Chemistry, National Research Council of Canada, Ottawa, Canada Telephone: 613-993-2518

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Nickel	<u>Z</u>	12 dia. x 1	(100), (111) and (112) Ni crystals of 99.997 wt.% purity. Characterized by electron and X-ray diffraction. Used in studies of the kinetics and mechanism of thin oxide film formation, and in studies of oxide film breakdown in aqueous environments.	+
Iron	<u>Z</u>	12x5x1	(112) Fe crystal of 99.997 wt.% purity. Characterized by electron and X-ray diffraction. Used in studies of the kinetics and mechanism of thin oxide film formation, and in studies of the structure of iron oxide films using Mössbauer spectroscopy.	+

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Name(s): C.M. HURD and S.P. McALISTER

Inst. & Address: NATIONAL RESEARCH COUNCIL OF CANADA, OTTAWA, CANADA K1A 0R6

Telephone: 613-993-2514

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cu, Ag, Au, Mg, Zn, Cd, Al sputtered and composite materials	Br	10 x 2 x 70 50 diam	Monocrystalline and polycrystalline samples prepared from 69-grade starting materials for basic studies of electron motion in metals and alloys. Studies of Hall effect, magneto-resistance, magnetic properties. Dopants include low melting point metals and especially 3d group transition metals.	

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Name(s): J. Rolfe

Inst. & Address: Physics Division, National Research Council, Ottawa, K1A 0S1, Canada

Telephone: (613)993-2046

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Alkali Halides: NaBr, NaCl, KCl, KBr, KI, RbCl, RbBr, RbI	Cz	15x15x75	Nominally pure, also doped with alkaline earths and some anionic impurities such as O_2^- , NO_3^- , CO_3^{2-} , OH^- .	+

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Name(s): Dr. I. M. Templeton

Inst. & Address: National Research Council, Division of Physics, Ottawa, Canada, K1A 0R6

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
AuAl ₂ , AuGa ₂ , AuIn ₂ , AuSn, PtSn, AuGa, Cu ₂ Sb, AuPb ₂ , CuZn, AgZn, InPd, MgTl, HgMg, YZn, AuMg, AgMg, Al ₂ Cu, NiSi, PdTe ₂ , Pt ₃ Sn	Br quench- anneal M	a few mm typically 2x2x6	de Haas-van Alphen effect and Fermi surface of intermetallic compounds, samples usually not available, information on preparation available	-
Cu, Ag, Au.	Br	Boule 10 diam x30 long	Pure metals and single xtal dilute alloys to ~1% with homovalent, heterovalent & transition metal solutes. de Haas-van Alphen studies: Fermi surface and electron scattering. Small samples occasionally available.	-
Cu, Ag, Au, Pt, Cu-Au, Ag-Au, CuPt	M	25x25x25	Polycrystalline pure metals and alloys. Ordering. Specific heat 0.4-30K studied.	-

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Name(s): R. Fletcher

Inst. & Address: Physics Dept., Queen's University, Kingston, Ontario, Canada K7L 3N6

Telephone: 613-547-6150

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cadmium	M	11 x 5 x 1½	RRR 4-9 x 104 C-axis perp. to length Magnetotransport R.J. Douglas & R. Fletcher, Phil. Mag. <u>32</u> , 73-81 (1975). R. Fletcher, J. Low Temp. Phys. <u>22</u> , 39-59 (1976).	+
Lead		25 x 8 x 1½	RRR ~ 10 ⁵ cubic axis perp. to length Magnetotransport R. Fletcher & M.R. Stinson, J. Low Temp. Phys. <u>27</u> , 787-800 (1977).	+
Tungsten	M	12 x 5 x 1½	RRR ~ 3 x 10 ⁴ cubic axis perp. to length Magnetotransport R.J. Douglas & R. Fletcher, Phil. Mag. <u>32</u> , 73-81 (1975). R. Fletcher, Phil. Mag. <u>32</u> , 565-76 (1975).	+
Indium		70 x 3.6 x 1	RRR ~ 10 ⁵ cubic axis perp. to length Magnetotransport B.J. Thaler & R. Fletcher, J. Low Temp. Phys. <u>30</u> , 773-97 (1978).	-
Molybdenum		12 x 5 x 1½	RRR ~ 6000 cubic axis perp. to length Magnetotransport R. Fletcher, Phy. Rev. <u>B14</u> , 4329-37 (1976). R. Fletcher, J.L. Opsal & B.J. Thaler, J. Phys. (F) <u>7</u> , 1489-96 (1977).	+

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Name(s): Professor Reginald W. Smith

Inst. & Address: Nicol Hall, Queen's University, Kingston, Ontario, CANADA K7L 3N6

Telephone: 613-547-6226
613-547-5877

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Metals and Alloys	M, Z Cz, Br	2.5x2.5x 20.0 cm.	99.9999 + controlled orientations B.C.C. alloys, Li-Mg Cu-Zn and its analogues, electronic structure via positron annihilation techniques. Eutectic alloys morphological characterisation, mechanical properties, electro-magnetic properties.	+ + +

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Name(s): Professor F.W. Boswell

Inst. & Address: Dept. of Physics, University of Waterloo, Waterloo, Ontario, CANADA N2L 3G1 Telephone: (519)885-1211

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cu_xNbS_2 ($x \leq 0.50$)	VP	6x6x0.1	Thin flakes. F.W. Boswell, A. Prodan and J.M. Corbett, Phys. Stat. Sol. (a) <u>35</u> , 591 (1976)	-
Fe_xNbS_2 ($x \leq 0.50$)	VP	5x5x0.1	Thin flakes. F.W. Boswell, A. Prodan, W.R. Vaughan and J.M. Corbett, Phys. Stat. Sol. (a) <u>45</u> , 469 (1978)	-
$\text{Mn}_{.25}\text{NbS}_2$	VP	5x5x0.1	Thin flakes. F.W. Boswell, W.R. Vaughan and J.M. Corbett, J. Sol. State Chem. <u>24</u> , 115 (1978).	-
Cu_xZrS_2 ($x \leq 0.45$)	VP	3x3x0.05	Thin flakes.	-
Cu_xHfS_2 ($x \leq 0.5$)	VP	3x3x0.05	Thin flakes.	-
Fe_xZrS_2 ($x \leq 0.45$)	VP	3x3x0.05	Thin flakes.	-
Fe_xHfS_2 ($x \leq 0.1$)		3x3x0.05	Thin flakes.	-
$\text{V}_{1-x}\text{Se}_2$ ($x \approx 0.10$)	VP	2x2x0.1	Thin flakes.	-

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Name(s): H.H. Schloessin

Inst. & Address: Dept. of Geophysics, University of Western Ontario
London, Ontario, Canada N6A 5B7

Telephone: 519-679-3147

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Calcite I, II, III - aragonite (optically clear single crystals) quartz-coesite/germanium dioxide (Single crystals + polycrystalline)		7 x 7 x 10	phase transitions; rate constants; nucleation and growth kinetics at P up to 6 GPa.	
Triglycine sulphate (single crystals) Pb (Zr, Ti)O ₃ ceramics BaTiO ₃ , CaTiO ₃ (single crystals)		1 x 7 x 7	ferroelectric phase transitions; dielectric constant measurements at P up to 6 GPa	
olivine enstatite calcite periclase spinel garnet quartz			anisotropic coefficients of lattice thermal cond., electrical conductivity, dielectric constant, sound velocity, at varying P and T	
single crystals			in situ-high pressure x-ray diffraction. x-ray diffraction topography	

298				
Name(s): Professor P.W.M. Jacobs				
Inst. & Address: University of Western Ontario, Department of Chemistry, London, Ontario, Telephone: (519) 679-2397 N6A 5B7, Canada.				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Alkali halides	M Br	10x10x5	Approx. 0.5 ppm purity; dopants M^{2+} , A^{2-} , M^+ Ionic conductivity ITC Optical absorption and Emission e.g. J. Phys. Chem. Solids <u>36</u> , 1375, 1383 (1975); J. Phys. (Paris) <u>34</u> , C9-437 (1973).	
Ammonium chloride	Sol	10x10x5	Purity unknown: dopants OH^- , H^+ , CrO_4^{2-} , Ni^{2+} ITC	

299				
Name(s): Bohumil Kratochvíl				
Inst. & Address: Dept. of Inorg. Chem., Charles Univ., Albertov 2030, Telephone: 292051 128 40 Praha 2, Czechoslovakia				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ka_2TeO_4	M	0.1x0.2 x0.3	Crystal structure / Acta Crystallogr. B33,2596, 1977	-

300				
Name(s): Zdeněk Kotrba				
Inst. & Address: Geological Survey, Malostranské n. 19, Prague 1, Telephone: 22-88-34 Czechoslovakia				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Geological samples, silica- tes, semiconductors			electron probe X-ray microanalysis	ref.mat. +

301 Name(s): S. KADEČKOVÁ, B. ŠESTÁK Inst. & Address: Institute of Physics ČSAV, Na Slovance 2, 180 40 Prague 8, Czechoslovakia Telephone: 84 22 41				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability* + -
Fe, (zone refined)	R	$\varnothing = 4$ $l = 90$	S. Kadečková, B. Šesták: Kristal und Technik 2 (1967), 191 S. Kadečková, B. Šesták: Can. J. Phys. 45 (1967), 1041	+
Fe - 0,5 wt % Si	R	$\varnothing = 4$ $l = 90$	K. Z. Saleeb, S. Kadečková: Kristal und Technik 9 (1974), 1265 S. Kadečková, K. Z. Saleeb: J. Cryst. Growth 30 (1975), 335 S. Kadečková, J. Brádler: J. Cryst. Growth 43 (1973), 301	+
Fe - 3 to 7 wt % Si	Z	$\varnothing = 13$ $l = 200$	S. Kadečková, B. Šesták: Kristal und Technik 4 (1969), 243	+
			Studies of Plasticity: V. Novák, K. Z. Saleeb, S. Kadečková, B. Šesták: Czech. J. Phys. B 26 (1975), 565 N. Zárubová, B. Šesták: phys. stat. sol. (a) 30 (1975), 365; 479 *) Single crystals of the desired orientation are available	

302 Name(s): dr. Zdirad ZAK Inst. & Address: Department of Inorganic Chemistry, J. E. Purkyne University Kotlarska 2, 611 37 BRNO, Czechoslovakia Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiSO₃F Lithium Fluorosulphate	Sol	1x0.5x0.5	* crystal structure study	
Se₂O₅ Diseleniumpentoxide	Sol	5x1x1	* crystal structure study	

303 Name(s): Salah Arafa Inst. & Address: Materials Engineering and Physical Sciences Department, The American University of Cairo, 113 Kasr El-Aini Street, Cairo, Egypt Telephone: 22969 Cairo				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
TiO₂	Flu	0,5x0,5x20 rods	Containing vanadium and lead impurities	+
Al₂O₃	Flu	"	Magnetic Properties	+
NaCl	Br	10x10x50	High purity, color centers and ionic conductivity	+

304 Name(s): Lehto A. and Harjunmaa H. Inst. & Address: University of Helsinki, Department of Physics Siltavuorenpenger 20 D, SF- 00170 Helsinki 17 Telephone: 90-650211				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon	delayed break- down		Lifetimes of the Lower Excited s-States of Phosphorus in Silicon at Low Temperatu- res. Samples: 100 ohmcm (room temp) P- doped Silicon. Application: IR- detector. Publications: Lehto A. and Proctor W.G.: J. Phys. C: Solid St. Phys., <u>10</u> , L481-6 (1977) Lehto A.: J. Phys. C: Solid St. Phys., <u>10</u> , L 699- 703 (1977) Lehto A and Proctor W.G.: J. Phys. C: Solid St. Phys., issue of 14 June, in print (1978)	+

305 Name(s): Markus Pessa, Markku Ahonen, Tuomo Suntola, Gunnar Graeffe Inst. & Address: Tampere University of Technology Dept. of Physics, P.O. Box 527, SF-33101 Tampere 10, Finland Telephone: 931-162 111				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
II-VI compound semiconductors (thin films)	MBE	$10 \times 10 \times 10^{-3}$	Different orientations are obtainable depending on the substrate material used. Dopants are (usually) Mg and Sn. Properties studied are crystallographic structure of the surface, and electronic structure of the valen- ce band. Characterization of the films is carried out by using UPS/XPS/AES/LEED techniques. The films are prepared for use in optoelectronic devices.	+

306 Name(s): Dr. P. Paufler, Dr. K. Eichler Inst. & Address: TU Dresden, Sektion Physik, DDR 8027 Dresden, Mommsenstr. 13 GDR Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
MgZn ₂	Br	$\phi = 10 \text{ mm}$ $l = 80 \text{ mm}$	high purity; rod axis prallel (0001); dislocation distribution and mechanical properties as a function of composition within the range of homogeneity. Thesis K. Eichler; TU Dresden 1973	-

307				
Name(s): E. Krén				
Inst. & Address: CENTRAL RESEARCH INSTITUTE FOR PHYSICS H-1525 Budapest 114, P.O.B. 49, Hungary		Telephone: 166-440		
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gd ₃ Ga ₅ O ₁₂ /Gadolinium-Gallium-Garnet/	CZ	Ø 25x100	Purity: 4N Orientation: <111> Characteristics: dislocation free Application: Substrate material	+
/Y, Sm, Ca/ ₃ /Fe, Ge/ ₅ O ₁₂	LP	Ø 20 mm Thickness: 5 µm	Orientation: <111> Characterized defects, magnetic properties Application: bubble memory Publication: B. Keszei, M. Pardavi-Horváth; Coercivity of /YSmCa/ ₃ /FeGe/ ₅ O ₁₂ LPE Films, INTERMAG, Florence 1978.	+

308				
Name(s): K. Veerabhadra Rao,				
Inst. & Address: IIT, Kanpur-208 016, India		T.S. Narasimhamurthy, Department of Physics, Osmania University, Hyderabad-500 007, India. Telephone: 40066 ext. 296 (KVR) 220338 (TSN)		
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NH ₄ H ₂ PO ₄ Ammonium dihydrogen phosphate	S	90 x 70 x45	Purity of chemicals for all these crystals has been 99.9%. Properties studied include birefringence, linear electrooptic and photoelastic behavior. Determined all the stress optical (q _{ij}),	+
KH ₂ PO ₄ Potassium dihydrogen phosphate	S	55 x 25 x15	strain optical (p _{ij}), linear electro- optic (r _{ij}) constants (clamped and unclamped) and their dispersion by studying different orientations.	+
NaKC ₄ H ₄ O ₆ ·4H ₂ O Rochelle salt	S	80 x 50 x25	Applications are light modulators, deflectors etc., by electrooptic and acoustooptic techniques.	+
NaNO ₃ Sodium nitrate (p _{ij} /p _{kl} only)	Br ⁺ and S	70 (len) 12 (dia.) and 18 x 15 x8	J. Mater. Sci. (UK), 8, 577 (1973)* J. Mater. Sci. (UK), 10, 1019 (1975) J. Opt. & Quantum Electron. (UK), 8, 255 (1976) J. Phys. C: Solid State (UK), 11, 2343 (1978) Ph.D. Thesis (KVR), Osmania University, Hyderabad, India (1972).	-
+In collaboration with Dr. Kolluri V. Ramanaiah, PSG College of Technology, Coimbatore-641 004, India.			*In collaboration with Dr. Howard L. Petterson, Albion College, Albion, Michigan, 49224, USA.	

309 R.Y. Deshpande, G.L. Bhide, Malathi Ghosh, S.C. Sabharwal, G. Alexander, U.R. Marwah, S.M.D. Rao, Gurnam Singh, Name(s): R.V. Srikantiah, T. Mirza, S.C. Karandikar, S.S. Unki, K. Ramachandran, K.S. Keswani, M.N. Ahuja, M.K. Rajpurkar, G.P. Kothiyal and B. Ghosh.
Inst. & Address: Technical Physics Division, Bhabha Atomic Research Centre, Bombay-400085, Telephone: 523321/Ex. 309 INDIA.

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KCl, KBr, NaCl	M, Br	150 di x 150	Optran grade, random orientation, used as optical windows.	+
NaI	M, Br	80 di x 80	Scintillator quality, random orientation, Tl dopant, used as nuclear radiation detectors.	+
CsI	M, Br	50 di x 50	Scintillator quality, random orientation, used as nuclear radiation detectors.	+
Anthracene Stilbene	M, Br	35 di x 35	Optical quality, random orientation, used as nuclear radiation detectors.	+
Cu	M, Br	20 di x 100	OFHC quality, random orientation, dislocation studies.	+
Cu	Cz	30 di x 150	OFHC quality, <110> orientation, used for neutron diffraction.	+
Ge	Cz	35 di x 150	Optical filter and driftable quality <111> and <100> orientations, used for IR filters and Ge (Li) detectors.	+
LiF, CaF ₂	M, Br	30 di x 30	Optran grade, random orientation, Fu dopant, TL studies made, optical windows and membrane electrodes, Paper No. A 7/141 to be published in the Analyst.	+

310 R. Y. Deshpande, G. L. Bhide, Malathi Ghosh, S. C. Sabharwal, G. Alexander, U. R. Marwah, S.M.D. Rao, Name(s): Gurnam Singh, R. V. Srikantiah, T. Mirza, S. C. Karandikar, S. S. Unki, K. Ramachandran, K. S. Keswani, M. N. Ahuja, M. K. Rajpurkar, G. P. Kothiyal, and B. Ghosh
Inst. & Address: Technical Physics Division, Bhabha Atomic Research Centre, Bombay-400085, Telephone: 523321/Ex. 309 INDIA

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ADP, KDP	Soi	150x30x30	Optical grade, C axis along the longest dimension, radiation damage studies, Laser beam modulator and SHG, J. Rad. Effects, 30, 123 (1976).	+
As ₂ S _{2.15}	VP	3x1x0.2	Five N purity, needle axis is coincident with <110>, optical and electrical properties studied, J. Cryst Growth, 32, 29 (1976).	+
As ₂ S ₃	VP	2x1x0.2	Five N purity, needle axis coincident with <110>, optical and electrical properties.	-
As ₂ Se ₃	VP	4x2x0.1	Five N purity, optical, electrical and photoconducting properties.	+
As ₂ Te ₃	VP	2x0.5x0.5	Five N purity, optical, electrical and photoconducting properties.	+
666-701-418				

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Name(s): M. R. L. N. Murthy and Dr. N. S. Satya Murthy (Head, Solid State Physics Section)

Inst. & Address: Material Development Laboratory, Solid State Physics Section, Bhabha Atomic Research Center, Bombay 400 085, India Telephone: 523321 Ext. 354

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
<u>Single Crystals:</u>				
Cu_2MnAl	Br	6 mm dia, 40 mm long	Starting materials purity 99.99% Application - neutron polariser	-
YIG	Flu	10 mm cube	Starting materials purity 99.99% Application - neutron diffraction studies	+
Aluminum	Br	80 mm dia, 150 mm long	Starting material purity 99.99% Application - neutron monochromators	+
Copper	Cz	30 mm dia, 100 mm long	Starting material purity 99.99% Application - Fabrication of vertical focussing monochromators	+
Bismuth	Br	40 mm dia, 150 mm long	Starting material purity - zone refined Application - γ -ray filtering and increased transmission of low energy neutrons	+

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Name(s): M.R.L.N. Murthy and Dr. N.S. Satya Murthy (Head, Solid State Physics Section)

Inst. & Address: Material Development Laboratory, Solid State Physics Section, Bhabha Atomic Research Centre, Bombay 400 085, India Telephone: 52331, Ext. 354

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
<u>Polycrystalline Materials:</u>				
Alloys: Heusler alloys such as Pr_2MnGe , Pd_2MnSn , etc.; alloy of Fe-Ge, Co-Ga, Mn-Sn systems, etc.; ternary alloys FeMnGe , MnAlGe , GaCoCuFe , etc.; and carbides of Mn-Co, MnNi, etc.	Arc and induction melting, annealing	20-30 g of the alloy	Starting materials purity 99.99% to 99.999% Application: neutron scattering and Mossbauer effect	+
CoAl , FeAl , NiAl	Arc melting, annealing	10-15 g	Starting material purity 99.99% Application: Compton scattering studies	+
$\text{Ti}_{0.62}\text{Zr}_{0.38}$	Arc melting, extrusion	1 Kg charge	Starting material purity 99.9% Application: Cassettes for neutron scattering work at high pressures	+
MFe_2O_4 for M = Mn, Cu, Zn, etc.; $(\text{MN})\text{Fe}_2\text{O}_4$ for MN = (NiZn), (MgMn), (NiV), etc.	Sintering and wet methods		Starting materials are Analar grade chemicals Application: neutron scattering and Mossbauer effect	+
Chalcogenides $\text{CoS}_{1.4}\text{Se}_{0.6}$ and FeS	Annealing		Starting materials are Analar grade chemicals and metals of 99.99% purity Application: neutron scattering and Mossbauer effect	+

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Name(s): K.S.V.NAMBI

Inst. & Address: Health Physics Division, Bhabha Atomic Research Centre, Bombay 400085, INDIA Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. Calcium Sulphate - CaSO_4	Sol.	polycrystals	Doped with lanthanide rare earth elements at various concentrations ranging between 0.01 and 10% by wt.; studies on Thermoluminescence and Electron Spin Resonance characteristics; applications in Radiation Dosimetry - J.Phys.C Solid St. Phys.7(1974)4403-15.	+
2. Calcium Fluoride - CaF_2	Br.	9.55 x 50(L)	Doped with lanthanide rare earth elements at various concentrations- typically at 0.05% by wt. - either singly or in pairs of the dopant; studies on Thermo-luminescence characteristics - reference in Proc. of A Specialists Meeting on TL Dating, Oxford, 1978 (to be published in RADIATION Journal, Strasbourg).	+
3. Lithium Fluoride - LiF	Br.	9.55 x 50(L)	Doped with Mg, Ti, Mg/Ti and rare earth elements at a typical concentration of 0.1% by wt.; studies on Thermoluminescence emission characteristics - Proc. of National Symposium on Thermoluminescence, 1975, pp 365-72, Pub. by Bhabha Atomic Research Centre, Bombay.	+
4. Lithium Tetraborate - $\text{Li}_2\text{B}_4\text{O}_7$	M.	glassy butt.	Doped with rare earth elements at 1.6% by wt. conc.; studies on TL Characters - Proc. Fifth Int. Conf. on Lumines. Dosim. 1977, pp 197-206, Physik. Inst. Giessen.	-

314

Name(s): A. J. Singh, P. Suryanarayana, T. K. Das, G. S. Rao, M. D. Karkhanavala

Inst. & Address: Chemistry Division, Bhabha Atomic Research Centre, Bombay 400085, India Telephone: 523321/327

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ag, Silver	Br, Cz	7 x 7 x 80	Undoped, Basic research (diffusion studies)	+
Bi, Bismuth	Br	40 x 40 x 150	Undoped, Neutron attenuation	+
	Cz	20 x 20 x 65	Undoped, Research and development	+
Sb, Antimony	Z	25 x 25 x 10	Undoped, Basic research, Purity 99.999+	+
Te, Tellurium	Br	10 x 10 x 20	Undoped, Electro-optics (nonlinear), Purity 99.999+	+
PbI ₂ , Lead Iodide	Br	10 x 10 x 10	Basic research	-
Bi ₂ Te ₃ , Bismuth telluride	Br	15 x 15 x 25	Undoped, Basic research	-
PbSe, Lead selenide	Br	20 x 10 x 10	Undoped, Basic research	-
Pb _{0.8} Sn _{0.2} Se, Lead tin selenide	Br	20 x 20 x 50	Undoped, Basic research	-
InSb, Indium antimonide	Cz	6 x 6 x 80	Undoped, Research & development	-
GaAs, Gallium arsenide	Br	5 x 5 x 20	Undoped, Research & development	-
GaP, Gallium phosphide	Sol	1 x 5 x 5	Undoped, Research & development	-
CdTe, Cadmium telluride	Sol (THM)	10 x 10 x 10	Undoped, Basic research	-
PbTe, Lead telluride	Br	10 x 15 x 30	Undoped, Basic research	-

315 Name(s): G. Alexander Inst. & Address: Ore Extraction Section, Chemical Eng. Group, Technical Physics Division, Bhabha Atomic Research Center, Bombay, India Telephone: 523321 Ext. 602				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Germanium Ge	Cz	30 mm dia, 100 mm long	Purity - $\approx 10^{14}$ impurity atoms/cc Orientation - $\langle 111 \rangle$ and $\langle 100 \rangle$ Dopants - gallium Properties studied - Resistivity, etch and pit density, charge carrier lifetime, lithium precipitation rate, and lithium drift rate Application - Lithium-drifted γ -ray detector, infrared filters and detectors	+

316 Name(s): M. D. SASTRY Inst. & Address: Radiochemistry Division, Bhabha Atomic Research Centre, Trombay, Bombay 400 085, India Telephone: 523321 (456)				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ammonium Chloride NH_4Cl	Sol	3 x 3 x 3	Well developed {100} faces. Pd dopant. Structure of Pd complexes by electron paramagnetic resonance (EPR) J. Chem. Phys. <u>64</u> , 3957 (1976)	+
Potassium ferrocyanide trihydrate (KFCT) $\text{K}_4\text{Fe}(\text{CN})_6 \cdot 3\text{H}_2\text{O}$	Sol	2 x 2 x 5	Co dopant. Ferroelectric transition by EPR Phys. Rev. B <u>13</u> , 1156 (1976)	-

317 Name(s): DRS. M. D. Karkhanavala, V. Amrithalingam, and U. R. K. Rao Inst. & Address: Chemistry Division, Bhabha Atomic Research Centre, Bombay 400 085, India Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Na_2SO_4 (III) Orthorhombic (Sodium Sulphate anhydrous, orthorhombic) (metastable at RT)	Sol	Needles	99.99%, needle axis is C axis of the orthorhombic unit cell. Studied: (a) The topotactic phase transition of III to hexagonal (I), using single crystal x-ray technique; V. Amrithalingam, M. D. Karkhanavala, and U. R. K. Rao, Acta Cryst. A33, 522 (1977). (b) Optic orientation in Na_2SO_4 (III) determined and its relation to the topotactic phase transition to the uniaxial (I) determined; V. Amrithalingam, U. R. K. Rao, and S. F. Sethna, Symposium on Crystallography and Crystal Physics, Osmania University, Hyderabad, India, Dec. 1977.	-

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Name(s): Dr. A. R. Cetal and P. R. Sarode

Inst. & Address: Department of Physics, Indian School of Mines, Dhaubad 826004, India

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Metastable solid solutions in Lead-Tin Alloys	M	Ingots	A. R. grade Amount of Solute (Pb) Phase Lattice Parameters c/a At.% a(Å) c(Å) 15 Tetrag- 5.9503 3.3551, 0.5638 10 onal 5.9950 3.3857, 0.5648 20 " 6.0496 3.4511, 0.5706 25 " 6.1004 3.4849, 0.5712 30 " 6.1325 3.5204, 0.5760 Lattice Parameters as given above were determined. Publication: Curr. Sci. <u>43</u> (11), 339 (1974)	Not availabl

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Name(s): S. Dhar.

Inst. & Address: Institute of Radio Physics and Electronics
92, Acharya Prafulla Chandra Road, Calcutta-700009, India.

Telephone: 35-9115

35-9116

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. ZnGeP_2 , Zinc Germanium Phosphide	Br	$5 \times 5 \times 3 \text{ mm}^3$	Etching solutions and Crystal orientation. Publication :- "Etching Solutions for the determination of the Crystallographic axes of ZnGeP_2 ". Journal of Crystal Growth <u>42</u> (1978), 120.	+
2. Si, Silicon	Br	$3 \times 3 \times 0.5 \text{ mm}^3$ in general	Contact properties. Publication :- "A Pulse Method for the Measurement of contact-Resistance and Bulk Resistance of Semiconductor Samples" 'The Journal of the Electrochemical Society' <u>125</u> (1978), 508.	+
3. (A) Si, Silicon (b) Ge, Germanium (c) GaAs, Gallium Arsenide (d) InAs, Indium Arsenide (e) InSb, Indium Antimonide (f) ZnGeP_2 , Zinc Germanium Phosphide.	Br	$10 \times 5 \times 1 \text{ mm}^3$ in general	Conductivity type testing. Publication :- "Cold Probe Semiconductor Type Tester." 'Review of Scientific Instruments', <u>48</u> (1977), 1720.	+

320

Name(s): DR. DIPAK CHAKRABORTY

Inst. & Address: Central Glass & Ceramic Research Institute,
P.O. Jadavpur University, Calcutta - 700 032, India

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Prepared : Synthetic Quartz Single Crystals (Colourless) - SiO_2 Studied : Natural Quartz Single Crystals (Colourless), Natural Amethyst, Natural Citrine, Smoky Synthetic Quartz Single Crystals - SiO_2	Hyd	150 x 25 x 15 mm ³	Total Impurities- 135 ppm; Y-bar second Li salts; Mechanical Q; IR-spectra; Hydro- gen concentration, distribution and orien- tation; Metallic impurities; X-ray irradi- ation and Heat-treatment. 1. D. Chakraborty and P. Saha, Indian J. Tech- nol. 11(1973) 127 2. D. Chakraborty and P. Saha, Indian J. Tech- nol. 12(1974) 189 3. D. Chakraborty and P. Saha, Indian J. Phys. 48 (1974) 432 4. D. Chakraborty and G. Lehmann, J. Sol. State Chem. 17 (1976) 305 5. D. Chakraborty and G. Lehmann, Physica Status Solidi (a) (1976) 487 6. D. Chakraborty, J. Crystal Growth 36/1 (1976) 188. 7. D. Chakraborty & G. Lehmann, Neues Jahr. Miner. Monats. 7 (1977) 289 8. D. Chakraborty, J. Crystal Growth 41/1 (1977) 177 9. D. Chakraborty & G. Lehmann, Z. Naturforsch- ung 33(a) (1978) 290 10. D. Chakraborty, Indian J. Phys. (In print) 11. D. Chakraborty, J. Crystal Growth (In print) 12. D. Chakraborty, J. Materials Science (=do=) 13. D. Chakraborty, (Under publication) 14. D. Chakraborty, (Under publication)	Not for sale

321

Name(s): Dr. P. N. Kotru

Inst. & Address: Department of Physics, University of Jammu, Jammu 180001, India

Telephone: 6340

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Quartz (SiO_2) (studied)	Hyd	150x50x40	Surface characterization, growth mechanism, etching and imperfection studies (contact for publications)	Inquire
Lanthanum Borate (LaBO_3) (prepared and studied)	Flu	10x3x1	Orientation, characterization (contact for publication)	-
Dysprosium Vanadates (DyVO_4) (prepared)	Flu	20x2x2	Study to be taken up	-
Rare Earth compounds particularly RFeO_3 , RCrO_3 , RAIO_3 , RMnO_3 , RNbO_4 (R = Rare earth) (study)	Flu	Varying sizes	Surface characterization, growth mechanism, etching and imperfection studies (taken up in collaboration with Crystal Growing Group of Clarendon Laboratory, University of Oxford)	-
KMgF_3 (Pure & Doped with Ni) (study)	Cz	Boules 15x12x12	Ditto	-
Glycine, Potassium Nitrate, Potassium Nitrite, Potassium Phthalate (prepared)	Sol	5x4x4; 25x4x3; 15x1x1, 5x3x3	Growth kinetics and characterization studies	-
Zinc Sulfate (prepared)	Gel	6x5x3	Ditto	-
Amethyst (SiO_2), Natro-lite (study) & Apophyllite (study)	Nat*	15x10x6 50x2x1 20x15x15	Surface characterization growth mechanism, etching & imperfection studies (Contact for publications)	- - -
*Natural variety				

322 Name(s): Dr. Ramji Srivastava Inst. & Address: School of Studies in Physics, Jiwaji University, Gwalior, India 474002 Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Thin films of Tungsten Oxide (WO_3)	VP	5 x 5 mm x 5000 Angstroms	Electrical conduction, breakdown and dielectric properties, films characterized for WO_3 by electron diffraction	
Thin films of Molybdenum Oxide (MoO_3)	VP	5 x 5 mm x 5000 Angstroms	Electrical conduction, breakdown and dielectric properties, films characterized for MoO_3 by electron diffraction Publications: J. Phys. Soc. Japan 39, 1316 (1975) Thin Solid Films 28, L-13 (1975) Thin Solid Films 30, 319 (1975)	

323 Name(s): Dr. M.A. ITTYACHEN Inst. & Address: Department of Physics, University of Kerala, Kariavattom, Trivandrum, Kerala, INDIA. Telephone: 8420				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$PbMoO_4$ (Lead molybdate)	Gel	2 x 2		+
$BaMoO_4$ (Barium molybdate)	Gel	1 x 2		+
$CaCO_3$ (Calcium carbonate)	Gel	1 x 2		+
$PbCO_3$ (Lead carbonate)	Gel	2 x 2	Dislocation studies, X-ray & I.R. studies doped with different materials.	+
$MnCO_3$ (Manganese carbonate)	Gel	2 x 2		+

324 Name(s): Dr. Ingle Sharadchandra Govind Inst. & Address: Laxminarayan Institute of Technology, Nagpur 440010, India Telephone: 31659				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$KNbO_3$ (Potassium Niobate)	Flu	1cm X 1cm X 1cm	Suitable for domains study. Domain structure highly susceptible to defects like impurity ions and dislocations. Properties studied: domain structure distinction between thin and thick crystals, influence of defects. Publications - J. Phys. D, Appl. Phys. (U.K.) 4, 124 (1971); 4, 1633 (1971); 5, 1474 (1972); 6, L103 (1973); 6, 1816 (1973); 10, L149 (1977). J. Appl. Phys. (USA) 45, 5152 (1974). Ind. J. Pure Appl. Phys. 10, 29 (1972); 10, 353 (1972); 10, 839 (1972); 10, 881 (1972); 15, 323 (1977). Curr. Sci. 39, 368 (1970).	-
$PbNb_2O_6$ (Lead Metaniobate)	M	1mmX1mmX0.4 mm	Properties studied: As above Publications: Pramana 10, 163 (1978); 10, (1978) in press. Ind. J. Pure Appl. Phys. 16, (1978) in press.	-
(NaCl + KCl) (Two-phase mixed crystals of NaCl + KCl)	Sol	0.5cm X 0.5 cm X 0.3cm	These two-phase mixed crystals are different from the melt grown mixed crystals that are crystallographically the solid solution of NaCl and KCl and hence single phase. Properties studied: X-ray studies of structure; etching studies of dislocation substructure, growth mechanism; impurity ion diffusion across the phase boundaries; phase changes at higher temperatures, etc. Publications: J. Phys. D, Appl. Phys. (1978) in press.	+

Telephone: 414323, Ext. 11

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KCl, NaCl single crystals	Melt	50 x50 x50	BDH quality Dopants: Ca^{2+} , Sr^{2+} , Ag^+ , etc. Electronic properties are being studied by positron annihilation techniques	+
Iron-Nickel alloys	Induct- tion furnace melting	10x10x5	Magnetic properties on order-disorder transformation are being studied by Mossbauer effect and positron annihilation techniques	+
ZrMoFe ₂ intermetallic compounds	Arc melting		Mossbauer studies	+
Polycrystalline spinel ferrites				
$\text{Ni}_x\text{Cu}_{1-x}\text{Fe}_2\text{O}_4$ $\text{Mn}_x\text{Cu}_{1-x}\text{Fe}_2\text{O}_4$ $\text{Ni}_x\text{Cd}_{1-x}\text{Fe}_2\text{O}_4$	$\left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} x = 0, 0.2, 0.4, \\ 0.6, 0.8, \text{ and} \\ 1.0 \end{array}$ Ceramic sinter- ing	Polycrystal- line powder	Magnetic and electrical properties are studied by Mossbauer, dielectric, and conductivity measure- ments	+

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$(As_2S_3)_{1-x}(PbS)_x$ Arsenic-Lead Sulfosalts	M Sol	3-8 mm dia. Disc/chips	Prepared from high purity semiconducting grade As_2S_3 and PbS or constituent elements. Glassy semiconductors. Properties studied: Thermal analysis (DSC, TGA), microstructure studies by electron-microscopy. Far-infrared transmission, electrical conductivity and photoconductivity, elastic constants by ultrasonic propagation Publications: J. Non-Crystalline Solids 27, 399 (1978) Solid State Comm. 22, 789 (1977) J. Phys. C (Solid State) 11, L479 (1978) Phys. Chem. Glass. (August 1978) (in press)	

327 Name(s): P. R. SARODE Inst. & Address: Physics Department, Nagpur University, Nagpur, India. Present Address: Telephone: 31946 Solid State & Strl. Chem. Unit, Indian Institute of Science, Bangalore 12, India				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
RCO_2 , RCO_3 , RCO_5 (where R stands for a rare earth or yttrium), Metastable lead-tin alloys	M	--	X-ray diffraction, X-ray emission, fluorescence and absorption spectra, electronic structure, and nature of chemical bonding Publications: P. R. Sarode and A. R. Chetal, J. Phys. C: Solid State Phys. <u>9</u> , 153 (1977) P. R. Sarode and A. R. Chetal, J. Phys. Soc. Japan <u>40</u> , 1637 (1976) P. R. Sarode and A. R. Chetal, Curr. Sci. (India) <u>43</u> , 339 (1974) P. R. Sarode and A. R. Chetal, J. Phys. F: Metal Phys. <u>7</u> , 1103 (1977)	-

328 Name(s): Dr. L.C.GUPTA Inst. & Address: Tata Institute of Fundamental Research Telephone: 219111 Ext.356 Homi Bhabha Road, Colaba, Bombay 400 005				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. REX_2Si_2 (RE = La, Ce, Pr; X = Cu, Ag, Au)	Arc melt- ing	Powder	99 % ; powder property studied Knight shift of Si; valence fluctuation of Cerium characterization X-ray Publications under preparation	+
2. $\text{Bi}_4(\text{SiO}_4)_3$; $\text{Bi}_4(\text{GeO}_4)_3$	CZ	irregular, volume more than 2cc.	Purity - Not known Properties studied NQR of Bi Publications: K.V.Gopalakrishnan, L.C.Gupta and R. Vijayaraghavan Pramana, <u>6</u> , 343 (1976)	-

329

Name(s): HERZL AHARONI

Inst. & Address: DEPT. OF ELECTRICAL ENGINEERING, BEN-GURION UNIVERSITY OF THE NEGEV
P.O. BOX 653 BEER-SHEVA - ISRAEL

Telephone:

TEL: 64111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. Heteroepitaxial growth of single crystal solid solutions of Si-Ge grown from the vapour on Si substrates.	GE	Diameter - - 25 mm Si substrate thickness - - 0.25 mm Ge-Si layer thickness - - $12 \cdot 10^{-3}$ mm.	Basic materials: SiCl ₄ , GeCl ₄ (liquid) 99.9995% Dopants: PH ₃ , B ₂ H ₆ diluted in H ₂ (5 p.p.m) high purity electronic grade. Growth orientations: (111); (110); (100). Properties studied: study of growth processes, calibration and Transfer Curves, Changes in lattice constants as function of the layer composition, surface oxidation, characterization of various defects. Interface problems between the hetero-epitaxial layer and the substrate. The Ge content in the (111) orientation reached up to 30%. Publications: 1) Thin Solid Films Vol. 12, No. 1, Sept. 1972. 2) J. Cryst. Gr. 17, 1972 (2 papers). 3) Israel J. Tech. Vol. 14, No. 4/5, 1977. 4) Proc. Symposium of Joint Chapters S/AP/MTT and G-ED/PHP - IEEE in Israel, April 1976, pp. 18-31.	
2. Epitaxial growth of single crystal Si layers grown on Si substrates.	GE	Diameter - - 25 mm Si substrate thickness - - 0.25 mm Si layer thickness - - $20 \cdot 10^{-3}$ mm.	Basic material - SiCl ₄ , dopants, orientations as above. Properties studied: Influence of growth parameters on layer quality and their defects. Publications: 1) Vacuum, Vol. 24, No. 2, 1974. 2) Vacuum, Vol. 26, No. 4/5, 1976 (3 papers). 3) Proc. 7th Natl. Conv. IEEE, Israel, pp. 99-109, April 1971.	

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Name(s): SIDNEY B. LANG

Inst. & Address: DEPARTMENT OF CHEMICAL ENGINEERING
BEN GURION UNIVERSITY OF THE NEGEV, BEER SHEVA, ISRAEL

Telephone: (057) 61330

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
PYROELECTRIC COEFFICIENT MEASUREMENT			STATIC AND DYNAMIC COEFFICIENTS OVER 90 -380K TEMPERATURE RANGE.	

331 Name(s): Zelingher, Flicstein, Zangvil Inst. & Address: Ben Gurion University of the Negev, Faculty of Technology, Beersheva. Telephone: 057-				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NaCl, NaBr, NaNO_3 , $\text{NaC}_2\text{H}_3\text{O}_2$, KCl, KBr, KNO_3	Ze	150 x 0.001 x 0.001 mm.	Purity: 99% Orientation: [100] (KCl, NaCl) Characterization: Laue X-Ray Technique, Electron Microscope Diffraction, Electron Microprobe Microanalyzer. Applications: In Study Publications: 1) N. Zelingher, J. Flicstein, A. Zangvil, "Sodium Acetate Tri-Hydrate Whisker", Journal of Crystal Growth 42(1977) 253-258. 2) A. Zangvil, N. Zelingher, J. Flicstein, "Growth Mechanism of NaCl and KCl Whiskers Grown by Zelingher's Method", Ben Gurion University of the Negev, Department of Materials Engineering, Internal Report 12/77. 3) N. Zelingher, "Mass Transport Trough Cellulose Acetate Membrane - A New Concept". Proceedings of the Sixth International Symposium on Fresh Water from the Sea, Las Palmas, Gran Canaria, Spain, Sept. 1978.	+
Ze = Zelingher's Membrane Method.				

332 Name(s): J. Shamir Inst. & Address: The Hebrew University, Dept. of Inorganic & Anal. Chemistry, Jerusalem, Israel Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Pb}_{1-x}\text{Sn}_x\text{Te}$ (LTT)	PJux	20-100 um	Polycrystalline material to be used as raw material for single crystal growth or epitaxial film preparation. The materials were characterized by powder X-rays, the cell constants following Vegard law. Particle size determined by electron microscope scanning. Preparation by reacting together all the elements, lead, tin, tellurium and iodine, forming the lead iodide flux in situ. The purity of the products similar to those of the original materials used for the reaction. (submitted for publication).	

333 Name(s): M. Schieber, I. Beinglass, G. Dishon, I. Mcloslavski, A. Holzer Inst. & Address: School of Applied Science, Hebrew University of Jerusalem, Israel Telephone: (02) 30211 x 693				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
H I_2 S^{+2}	Sub.	20 X 10 X 10	can be cleaved into (001) plates for nuclear detectors. See e.g., Schieber et al., Nucl. Inst. and Methods <u>150</u> , (1978), 71-77.	+

334

Name(s): Prof. Y. Yacoby

Inst. & Address: The Hebrew University of Jerusalem, The Racah Institute of Physics, Jerusalem, Telephone: 972-2-584622
ISRAEL.

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KTaO ₃ (POTASSIUM TANTALATE)	Flu	2x2x5	dopants: Li, Na, Nb properties studied: dynamical properties and phase transitions using Raman spectroscopy publications: Y. Yacoby and A. Linz Phys. Rev. <u>9</u> 2723 (1974) Y. Yacoby and S. Just Sol. Stat. Comm. <u>15</u> 715 (1974) Y. Yacoby, Z. Physik B24 397 (1976) Y. Yacoby, Z. Physik, to be published	+
SrTiO ₃ (STRONTIUM TITANATE)	Ver	2x2x5	properties studied: critical behavior of the soft mode. publication: Y. Yacoby, W. W. Kruhler and S. Just Phys. Rev. B13 4132 (1976).	+

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Name(s): Y. Kumashiro and E. Sakuma

Inst. & Address: Electrotechnical Laboratory. 5-4-1 Mukodai, Tanashi, Tokyo, Japan. Telephone: 0424-61-2141

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
TiC	Z	10x10x90	.Purity > 99.9% .Elastic constants and Ultrasound absorp- tion experiment.	+
VC	Z	10x10x90	.High temperature properties (Electric resistivity, Thermoelectric power, Work function and Hardness)	+
NbC	Z	7x7x50	.Chemical analysis, X-ray fluorescence analysis, EMX, ESCA, Lattice parameter, Optical microscopy and X-ray topography.	-
ZrC	Z	5x5x5	.Point emitter .J. Less-Common Metals. <u>32</u> (1973) 21. .Japan. J. Appl. Phys. <u>15</u> (1976) 921: <u>12</u> (1973) 1462: Suppl. 2, Part 1 (1974) 289. .Ceramics (in Japanese) <u>11</u> (1976) 1073. .Phys. Rev. B <u>12</u> (1975) 5465. .Internal Friction and Ultrasonic Attenuation in Solids. (University of Tokyo Press, Tokyo, 1977) p. 395. .Oyo Buturi (in Japanese) <u>45</u> (1976) 607. .Surface Science. <u>62</u> (1977) 313. .J. Mater. Sci. <u>12</u> (1977) 595.	-
NbN	Z	10x10x90	.Superconducting device	+

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Name(s): Michio Tajima

Inst. & Address: Electrotechnical Laboratory, Tanashi, Tokyo 188, Japan

Telephone: 0424-61-2141

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Si, Silicon (single crystal)	Z, Cz		Characterization of impurities by photo-luminescence (PL) method. ► Nondestructive detection of extremely small amount of impurities. (detection limit; $1 \times 10^{11} \text{ cm}^{-3}$ for B, $5 \times 10^{11} \text{ cm}^{-3}$ for P) ► Analysis of compensating impurities. ► Determination of impurity concentration from PL spectra. (calibration curve for B and P; $10^{11} - 10^{15} \text{ cm}^{-3}$) ► Ref.: Appl. Phys. Letters, <u>32</u> , 719.	

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Name(s): Kohzo SUGIYAMA, Yasutaka TAKAHASHI, and Seiji MOTOJIMA

Inst. & Address: Department of Synthetic Chemistry, Faculty of Engineering,
Gifu University, Kakamigahara, Gifu 504, Japan

Telephone: 0583-82-1201

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
TiB ₂ whiskers bulk single crystal	VP "	~250 μm 0.1x5x5mm max.	Au doped, <100>, Bull. Chem. Soc. Japan, 48(1975)3156. By modified hot-filament method. J. Cryst. Growth, 43(1978)533.	- -
ZrB ₂ whiskers	"	~500 μm	Au doped, <100>, Denki Kagaku, 43(1975)323.	-
Nb ₂ coatings	"		J. Cryst. Growth, 30(1975)233.	-
LaB ₆ whiskers and bulk single crystal	"	~5 mm long	J. Cryst. Growth, 44(1978)106.	-
TiP whiskers	"	~10 mm long	Au, Pt doped, <001>, J. Electrochem. Soc., 123(1976)290.	-
ZrP whiskers	"	~5 mm long	Au, Pt doped, <100>, J. Cryst. Growth, 30(1975)1.	-
NbP whiskers	"	~5 mm long	Au, Pt doped, <001>, Bull. Chem. Soc. Japan, 49(1976)2122.	-
BP whiskers	"	~4 mm long	Ni, Pt doped, <111>, Bull. Chem. Soc. Japan, 48(1975)3161.	-
TiN bulk single crystal	"	2x2x2mm max.	By modified hot-filament method. J. Cryst. Growth, 32(1976)141.	-
ZrN bulk single crystal	"	3x3x3mm max.	Ditto	-
TiC bulk single crystal	"	6x6x6mm max.	Ditto	-
ZrC bulk single crystal	"	2x2x2mm max.	Ditto	-
ZrS ₂ pillar-like crystals	"	~2 mm long	Hollow crystals. J. Cryst. Growth, 33(1976)116.	-
TiS ₂	"		Bull. Chem. Soc. Japan, 51(1978) to be published.	-
NbO whiskers	"	~1 mm long	Pd doped, <100>, Yogyo Kyokai-Shi, 83(1975)510.	-
B ₄ Si whiskers	"	~2 mm long	Au doped, Bull. Chem. Soc. Japan, 48(1975)1463.	-

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Name(s): T.Sekiya, G.Nishiyama, H.Hayashi and H.Okuda

Inst. & Address: Government Industrial Research Institute, Nagoya
Hirate-cho, Kita-ku, Nagoya 462, Japan

Telephone: 052-911-2111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cr_2O_3 Chromic oxide	VP	10x10x.02	This material is very thin(transparent), uniformly thick and highly perfect plate, therefore, this can be used as Faraday rotator. Published in J. Crystal Growth, 43(1978)645	+

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Name(s): Koichi WATANABE

Inst. & Address: Department of Chemistry, Faculty of Technology, Gunma University
Kiryu, Gunma 376 JAPAN

Telephone: 0277(22)3181

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Al_2O_3 , Alumina, Corundam, Aluminum Oxide	Flu	10x50x50	about 98% purity, $[10\bar{1}2]$ plate-like crystal Fe_2O_3 , Cr_2O_3 , La_2O_3 , Nd_2O_3 , Yb_2O_3 , Dy_2O_3 , Y_2O_3 , TiO_2 , Ni_2O_3 . etch pits, surface topography, inclusion twin, habit change. J. Crystal Growth 24/25 666(1974), 32 316 (1976), 41 1(1977), 42 293(1977).	-
BaFe_2O_4 , Barium Ferrite	M	0.1x1x2	99% up, plate-like, surface micro- topography	-

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Name(s): Hideo Izumi

Inst. & Address: Happy Science LTD, 5-28-10 Inogashira, Mitaka, Tokyo, 181, Japan

Telephone: 0422-48-3030

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
BP (Boron Phosphide)	GE*	10x10x0.05	Orientation = (100) Si Doped n-type $\rho \approx 0.5 \Omega \text{ cm}$ electron density $\approx 10^{17} \text{ cm}^{-3}$ (non doped) 10^{19} cm^{-3} (doped) * Grown on Si and Si-removed. ** if neccessary, it is available, but expensive.	(-)**

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Name(s): Mitsuru Ishii and Takao Nakada

Inst. & Address: Ibaraki Res. Lab., Hitachi Chemical Co. Ltd.

Telephone: (0292) 85-1101

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Bi}_4\text{Ge}_3\text{O}_{12}$ (BGO)	Cz	$38\phi \times 250\text{L}$	None dope, Extra High purity for X-ray and Gamma-ray SCINTILLATORS.	+
LiTaO_3	Cz	$38\phi \times 250\text{L}$	High purity for Surface Acoustic Wave Devices.	+

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Name(s): Sakichi Ashida, Yoshio Furuhashi.

Inst. & Address: Central Research Laboratory, Hitachi, Ltd. Kokubunji Tokyo Japan Telephone: 0423-23-1111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiNbO_3	Cz	$75\phi \times 200$	Nondoped [Z], [X], [Y] direction.	-
	Cz	$10\phi \times 150$	Rh, Fe, Rh+Cu doped. [Z] direction.	-
$\text{Bi}_{12}\text{SiO}_{20}$	Cz	$10\phi \times 200$	[001] direction.	-
$(\text{Sr}_{0.75}\text{Ba}_{0.25})\text{Nb}_2\text{O}_6$	Cz	$30\phi \times 200$	[001] direction. Ce doped. Congruent melt.	-
TeO_2	Cz	$10\phi \times 100$	[001] direction. Nondoped.	-
$\text{Gd}_2(\text{MoO}_4)_3$	Cz	$25\phi \times 200$	[001], [100], [110] direction. Nondoped, Er, Cr, Dy doped.	-
PbMoO_4	Cz	$15\phi \times 200$	[001] direction.	-
CaF_2	Cz	$10\phi \times 100$	[001] direction. Nondoped, Sm doped.	-
$\text{Li}_2\text{Nd}_4(\text{MoO}_4)_3$	Cz	$15\phi \times 70$	[001] direction. Nondoped.	-
NaNO_3	Br	$200\phi \times 100$		-
ZnS	Br	$15\phi \times 70$		-
CdS	Br	$15\phi \times 70$		-
BaTiO_3 , $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}:\text{Cr}$, $\text{Y}_3\text{Al}_5\text{O}_{12}:\text{Nd}$, VO_2 , $\text{PbFe}_{12}\text{O}_{19}$, NiCr_2O_4 , Al_2O_3 , $\text{Y}_3\text{Fe}_5\text{O}_{12}$, $\text{NdP}_5\text{O}_{14}$, YP_5O_{14}	Flu	$\sim 10 \times \sim 10 \times$ ~ 10		-

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Name(s): Sakichi Ashida, Yoshio Furuhashi				
Inst. & Address: Central Research Laboratory, Hitachi, Ltd. Kokubunji Tokyo Japan. Telephone: 0423-23-1111				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SbSI	Vp			-
CdS	Vp			-
$\text{Li}(\text{N}_2\text{H}_5)\text{SO}_4$	Sol	30x20x10		-
$\text{Ca}_2\text{Sr}(\text{CH}_3\text{CH}_2\text{COO})_6$	Sol	30x30x20		-
$\text{C}(\text{NH}_2)_3\text{Al}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$	Sol	30x30x5	GASH.	-
$(\text{NH}_4)_2\text{SO}_4$	Sol	20x20x10		-
$\text{KNaC}_2\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$	Sol	30x30x10		-
$(\text{NH}_2\text{CH}_2\text{COOH})_3\text{H}_2\text{SO}_4$	Sol	100x100x10	b-plate. TGS.	-
		70x50x30	Euhedral crystal. TGS.	-
KH_2PO_4	Sol	50x20x20	KDP.	-
NH_4Cl	Sol	15x15x5		-
HIO_3	Sol	15x15x5		-

343				
Name(s): M. Futamoto, T. Aita, and U. Kawabe				
Inst. & Address: Central Research Laboratory, Hitachi, Ltd., Kokubunji, Tokyo, Japan				
Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CaB_6 , SrB_6	Flu	0.2x0.2x15	Hexaboride and tetraboride single crystals were grown by a molten aluminum flux method	
BaB_6	Flu	0.2x0.2x3		
LaB_6 , CeB_6 , PrB_6	Flu	0.2x0.2x15	Properties studied: Hardness	
NdB_6 , SmB_6 , EuB_6	Flu	0.1x5x5	Electron emission properties	-
YB_4 , GdB_4 , DyB_4 , HoB_4 , ErB_4 , TmB_4	Flu	0.3x0.3x3	Publications: T. Aita, U. Kawabe, and Y. Honda, Japan J. Appl. Phys. 13 (1974) 391 M. Futamoto, T. Aita, and U. Kawabe, Japan J. Appl. Phys. 14 (1975) 1263 H. Okano, M. Futamoto, S. Hosoki, and U. Kawabe, Shinku 20 (1977) 127 M. Futamoto, S. Hosoki, H. Okano, and U. Kawabe, J. A. P. 48 (1977) 3541	

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Name(s): F. Ishida, N. Ohta, T. Ikeda, Y. Sugita

Inst. & Address: Hitachi Central Research Laboratory, Kokubunji Tokyo, 185 JAPAN

Telephone: 0423-23-1111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$(Y Sm)_3(Fe Ga)_5O_{12}$	<u>LE</u>	Dia.: 50mm Thickness: 2~10 μ m	(111), Magnetic Bubble, Bubble Dia. 5 μ m	+
$(Y Sm Lu)_3(Fe Ga)_5O_{12}$	<u>LE</u>	//	(111), Magnetic Bubble, Bubble Dia. 2~5 μ m	+
$(Y Sm Lu Ca)_3(Fe Ge)_5O_{12}$	<u>LE</u>	//	//	+
$(Y Eu Lu Ca)_3(Fe Ge)_5O_{12}$	<u>LE</u>	//	//	+
$(Y Er Ca)_3(Fe Ge)_5O_{12}$	<u>LE</u>	//	//, g-factor > 10, high wall velocity	-
$(La Tm Ca)_3(Fe Ge)_5O_{12}$	<u>LE</u>	//	//, g-factor > 10, high wall velocity	-

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Name(s): Norio Ohta

Inst. & Address: Hitachi Central Research Laboratory, Kokubunji Tokyo, 185 Japan

Telephone: 0423-23-1111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$Nb_{1-x}Ta_x$ (x=0.0-0.3)	Stress anneal	20x20x20	Superconductor, Single crystal, (110) cut, H_{c2} anisotropy J. Low Temp. Phys. <u>15</u> , 231, (1974)	-
Nb	<u>Z</u>	8 mm dia. sphere	Superconductor, Single crystal, H_{c1} anisotropy To be published in J. Phys. Soc. Japan (1978)	-
V	<u>Z</u>	//	//	-
$Pb_{1-x}In_x$ (x=0.0 - 0.2)	<u>Br</u>	6mm dia x 10 l.	Superconductor, Single crystal, H_{c2} anisotropy, T_c J. Low Temp. Phys. <u>27</u> , 125, (1976)	-

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Name(s): K. Takagi, T. Fukazawa, S. Akiyama, T. Kobayashi and T. Oi

Inst. & Address: Central Research Laboratory, Hitachi Ltd.
Kokubunji, Tokyo 185, Japan

Telephone: 0423-23-1111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$Gd_3Ga_5O_{12}$ (GGG)	Cz	50 in dia. x50	Boule axis: $\langle 111 \rangle$ $\langle 100 \rangle \langle 110 \rangle$ Dislocation: less than 3 per wafer Inclusion free	+ -
$Bi_4Ge_3O_{12}$ (BGO)	Cz	35 in dia. x 50	Boule axis: $\langle 100 \rangle$ Application: scintillator (photon peak pulse height: 12% vs NaI)	+
LaB_6	Z	1 in dia. x 30	Axis: $\langle 100 \rangle$ Application: electron emission source (A LaB_6 single crystal is grown by a laser heated floating zone method.) Publications: (GGG) J. Crystal Growth <u>32</u> (1976) 89. J. Crystal Growth <u>38</u> (1977) 206. (LaB_6) J. Crystal Growth <u>40</u> (1977) 1	-

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Name(s): Kohei Ito, Fumio Nitanda, Juro Endo.

Inst. & Address: Hitachi Metals, Magnetic and Electronic Mats. Res. Lab. .
5200 Mishiri, Kumagaya, Saitama, Japan.Telephone: 0485
32-22]]

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENT: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$Gd_5Ga_3O_{12}$ (GGG)	Cz	ϕ 50 x 50	Orientation [111]	+
$Gd_2(MoO_4)_3$ (MOG)	Cz	ϕ 30 x 200	Orientation [110]	+

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Name(s): Takashi Sambongi Department of Physics, Hokkaido University

Inst. & Address: Kita-Ku, Sapporo 060, Japan

Telephone: 011-711-2111-
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MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tantalum-triselenide; $TaSe_3$	VP	$(\sim 10^{-2})^2 \times 10$	resistance ratio ~ 100 , $\langle 010 \rangle$ superconducting properties	+
Tantalum-trisulphides; TaS_3	same	same	$\langle 001 \rangle$ semiconducting below ~ 200 K	+
Niobium-triselenide; $NbSe_3$	same	same	resistance ratio ~ 100 , $\langle 010 \rangle$ conductivity anomalies	+
Niobium-trisulphide; NbS_3	same	same	semiconducting	+

349 Name(s): Itaru Tatsuzuki Inst. & Address: Research Institute of Applied Electricity Hokkaido University, Sapporo, JAPAN Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$Kl_3(SeO_3)_2$ $KD_3(SeO_3)_2$ $RbH_3(SeO_3)_2$ $RbD_3(SeO_3)_2$ KDP DKDP	Sol	More than 10 x 10 x 10		

350 Name(s): S. Shimada, K. Kodaira and T. Matsushita Inst. & Address: Department of Applied Chemistry, Faculty of Engineering, Hokkaido University Sapporo, 060 Japan Telephone: (011)711-2111				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GeO_2	VP	6x0.01x0.01	S.Ito, K.Kodaira, T.Matsushita, Mat.Res.Bull, 13(2) 97 (1978)	
SnO_2	VP	47x0.08x0.08	T.Matsushita, H.Kataoka, A.Tsunashima, K.Kokaira, Yogyo Kyokai shi, 84(3), 147 (1976)	
Zn_2GeO_4	VP	15x0.04x0.04	S.Ito, K.Kodaira, T.Matsushita, J.Chemical Society of Japan, Chemistry and Industrial Chemistry, No.9 (1976) pp.1372	
Zn_2SnO_4	VP	50x0.01x0.01	R.Yoshida, Y.Yoshida, I.Yamai, K.Kodaira, T.Matsushita, J.Crystal Growth, 36(1), 181 (1976)	

351 Name(s): S. Shimada, K. Kodaira and T. Matsushita Inst. & Address: Department of Applied Chemistry, Faculty of Engineering, Hokkaido University Sapporo, 060 Japan Telephone: (011)711-2111				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SnO_2 (tin oxide)	Flu	15x0.4x0.4	Chem. Lett., (1976) pp.235	
TiO_2 (rutile)	Flu	12x0.1x0.1	growth direction c-axis, main faces {110}	} unpublished
ZrO_2 (zirconia)	Flu	4x0.3x0.3	growth direction c-axis, main faces {100}{110}	
HfO_2 (hafnia)	Flu	11x0.6x0.6	growth direction c-axis, main faces {100}	
TiO_2	Flu	25x3x3	} J. Crystal Growth, 41(2), 317 (1977)	
$Bi_4Ti_3O_{12}$	Flu	15x1x1		
$Bi_2Ti_4O_{11}$	Flu	15x1x1		
$Bi_2Ti_2O_7$	Flu	10x10x6		
SnO_2	Flu	1x1x0.1	growth direction c-axis	} unpublished
Zn_2SnO_4	Flu	3x3x3	main faces {111}	
Mg_2SnO_4	Flu	1x1x1	main faces {111}	
$Bi_2Sn_2O_7$	Flu	4x4x4	main faces {111}	

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Name(s): K.Kuriyama, T.Kamijoh, and K.Iwamura

Inst. & Address: College of Engineering, Hosei University, Kajinocho, Koganei, 184, Tokyo, Japan Telephone: 0423-81-5341

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Li ₂ CdPb	M	10x10x15	polycrystalline: crystal structure determination Japan. J. Appl. Phys. <u>12</u> 743 (1973)	-
LiIn	Br	6x5x5- 10x10x15	(111) orientation: single crystal growth J. Cryst. Growth <u>23</u> 160 (1974) Field emission study (Fowler-Nordheim plots) J. Appl. Phys. <u>47</u> 3754 (1976) Elastic properties (Elastic const. Debye temp.) Phys. Rev. <u>B13</u> 1528 (1976)	-
LiAl	M	10x10x15	Neutron diffraction study Acta Crystallogr. <u>B31</u> 1793 (1975) Applications: solid blanket for nuclear fusion, and electrode material	+
NaTl-structure (NaTl, NaIn, LiIn, LiAl, and LiZn)	M	10x10x15	Elastic properties (Elastic moduli, Debye temp.) (to be published)	-
Li ₂ ZnGe	Sol	5x5x8	(111) orientation: single crystal growth, semiconductor J. Cryst. Growth (in press)	+

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Name(s): Kiyoyasu Imai

Inst. & Address: Ishikawa Technical College, Tsubata, Ishikawa 929-03, Japan Telephone: 07628-9-3111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KDP	Sol.	20x20x40	Properties Studied	
TGS	Sol.	40x40x20		
ADP	Sol.	20x20x40		
RS	Sol.	20x30x50		

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Name(s): Kôki Ikeda

Inst. & Address: Department of Metallurgy, Faculty of Engineering, Iwate University Morioka, Japan Telephone: (0196) 23-5171

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Cu-Al Alloy Single Crystal 0.1 at.% Al 0.2 at.% Al 0.3 at.% Al 0.4 at.% Al 0.8 at.% Al 1.6 at.% Al	Cz	about 30φ x 60	Materials 99.99 %Cu Zone Melting (three times) 99.999% Al Yield stress is measured at 1.4~300 K.	+

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Name(s): Kazuo GESI

Inst. & Address: Japan Atomic Energy Research Institute, Tokai, Ibaraki 319-11, Japan

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$	<u>sol</u>	30	For dielectric studies.	-
$\text{SnCl}_2 \cdot 2\text{D}_2\text{O}$	<u>sol</u>	20	For dielectric and high pressure studies.	-
AgNO_2	<u>sol</u>	3 - 8	For dielectric studies.	-
$\text{AgNa}(\text{NO}_2)_2$	<u>sol</u>	5 - 10	For dielectric studies.	-
$\text{Ba}(\text{NO}_2)_2 \cdot \text{H}_2\text{O}$	<u>sol</u>	20	Hex. plsm. For pyroelectric studies.	-
X_2SnY_6 (X=K, NH ₄ , Y=Cl, Br)	<u>sol</u>	2 - 5	For studies of phase transitions.	-
$(\text{NH}_4)_3\text{H}(\text{SO}_4)_2$	<u>sol</u>	20 x 1	Hex. plates. For dielectric studies.	-
$(\text{ND}_4)_3\text{D}(\text{SO}_4)_2$	<u>sol</u>	10 x 1	Hex. plates. For dielectric studies.	-
$(\text{NH}_4)_3\text{H}(\text{SeO}_4)_2$	<u>sol</u>	10 x 0.5	Hex. plates. For dielectric studies.	-
Rb_2ZnY_4 (Y = Cl, Br)	<u>sol</u>	10 x 3	For neutron scattering studies.	-
$\text{SC}(\text{ND}_2)_2$	<u>sol</u>	30	Aggregation of oriented crystallites. For neutron scattering studies.	-
$\text{SC}(\text{ND}_2)_2$	<u>sol</u>	5 x 2	Transparent single crystals. For dielectric studies.	-
CsH_2PO_4	<u>sol</u>	10	For dielectric studies.	-
CsD_2PO_4	<u>sol</u>	8 x 2	For dielectric and high pressure studies.	-
$\text{NH}_4\text{H}_2\text{AsO}_4$	<u>sol</u>	10 x 3	For dielectric studies.	-

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Name(s): Katsuhiko YOKOTA, Susumu TAMURA and Saichi KATAYAMA

Inst. & Address: Faculty of Engineering, Kansai University, Suita, Osaka, JAPAN 564 Telephone 06-388-1121

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Ga}_{1-x}\text{Al}_x\text{As}$ film with uniform composition in depth	LE	0.1-10um in thickness	Electron density $< 10^{16} \text{ cm}^{-3}$	
$\text{InAs}_{1-x}\text{P}_x$ film	VP	2 - 10 um in thickness	Carrier density $< 10^{16} \text{ cm}^{-3}$	
CdTe on Te substrate	VLS	1 - 100um in thickness	single crystal with (110) on Te substrate with (100) direction. Surface carrier density $< 10^{15} \text{ cm}^{-3}$	
ZnTe film	VP	1 - 10 um in thickness	This work is started at May.1978.	

357 Name(s): H. Kasahara M. Yamada K. Yamamoto K. Abe Inst. & Address: Kobe University The Faculty of Engineering Rokko Nada Kobe Japan Telephone: (078) 881-1212 ext. 5101				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdIn_2S_4 CdIn_2Se_4 AgGaS_2 CuInS_2	Br	$W \times H \times L$ $10 \times 5 \times 30$	Far-Infrared Spectroscopy Normal and Resonant Light Scattering Lattice Vibration in CdIn_2S_4 K. Yamamoto, T. Murakawa, Y. Ohbayashi, H. Shimizu and K. Abe; J. Phys. Soc. Japan 35, 1258 (1973) Lattice Vibration in Spinel-Type CdIn_2S_4 H. Shimizu, Y. Ohbayashi, K. Yamamoto, K. Abe ; J. Phys Soc. Japan 38, 750 (1975) Single Crystal Growth of CdIn_2S_4 H. Kasahara, K. Abe, K. Yamamoto; Memoirs of the Faculty of Engineering, Kobe University No. 23 (1977)	

358 Name(s): Seichi Hasegawa Inst. & Address: Kanazawa University, Dept. of Electronics, Fac. of Technology, Kanazawa 920, Japan Telephone: Kanazawa 0762-61-2101				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
sputtered amorphous Si. sputtered or evaporated amorphous Ge. evaporated amorphous $\text{Ga}_{1-x}\text{P}_x$		$50 \times 50 \times 10^{-2}$	dopant: Al, P. properties studied: ESR, conductivity optical absorption, photo-conductivity, thermo- electric-power.	

359 Name(s): Associate Prof. S. Anzai Inst. & Address: Yagamidai-Campus, Faculty of Engineering, Keio University, 3-14-1, Hiyoshi, Yokohama-shi, 223 JAPAN Telephone: 044-63-1141				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NiS (Beta Nickel Sulfide)	<u>M</u>	5x5x5	Compressibility and Metal-Nonmetal Tran- sition Point (S. Anzai and K. Ozawa, J. Appl. Phys. 6 2139 (1977).)	-
Ni_{1-x}S (Beta Nickel Sulfide)	<u>Sub</u>	3x3x(<0.1)		-

360 Name(s): Toshio Harada Inst. & Address: Institute of Atomic Energy, Kyoto University, Gokasho, Uji-shi, Kyoto 611, JAPAN Telephone: 0774-32-3111 ext. 283				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ag	GE		Crystal growth mechanism	-

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Name(s): Saburo Magari
 Inst. & Address: Institute of Atomic Energy, Kyoto University,
 Gokasho, Ujishi, Kyoto 611, JAPAN

Telephone: 0774-32-3111
 ext. 280

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZrO ₂	Elec- tro- depo- sition from plasma	0.5φ×1.0	In Vacuum; ZrO ₂ (monoclinic)+ Zr(α), O/Zr~1.95 In Oxygen; ZrO ₂ (monoclinic) transparent, O/Zr~2.00 Asahi Garasu Kogyo Gijitsu Shorei-kai, Kenkyu Hokoku 29 30(1977)	- -
HfO ₂	Elec- tro- depo- sition from plasma	0.5φ×0.2	In Vacuum; HfO ₂ (monoclinic)+ ? , O/Hf~1.95 In Oxygen; HfO ₂ (monoclinic) transparent, O/Hf~2.00 Proc. of the 2nd Symp. on Ion Source and Application Technology P.165(1978)	- -

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Name(s): Dr. Hiroyuki Matsunami, Dr. Tetsuro Tanaka

Inst. & Address: Dept. Electronics, Kyoto Univ. Yoshidahonmachi, Sakyo, Kyoto, Japan

Telephone: (075)
 751-2111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. Silicon Carbide (SiC)				
6H-SiC	<u>LE</u>	5×5×0.02	(0001), Al, Ga, B, N, photoluminescence, Hall effects, blue light-emitting-diodes.	-
6H-SiC	<u>VP</u>	5×5×0.02	(0001), Al, N, electrical properties, photoluminescence.	-
3C-SiC	<u>VP</u>	10×10×0.005	(100), (110), (111), heteroepitaxy, electrical properties, optical properties, photoelectric devices.	-
2. PLZT((Pb,La)(Zr,Ti)O ₃)	<u>GE</u>	10×10×0.005	(100), (111), heteroepitaxial growth, optical properties, electro-optic effects, ferroelectricity, optical waveguide, light modulator.	+

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Name(s): Junji Saraie, Tetsuro Tanaka

Inst. & Address: Department of Electronics, Kyoto University
Kyoto, JAPANTelephone: 075-751-2111
ext. 5339

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe Cadmium Telluride	<u>Br</u>	13 ^φ × 40	6N, doped with Al and P, electrical properties, photoluminescence deep level spectroscopy	-
	<u>LE</u>	10 × 10 × 0.05	(111), (100), doped with Cl photo and electroluminescence J. Crystal Growth 43(1978)13	-
ZnS Zinc Sulfide	<u>GE</u>	5 ^φ × 0.05	(111), (100), photoluminescence to be reported at ICVGE (1978), Nagoya, JAPAN	-
ZnSe Zinc Selenide	<u>GE</u>	5 ^φ × 0.05	(111), (100), photoluminescence	-

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Name(s): Akio Kato

Inst. & Address: Kyushu University, Hakozaki, Higashi-ku, Fukuoka-shi, Japan

(092)
Telephone: 641-1101
ex. 3195

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
TiN	VP	whisker	J. Cryst. Growth <u>29</u> , 55(1975)	-
TiC	VP	whisker	J. Cryst. Growth <u>37</u> , 293(1977)	-
ZrN	VP	whisker	Nippon Kagaku Kaishi, <u>1978</u> , (66)(1978)	-
ZrC	VP	whisker	{ J. Less-Common Metals, <u>58</u> , 147(1978) Nippon Kagaku Kaishi, <u>1977</u> , 650(1977)	-

365

Name(s): Y. Oishi

Inst. & Address: Dept. of Nuclear Eng., Kyushu Univ.
Fukuoka-city 812, JapanTelephone: (092) 641-1101
Ext. 3709

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Li ₂ O	M	3 × 3 × 1	Measurement of self-diffusion coefficients	(-)

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Name(s): Noboru Kitamura, Masahiro Kakehi and Takao Wada
 Inst. & Address: Depart. Elec. Engin., Fac. Engin., Mie Univ.
 Tsu, Mie 514 JAPAN

Telephone: 0592-32-1211

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnTe	VP	13×4×3	undoped P-type crystal in a closed tube with iodine as a transport agent $\rho=10^4\sim10^6\Omega\text{cm}$ at 300K Japan. J. appl. Phys. <u>16</u> (1977) 1541	-
$\text{Ga}_x\text{Al}_{1-x}\text{Sb}$	VP	5×5×0.15	undoped P-type epi-layer on (111) GaSb in a closed tube with iodine as a transport agent $P\approx 10^{18}\text{cm}^{-3}$ $0.1 < x \leq 1$ Japan. J. appl. Phys. <u>17</u> (1978) 739	-

367

Name(s): Kunio TAKEI
 Inst. & Address: Dept. of Engineering Chemistry, Miyazaki University,
 Kirishima - 1, Miyazaki, Japan

Telephone: 0985-26-3155
ext. 238

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
HgS (black, metacinnabarite)	VP	0.4×0.5 × 0.1	unknown	-

368

Name(s): Macafumi Yamaguchi and Akio Yamamoto
 Inst. & Address: Ibaraki Electrical Communication Laboratory, N.T.T.,
 Tokai, Ibaraki, Japan

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnSe	Br	8 ϕ x 60	Purity of starting material; 99.999 % (Merck's powder). Impurity in undoped crystals; (a) Mg, Fe, Ni and Cu were detected by atomic absorption analysis. (b) Cr (0.6ppm), Co (8ppb), Cu (0.4ppm), Ag (30ppb), Sb (6ppb) and Au (0.07ppb) were determined by substoichiometric neutron activation analysis. Dopants; Al, In and Ga. Properties studied; photoluminescence** and electroluminescence.** Characterization; crystal imperfections (lineages, voids and twins),*** resistivity, carrier concentration. Applications; blue-LED** Publications; *Japan. J. appl. Phys. <u>17</u> (1978) 335. **J. Appl. Phys. <u>48</u> (1977) 196, 5237. Japan. J. appl. Phys. <u>16</u> (1977) 77. *** J. Cryst. Growth <u>41</u> (1977) 103.	+

369

Name(s): Kiyomasa Sugii

Inst. & Address: Musashino Electrical Communication Lab., NTT,
Musashino-shi, Tokyo 180, Japan

Telephone: 0422(59)2103

Musashino-shi, Tokyo 180, Japan										Material			
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.							Avail- ability + -			
InP, Indium phosphide	Sol	30 ^φ x 25	undoped single crystals							-			
			1. Purity: mass spectroscopic data (ppm).										
			Ca	Fe	Cu	Zn	Cl	Al	O		C	N	Si
			0.1	0.04	0.09	0.04	0.03	0.08	5		1	0.1	0.1
			2. Electrical properties.										
							77°K	R.T.					
			N _D -N _A (cm ⁻³)		3.1 x 10 ¹⁵		4.0 x 10 ¹⁵						
			Mobility (cm ² /Vsec)		4.0 x 10 ⁴		4.5 x 10 ³						
			3. Crystallinity.										
			The cell structure was observed: low-angle boundaries surrounded areas almost free from dislocations.										

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Name(s): T. Waho, S. Ogawa and S. Maruyama

Inst. & Address: Musashino Electrical Communication Laboratory, NTT
Musashino, Tokyo 180, Japan

Telephone: (0422)59-2336

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs _{1-x} Sb _x	MBE*)	10 x 12 x 3 μm	GaAs _{1-x} Sb _x (001) on GaAs(001) substrates MEED, AES, X-ray diffraction, 0.3<X<0.9 Japan. J. Appl. Phys. <u>16</u> (1977) 1875 *) Molecular Beam Epitaxy	

371 Name(s): M. Aono, C. Oshima, R. Nishitani, T. Tanaka, E. Bannai, and S. Kawai Inst. & Address: National Institute for Research in Inorganic Materials Namiki 1-1, Sakura-mura, Niihari-gun, Ibaraki 300-31, JAPAN Telephone: 0298-51-3351				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LaB ₆ (Lanthanum hexaboride) SmB ₆ (Samarium hexaboride)	<u>Z</u> <u>Z</u>	5×5×40 mm ³	LaB₆ is an excellent electron emitter because of its unusually low work function, and SmB₆ is a remarkable compound in that it contains both Sm³⁺ and Sm²⁺ ions in a ratio of ~7:3. The atomic and electronic structures of the (001) surfaces of LaB₆ and SmB₆ have been studied by angle-resolved X-ray and ultraviolet photoelectron spectroscopy, low-energy electron diffraction, work-function measurements, and low-energy ion scattering spectroscopy: M. Aono, T. Tanaka, E. Bannai, C. Oshima, and S. Kawai, Phys. Rev. B <u>16</u>, 3489 (1977). M. Aono, T. Tanaka, E. Bannai, and S. Kawai, Appl. Phys. Lett. <u>31</u>, 323 (1977). M. Aono, C. Oshima, T. Tanaka, E. Bannai, and S. Kawai, J. Appl. Phys. (April, 1978). M. Aono, R. Nishitani, C. Oshima, T. Tanaka, E. Bannai, and S. Kawai (Proc. Intern. Conf. Solid Films and Surfaces, Tokyo, 1978).	

372 Name(s): Noboru Kimizuka Inst. & Address: National Institute for Researches in Inorganic Materials Sakuramura, Niihari-gun, Ibaraki, 300-31, Japan Telephone: 0298-51-3351				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
YFe ₂ O ₄ , HoFe ₂ O ₄ , ErFe ₂ O ₄ , YbFe ₂ O ₄ , TmFe ₂ O ₄ , LuFe ₂ O ₄ , Yb ₂ Fe ₃ O ₇ , Lu ₂ Fe ₃ O ₇	<u>M</u> <u>VP</u>	1x1x1	magnetic property, catalytic reaction, new materials,	+
Bi ₂ ReO ₅ , Bi ₂ ReO ₆ , Bi ₂ WO ₅ , Bi ₂ WO ₆ , Bi ₂ MoO ₅ , Bi ₂ MoO ₆ , Bi ₂ LnO ₆ (Ln : Rare earth elements) Ln ₂ ReO ₆ , Ln ₂ MoO ₆ , Ln ₂ WO ₆	<u>Sub</u>	2x1x1	catalytic reaction, new materials, ferroelectric materials	+

373 Name(s): S. Kimura Inst. & Address: National Institute For Researches in Inorganic Materials 1-1, Namiki, Sakura-mura, Niihari-gun, Ibaraki-ken, Japan 300-31 Telephone: 0298-51-3351				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Y ₃ Fe ₅ O ₁₂ (YIG)	Z	8φ x 50	J. Crystal Growth <u>41</u> (1977) 192	-
" (")	Z	8φ x 5		+
YFe ₂ O ₄	Z	3x1.5x0.5	Mat. Res. Bull. <u>11</u> (1976) 637	+
YFeO ₃ (Yttrium Orthferrite)	Z	8φ x 50		+
Fe ₃ O ₄ (Magnetite)	Z	8φ x 20	Cation Purity: 99.9%, [110]	+
BaFe ₁₂ O ₁₉ (Ferroxdure, M)	Z	6φ x 20		-

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Name(s): C. Oshima, M. Aono, T. Tanaka and S. Kawai

Inst. & Address: National Institute for Researches in Inorganic Materials, Namiki, Sakuramura Telephone: 0298-51-3351
Niiharigun, Ibaraki, Japan

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Lanthanum Hexaboride (LaB_6)	Z	10 mm ϕ x 50 mm	Thermionic work functions and Richardson constants for LaB_6 single crystals are measured. They are 2.86 eV for the (100) plane, 2.68 eV for the (110) plane and 3.4 eV for (111) plane. The (111) faceting on the (110) plane has an important effect on the reduction of thermionic current at 5×10^{-6} Torr. C. Oshima et al. J Appl. Phys. 48(1977) 3925 Carbon atoms segregated on the LaB_6 (001) surface at elevated temperatures formed an epitaxial graphite layer and carbide-like layer. The properties of the layers has been discussed. C. Oshima et al. Japan. J. Appl. Phys. 16(1977)965. The values of work function of borides are much correlated with its electron density as similar to the case of metallic element and is explained in view of the theory proposed by Lang and Kohn. The change of surfaces due to ion sputtering and oxygen chemisorption are also discussed. Three dimensional frame work of boron at LaB_6 surfaces is important for stability of the surface. C. Oshima et al. Ohyo Buturi (Appl. Phys.) 45 (1976)600. (in Japanese). Shinkuh (Vacuum) 20(1977)46. (in Japanese).	

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Name(s): Manabu Saji

Inst. & Address: Nagoya Institute of Technology, Gokiso-cho, Showa-ku, Nagoya, Japan

Telephone: (052) 732-2111
Ext. 442

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdTe	Br	10 x 70	5N, dopants (undoped, In, Cl, P)	
	THM			
ZnTe	Br	10 x 70	5N, undoped	
	GE		Heterojunction (on CdTe)	
GeS_2	Sub	2x5x0.05	5N, undoped	

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Name(s): Takaho Tanaka

Inst. & Address: Nat. Inst. for Res. in Inor. Mat., Namiki, Niihari-gun, Ibaraki, Telephone:
Japan

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
REB_6 , Rare Earth (La, Ce, Nd, Sm, Eu, Gd) Hexaborides.	Z	9mm diam. x60mm l.	Purities about 5N. Electron emission, electrical and magnetic properties studies.	-
RB_2 , Transition metal (Cr, Ti, Zr) Diborides.	Z	9mm diam. x50mm l.	Purities about 4N. Electronic properties.	-
Transition metal carbides.	Z	8mm diam. x100mm l.	Purities about 3N. Electronic and Mechanical properties.	-

377

Name(s): Haruki Kawamura

Inst. & Address: National Research Institute for Metals,
2-3-12, Nakameguro, Meguro-ku, Tokyo, Japan

Telephone: Tokyo 719-2271

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NbCl ₄	VP	3x0.2x0.2	Electrical property under high pressure	
NbI ₄	VP	3x0.2x0.2		
Nb ₃ I ₈	VP	3x3x0.1		
Nb ₃ Ge	CVD		Superconducting property	
Nb ₃ Si	CVD			

378

Name(s): Isao Nishida

Inst. & Address: National Research Institute for Metals
2-3-12, Nakameguro, Meguro-ku, Tokyo 153, Japan

Telephone: (719) 2271

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CrSi ₂	Z	8x 35	At room temp. hole concentration: $6.25 \times 10^{20} \text{ cm}^{-3}$ hole mobility: $15 \text{ cm}^2/\text{V s}$ (a-axis) $6.7 \text{ cm}^2/\text{V s}$ (c-axis) I. Nishida, J. Mater. Sci. <u>7</u> (1972) 1119.	+
MnSi _{~1.73}	Br	16x 70	Growth rates: 0.33~6.0 mm/h Chemical composition: MnSi _{1.713~1.715} All crystals contain plate-like MnSi pre- cipitates. All of the MnSi are parallel to the c-plane in the matrix of Mn ₁₅ Si ₂₆ . I. Kawasumi, I. Nishida, K. Masumot and M. Sakata, Jap. J. Appl. Phys. <u>15</u> (1976) 1405.	
Mn ₁₅ Si ₂₆	VP	1.5	Transport agent: CuCl ₂ T. Kojima and I. Nishida, Jap. J. Appl. Phys. <u>14</u> (1975) 141.	
MnSi	Br	18x 80	Growth rate: 1.0~6.0 mm/h Chemical composition: MnSi _{1.021~1.043}	

379 Name(s): Hisashi Sekine, Kazumasa Togano, Kyoji Tachikawa Inst. & Address: National Research Institute for Metals, 2-3-12, Nakameguro Meguro-ku, Tokyo, Japan Telephone: 03(719)2271 ext. 284				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
V ₃ (Ge,Al)	Dif- fusion		Properties studied: Superconducting transition temperature, T _c =7.0~12.5K, superconducting upper critical field, H _{c2} =2.0~18.0T. Characterization: Polycrystal Publication: Transition temperature and upper critical field of superconducting V-Al/Cu-Ge composite tapes, Appl. Phys. Lett. <u>32</u> (1978) 582.	+

380 Name(s): Yoshimi Tsuchiya Inst. & Address: Department of Physics, Faculty of Science, Niigata University Niigata 950-21 JAPAN Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ag ₂ S	<u>M</u>	2×20×30	J. Phys. C: Solid State Physics., <u>11</u> (1978) 651.	-
Ultrafine particles of metals: Al, Zn, Ge, Cd, Pb, Cu, Ag, Au, Mn, Fe, Co, Ni, and Pd	<u>VP</u>	100×10 ⁻⁸	J. Phys. Soc. Japan, <u>39</u> (1975) 1586. J. Phys. Soc. Japan, <u>44</u> (1978) 1547.	-

381 Name(s): Taketoshi Hibiya, Hiroshi Makino, Kozaburo Suzuki Inst. & Address: Central Research Laboratories, Nippon Electric Co., Ltd., Kawasaki, Japan Telephone: 044-855-1111 Ext. 2169				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
(Y,Sm,Tm,Ca) ₃ (Fe,Ge) ₅ O ₁₂ (Y,Sm,Lu,Ca) ₃ (Fe,Ge) ₅ O ₁₂ (Y,Eu,Tm,Ca) ₃ (Fe,Ge) ₅ O ₁₂ (Y,Eu,Lu,Ca) ₃ (Fe,Ge) ₅ O ₁₂ (Y,Sm,Tm) ₃ (Fe,Ga) ₅ O ₁₂ (Y,Sm,Lu) ₃ (Fe,Ga) ₅ O ₁₂ (Y,Eu,Tm) ₃ (Fe,Ga) ₅ O ₁₂ (Y,Eu,Lu) ₃ (Fe,Ga) ₅ O ₁₂	LE	50.0	Grown on [111] oriented Gd ₃ Ga ₅ O ₁₂ for use in magnetic bubble memory devices	

382 Name(s): Fumio SHIMURA Inst. & Address: Central Res. Labs., Nippon Electric Company, Ltd., Miyazaki, Takatsu, Kawasaki, Japan Telephone: 044-855-1111 Ext. 2095				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Ni_5TiO_7 ; Nickel titanium oxide	Flu	10,0.5,0.5	F. Shimura and T. Kawamura, Japan. J. Appl. Phys. <u>15</u> (1976), 1403	
$\text{LiNb}_{1-y}\text{Ta}_y\text{O}_3$; INT	Cz	40, 10 ⁶	F. Shimura and Y. Fujino, J. Crystal Growth, <u>38</u> (1977), 293 F. Shimura, J. Crystal Growth, <u>42</u> (1977), 579	

383 Name(s): Tsutomu MINAMI Inst. & Address: University of Osaka Prefecture, Department of Applied Chemistry Telephone: 0722-52-1161 Sakai-Shi, Osaka-Fu 591, JAPAN ex.2304				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Vitreous semiconductors (chalcogenide glass system)			Electrical, optical, thermal, thermoelec- trical, photoelectrical, and photovoltaic properties. J. Amer. Ceram. Soc. <u>59</u> , 461 (1976). J. Non-Crystalline Solids, <u>23</u> , 431 (1977). Japan. J. Appl. Phys. <u>16</u> , 977 (1977). Phys. Chem. Glasses, <u>18</u> , 112 (1977). Thin Solid Films, to be published (1978).	-
Superionic conducting glasses (silver halide containing glass system)			Electrical, optical, thermal, thermoelc- trical properties. J. Amer. Ceram. Soc. <u>60</u> , 283 (1977) J. Amer. Ceram. Soc. <u>60</u> , 467 (1977) J. Electrochem. Soc. <u>124</u> , 1659 (1977). J. Non-Crystalline Solids, in press (1978).	-

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Name(s): Yasumitsu MORIKAWA and Koichi NAGAMI

Inst. & Address: Department of Electronics, University of Osaka Prefecture
Mozu, Sakai, Osaka, JapanTelephone: 0722-52-1161
(2287)

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
n-type Silicon single crystal 1) Chemical etching for mirror surface Anodic oxidation to form SiO_2 2) Inert gas ion hot implantation through SiO_2 3) Removal of SiO_2 for the electrical measurement Redistribution of dopants in Si by the vacancy flow made by the inert gas ion implantation (the dopants were uniformly distributed before ion bombardment)	<u>Z</u> , <u>Cz</u>	7x7x0.3	4×10^{13} - $1 \times 10^{16} \text{ cm}^{-3}$ (P doped) [111] 1×10^{15} (Sb doped) [100] Carrier concentration profile measurement both by the C-V and the Copeland method The carrier decreased region was observed below the initial dopant level The strong electric field was generated near the surface to extract the excess electrons toward the implanted side (Applications) Novel impurity concentration (e.g. $n^+ - n^-$, $n^+ - p^- - n^-$; the latter may be obtained in the case where the compensated (n^-) substrates were used.) Gettering in the video camera Japan. J. appl. Phys. <u>12</u> (1973) 472 Japan. J. appl. Phys. <u>16</u> (1977) 1281	+

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Name(s): Hajimu Sonomura

Inst. & Address: University of Osaka Prefecture, Sakai, Osaka, Japan

Telephone: 0722 52 1161

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Al}_x\text{Ga}_{1-x}\text{P}$ and $\text{Al}_x\text{Ga}_{1-x}\text{Sb}$	<u>Sol</u>	4X3X0.5	Dopants; Si and Zn. Properties studied; Energy gap, Emission spectrum, Excitation spectrum, Photovoltaic effect and Resistivity. Applications; LED for green light, and modulator and laser for optical communication. Publications; APL <u>22</u> 532 (1973), APL <u>24</u> 77 (1974), JAP <u>45</u> 5109 (1974) and JAP <u>46</u> 3693 (1975).	+

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Name(s): T.Kataoka and T.Yamada

Inst. & Address: Department of Precision Engineering, Faculty of Engineering, Telephone: 06-877-5111
Osaka University, Suita, Osaka 565, Japan

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KCl-KBr Potassium chloride- potassium bromide solid solution	Cz	50 dia x 50	The mechanical properties have been studied at low temperatures. This material shows a solid solution softening near the liquid helium temperature. Ref. T.Kataoka, T.Uematsu and T.Yamada: Japan. J. appl. Phys. <u>17</u> (1978) 271.	+

387

Name(s): A.Mitsuishi. S.Nakashima. A.Manabe. T.Hattori.

Inst. & Address: Department of Applied Physics, Faculty of Engineering, Osaka University, Telephone: (06)877-5111
Yamada-kami, Suita, Osaka Japan

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnTe (Zinc telluride)	<u>M</u>	10 ϕ x 50	undoped dopant : As, ($10^{17} \sim 10^{18} \text{cm}^{-3}$) P ($4 \times 10^{16} \sim 10^{18} \text{cm}^{-3}$) Li, ($4 \times 10^{16} \sim 5 \times 10^{17} \text{cm}^{-3}$)	-
PbO (yellow lead oxide)	<u>M</u>	10 x 5 x 2	dopant : Bi	+
HgSe	<u>Br</u>	9x9x20mm ³	Far-infrared reflectivity	-
Cd _x Hg _{1-x} Se	<u>Br</u>	9x9x20mm ³	0 < x < 0.3, Far-infrared reflectivity	-
Zn _x Hg _{1-x} Se	<u>Br</u>	9x9x20mm ³	0 < x < 0.2, Far-infrared reflectivity	-
RbAg ₄ I ₅	<u>Br</u>	5x5x10mm ³	Raman scattering, Brillouin scattering and infrared measurements	-
RbCoBr ₃ , RbMnBr ₃	<u>Br</u>	3x3x10mm ³	Raman scattering, dopant : Co, Ni.	-
GaSe, GaTe	<u>Br</u>	5x5x10mm ³	Raman scattering non-linear optics	-

388 Name(s): Youichi Ohno Inst. & Address: ISIR. Osaka University, Yamadakami, Suita, Osaka, Japan. Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs (Gallium Arsenide)	<u>VLS</u>	triangular needle L.10-W.0.4 (mm)	orientation - [111] nearly perfect crystal dopant - Au resistivity - 0.004 Ω cm application - as samples of field emitters of FIM, FEM and APFIM	+
GaP (Gallium Phosphide)		triangular and hexag- onal needles L.16-W.0.5 (mm)	publication, Surf. Sci. 69 (1977) 521	

389 Name(s): Takanori Tomita, Dr Inst. & Address: Department of Physics, Faculty of Science, Saitama University, Urawa, 338 JAPAN Telephone: 0488-52-2111				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CaF ₂ single crystals Si very pure single crystals SiC various polytypes pure single crystals All the crystals were offered by the producers and only studied by the X-Ray topography, X-Ray energy analyser and EPR sometimes at Liq. He temperature			Sr, Ba other impurity Ti 0.05% or less Neutron or Co- irradiation effect very few dislocations Energy spectrum of diffracted X-rays at special angles of incidence should be asked to Norton Co., Canada General Electric Co., U.S.A. Semi-Elements Inco., U.S.A. A. F. C. R. L., U.S.A. Topographic study of Dislocations Determination of polarity by anomalous dispersion of X-rays Publications; X-Ray Study of -SiC Proceedings VI Int. Conf. X-Ray Optics & Microanalysis (1972) 685 Inclusions Found in Solution-Grown -SiC Proceedings III Int. Conf. SiC (1973) 141 A Concept for The Growth of Twinned Crystals of fcc Structure J. Crystal Growth 24/25 (1974) 331 Rotational Twin of The Hollow Crystal Found for Solution-Grown -SiC Acta Cryst. S (1975) (abstract)	- - -

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Name(s): J. Hatano

Inst. & Address: Department of Applied Physics, Faculty of Science, Science University of Tokyo, Shinjuku-ku, Tokyo 162, Japan

Telephone: 03-260-4271

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Ca}_2\text{Sr}(\text{C}_2\text{H}_5\text{COO})_6$: Dicalcium Strontium Propionate(DEP)	Sol	20x20x20	Bipyramidal	Ferroelectricity(Domain, Hysteresis, Dielectric properties) has been studied.
$(\text{CH}_3\text{H}_6)\text{Al}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$: Guanidine Almi- num Sulfate Hexahydrate(GASH)	Sol	55x55x25	Hexagonal Prism	
$\text{NH}_2\text{CH}_2\text{COOH}$ AgNO_3 : Glycine Silver Nitrate(GSN)	Sol	5x5x20	Plastic	
$\text{LiH}_3(\text{SeO}_3)_2$: Lithium Trihydrogen Selenite(LMS)	Sol	40x25x25	Yellowish	
$\text{SC}(\text{NH}_2)_2$: Thiourea	Sol	15x10x10	Sixsided Prism	
$(\text{NH}_2\text{CH}_2\text{COOH})_3 \cdot \text{H}_2\text{SO}_4$: Triglycine Sulfate(TGS)	Sol	15x15x20	Transparent	
$(\text{NH}_2\text{CH}_2\text{COOH})_3 \cdot \text{H}_2\text{SeO}_4$: Triglycine Selenate(TGSe)	Sol	30x30x10	"	
$(\text{NH}_2\text{CH}_2\text{COOH})_3 \cdot \text{H}_2\text{BeF}_4$: Triglycine Fluoberyllate (TGFB)	Sol	10x10x10	"	

391

Name(s): Shuichi Otake and Naoshi Matsuno

Inst. & Address: Department of Physics, Science University of Tokyo,
Shinjuku-ku Tokyo Japan

Telephone: 03(260)4271

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Bi Single Crystal	$\underline{Z} + \underline{C}_Z$	$4\phi \times 15$	Residual Resistivity $R_{300K} / R_{4.2K} \div 750$	

392

Name(s): Takeyo TSUKAMOTO, Hideo FUTAMA

Inst. & Address: Science University of Tokyo, Department of Applied Physics,
1-3 Kagurazaka, Shinjuku-ku, Tokyo, JAPAN

Tokyo

Telephone: 260-4271

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiIO_3	sol	10x10x50	dielectric and optical properties	
NH_4IO_3	sol	2 x2 x8	crystal growth, dielectric properties.	
CsIO_3 , RbIO_3	sol	1 x1 x1	crystal growth.	
$\text{SC}(\text{NH}_2)_2$	sol	5 x10x10 2 x30x30	dielectric properties, phase transitions, crystal growth mechanism.	
Rochelle salt	sol	30x30x50	optical properties, crystal and ferroelectric domain growth mechanism.	

393 Name(s): Koichi Matsumoto, Shoji Kaneko and Katsuki Takagi. Inst. & Address: Research Institute of Electronics, Shizuoka University, Hamamatsu 432, Japan. Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
SnO ₂	VP (CT)	2x2x2	J. of Crystal Growth, 40(1977)291	-

394 Name(s): Shoji Yamada and Michihiko Kitao Inst. & Address: Research Institute of Electronics, Shizuoka University, Hamamatsu 432, JAPAN Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
As ₂ Se ₃ (Arsenic Triselenide)	Sub	15x8x1.5 mm ³	* P ₂ /n (layered structure). * Optical gap is about 1.77eV. * $\sigma \approx 3 \times 10^{-14} \Omega^{-1} \text{cm}^{-1}$. * Japan. J. Appl. Phys. <u>8</u> (1969) 499.	-

395 Name(s): N. Watanabe, Y. Makino Inst. & Address: Sony Corporation Research Center 174 Fujitsukacho, Hodogayaku, Yokohama 240, Japan Telephone: Yokohama (045) 351-1271				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Bi ₁₂ GeO ₂₀	M,LE	40x40x800	Optical grade	+
LiNbO ₃	M	15x15x300	Optical grade	+
LiTaO ₃	M	15x15x300	Optical grade	+
Bi ₁₂ SiO ₂₀	M	20x20x400	Optical grade	-
GaAs	GE	20x30x0.01	FET, mixer diode on Cr doped substrates	-
GaAs-GaAlAs	LE	10x10x0.01	Laser diode	-

396

Name(s): Shinichi Akai, Takashi Suzuki

Inst. & Address: Sumitomo Electric Industries LTD, Semiconductor Dep.

Telephone: 06-461-1031

1 Shimaya Konohana-Ku, Osaka, Japan

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs poly crystal	Br	400x50x50	Purity: 6N up	+
GaAs single crystal	Br	350x50x25	Dopants: none, Si, Te, Sn, Zn, Cr-O	+
GaAs epitaxial wafer	GE	(~10 cm ²)	Applications: FET etc	+
GaP poly crystal	M	75 ^φ x 50	Purity: 6N	+
GaP single crystal	Cz	45 ^φ x 150	(111)/(100), Dopants: none, S	+
GaP epitaxial wafer	LE	(~15 cm ²)	Applications: LED (Green, Red etc)	+
InP poly crystal	M	75 ^φ x 50	Purity: 6N in use of source	+
InP single crystal	Cz	45 ^φ x 100	Dopants: S, (Fe)	+, (-)
InSb poly crystal	Z	1000x60x30	Purity: 6N	+
InSb single crystal	Cz	50 x 300	Applications: Detector, Hall-element etc	+
InAs poly crystal	Br	350x50x25	Purity: 6N in use of source	+

397

Name(s): T. Inoue, H. Komatsu.

Inst. & Address: The Research Institute for Iron, Steel and Other Metals,
Tohoku Univ. Sendai 980, JapanTelephone: 0222-27-6200
(ext. 2909)

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KCl	Cz	10x10x40	Purity: 99.99% (Merk.Co.) Orientation: [100] Dislocation density: $6 \times 10^4/\text{cm}^2$ Publications: Kristall und Technik, Heft 9 (September)/1978, (to be published).	+

398

Name(s): Yoshihiro Kokubun, Hideo Watanabe and Masanobu Wada

Inst. & Address: Department of Electronic Engineering, Faculty of Engineering,
Tohoku University, Aramaki Aza Aoba, Sendai 980, JAPAN

Telephone: 0222-22-1800

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
CdS (Cadmium Sulfide)	<u>Sub</u> <u>Br</u>	10x10x0.05 15φ x40	Dopants: group I, III, V and VII elements Properties studied: electrical and optical prop- erties Applications: light emitting diodes and solar cells Publications: Japan.J.appl.Phys. 14(1975) 1403 Japan.J.appl.Phys. 14(1975) 1733 Japan.J.appl.Phys. 15(1976) 2365 Japan.J.appl.Phys. 15(1976) 2463	No
CdSe (Cadmium Selenide)	<u>Sub</u> <u>Br</u>	16φ x40 15φ x40	Dopants: group I, III, V and VII elements Properties studied: electrical and optical prop- erties Applications: photoconductive cells and solar cells Publications: Japan.J.appl.Phys. 13(1974) 1393 Japan.J.appl.Phys. 14(1975) 1059 Japan.J.appl.Phys. 15(1975) 1387 Japan.J.appl.Phys. 17(1978) 1127	No

399

Name(s): Yoshihiro Kokubun, Hideo Watanabe and Masanobu Wada

Inst. & Address: Department of Electronic Engineering, Faculty of Engineering,
Tohoku University, Aramaki Aza Aoba, Sendai 980, JAPAN

Telephone: 0222-22-1800

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnS (Zinc Sulfide)	<u>Br</u>	15φ x40	Dopants: Al Properties studied: electrical and optical prop- erties Applications: light emitting diodes	No
ZnSe (Zinc Selenide)	<u>Br</u> <u>GE</u>	15φ x40 10x10x0.05	Dopants: group I, III, V and VII elements Properties studied: electrical and optical prop- erties Applications: light emitting diodes Publications: Japan.J.appl.Phys. 13(1974) 357	No
Zn _x Cd _{1-x} S (x = 0~0.7) (Zinc Cadmium Sulfide)	<u>GE</u>	10x10x0.05	Dopants: group VII elements Properties studied: electrical and optical prop- erties Publications: Japan.J.appl.Phys. 16(1977) 1455 Japan.J.appl.Phys. 16(1977) 2115	No
CuInSe ₂ (Copper Indium Selenide)	<u>Br</u>	3x3x3	Properties studied: electrical properties Applications: solar cells Publications: Japan.J.appl.Phys. 16(1977) 879	No

400 Name(s): Masaki Maeda Inst. & Address: Department of Applied Physics, Faculty of Engineering, Tohoku University, Sendai, JAPAN Telephone: 0222-22-1800 Ext. 4726				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{LiNH}_4\text{C}_4\text{H}_4\text{O}_6 \cdot \text{H}_2\text{O}$ (LAT)	Sol	10x30x60	For Cr^{3+} dope material. ESR measurement for pure material. Elastic properties ferroelectric and ferroelastic.	
$\text{NaKC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O} - \text{NaNH}_4\text{C}_4\text{H}_4\text{O}_6$ 4H ₂ O mixed crystal	Sol	10x20x30	Elastic properties (elastic compliance).	
$\text{K}_5\text{Nd}(\text{MoO}_4)_4$	Cz	15x 8x65	Dielectric constant, refractive indices, birefringence. Stoichiometric laser material.	

401 Name(s): Taneki TOKUDA Inst. & Address: Tokyo Metropolitan University, Fukazawa, Setagaya, Tokyo, Japan Telephone: 03-717-0111				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
PAA	<u>Sub</u>		P-V-T relations, transport.	
PBB	<u>Sub</u>		" "	
Cristobalite	<u>Flux</u>		Phase transition	
	<u>Hyd</u>			
MgO	<u>Hyd</u>		Topochemical properties.	

402 Name(s): Masaharu AOKI Inst. & Address: Dept. Electronic Eng., Fac. Eng., Univ. of Tokyo, Bunkyo-ku Tokyo, 113, JAPAN Telephone: 03-812-2111 ext. 3126				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnS	Sol	2 x 2 x 1 ^{mm}	Crystals are grown from Tellurium as a solvent	-

403 Name(s): Koichi Kitazawa & Kazuo Fueki Inst. & Address: Dept. Industrial Chem., Univ. of Tokyo, Tokyo, 113, Japan Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Y_2O_3 yttria	Z	3.5φ x 100	4N; 100, 111; Nd_2O_3 ; electrical conductivity, laser efficiency; dislocation density $2 \times 10^4/\text{cm}^2$; J. Crystal Growth, <u>39</u> 211 (1977)	-

404

Name(s): Ichiroh NAKADA

Inst. & Address: The Institute for Solid State Physics, The University of Tokyo,
Roppongi, Minato-ku, Tokyo, 106 Japan

Telephone: 03-402-6238

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
anthracene	Br	20 x 20 x 20	semiconducting grade; any orientation; photoconducting properties; J. Phys. Soc. Japan 17(1962)113, 24(1968)792.	+
anthraquinone	Br	10 x 10 x 10	semiconducting grade; any orientation; photoconducting properties; J. Phys. Soc. Japan 24(1968)798.	+
FeCr ₂ S ₄	VP	2 x 2 x 2	purity not yet identified; octahedron; ferrimagnetic semiconductor; to be published in Japan. J. appl. Phys.	+
CdCr ₂ Se ₄	VP	1.5x1.5x1.5	purity not yet identified; octahedron; ferromagnetic semiconductor;	+
Cr ₂ S ₃	VP	3.0x3.0x0.2	purity not yet identified; platelet; J. Crystal Growth 43(1978)711.	+
(SN) _x	VP	3 x 3 x 3	purity not yet identified.	+

405

Name(s): S.Yasuami

Inst. & Address: Toshiba Research and Development Center, Toshiba Corporation Telephone: 044-511-2111
Komukai-Toshibacho 1, Saiwai-ku, Kawasaki, 210, Japan

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Silicon	Cz		The swirl formation of microdefects in dislocation-free Cz crystals was observed by means of X-ray diffraction topography cooperated with heat treatment. J. Cryst. Growth 39, 227(1977)	-
	Z Cz		The structures and the dimensions of micro- defects in dislocation-free crystals were studied by means of diffuse X-ray scatter- ing. Appl. Phys. Lett. 32, 271(1978)	- -

406

Name(s): T. Fukuda

Inst. & Address: Toshiba Research and Development Center, Toshiba Corporation
Komukai-Toshibacho 1, Saiwaiku, Kawasaki-shi, Japan 210

Telephone: 044-511-2111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiTaO ₃ , Lithium Tantalate	C _z	65φ x 120	"SAW grade" quality. Purity: 4N except Rh, Rh < 150 ppm. X-axis grown boules. Surface acoustic wave (SAW) properties, piezoelectric properties and crystal homogeneity were studied for SAW, piezoelectric and pyroelectric devices applications. Proc. Ferro. Material appl. 1 (1977) 81 Proc. Ferro. Material appl. 1 (1977) 87	+
LiNbO ₃ , Lithium Niobate	C _z	50φ x 100	Optical quality. Purity: 5N. X, Y and Z axis grown boules. Piezoelectric, ferroelectric and mechanical properties were studied for high temperature acoustic sensor, AE sensor and piezoelectric devices applications.	+
	EFG	20x100x1.5	Purity: 5N pure and Cr-doped. Plate crystal, plate normal and pulling direction; (100)<001>, (001)<210>. Piezoelectric properties and impurity distribution were studied. X-ray and optical characterization were carried out. Mat. Res. Bull. 10 (1975) 801 J. Cryst. Growth 34 (1976) 350	-

407

Name(s): T. Fukuda

Inst. & Address: Toshiba Research and Development Center, Toshiba Corporation,
Komukai-Toshibacho 1, Saiwaiku, Kawasaki-shi, Japan 210

Telephone: 044-511-2111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
LiTa _x Nb _{1-x} O ₃ , Lithium Tantalate-Niobate	LE	15x30x0.005	LiNbO ₃ film on LiTaO ₃ substrate. Plate normal; (100) and (001). The film is grown by CLE technique using LiVO ₃ flux. Evaluation of the film is carried out by X-ray and optical waveguiding measurements for optical waveguiding and SAW applications. Appl. Phys. Lett. 28 (1976) 575	-
	C _z	20φ x 50	Purity: 5N. X, Y and Z axis grown boules; X = 0.025, 0.05, 0.1, 0.2, 0.7 and 0.95. SAW and crystallographic properties were studied. Compositional homogeneity is examined by X-ray and chemical analysis.	-
	EFG	20x40x1.5	Purity: 5N. Plate crystal, plate normal and pulling direction; (100)<001>: X = 0.05, 0.2. Compositional homogeneity are studied. J. Cryst. Growth 35 (1976) 127	-

408

Name(s): T. Fukuda

Inst. & Address: Toshiba Research and Development Center, Toshiba Corporation
Komukai-Toshibach 1, Saiwaiku, Kawasaki-shi, Japan 210

Telephone: 044-511-2111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
KNbO ₃ , Potassium Niobate	M, Sol.	40 x 40 x 20	Optical quality. Purity: 5N. "[001]" axis crystal grown by Kyropoulos technique. Dielectric, optical and Nonlinear optical properties were studied. X-ray and optical characterizations were carried out for nonlinear optical applications. Japan J. appl. Phys. 10 (1971) 507 12 (1973) 841 13 (1974) 1021 13 (1974) 1362 J. Cryst. Growth 24/25 (1974) 450 Solid State Commu. 17 (1975) 427 Izv AN SSR, ser fiz. 41 (1977) 548	

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Name(s): Kazuo Kondo

Inst. & Address: Faculty of Science, Toyama University, Toyama 930, Japan

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Au ₄ Mn	Br	12φ x 1	Au: 99.99%, Mn: 99.9% Magnetostriiction	

410

Name(s): Shigeo Okuda

Inst. & Address: Institute of Materials Science, University of Tsukuba
Sakura-mura, Ibaraki 300-31, Japan

Telephone: 0298-53-4995

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
W single crystals Mo single crystals		1 x 3 x 20	RRR ~ 2000 ~ 20000, <111> <110> <100> internal friction, elastic after effect, lattice constant and electrical resistivity after fast neutron irradi. at 5K.	-

411

Name(s): K. Matsuura, S. Kishida, F. Takeda, and I. Tsurumi

Inst. & Address: Department of Electronics, Faculty of Engineering,
Tottori University, Koyama, Tottori, Japan

Telephone: 0857-28-0321

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnS			Color centers, radiation damage, F ⁺ -center, optical absorption, luminescence	
ZnTe	M (THM)	5 x 5 x 5	p-Type, 120 cm ² /V·sec (RT) 1500 cm ² /V·sec (77K) Point defects	
CdTe	M (THM)	5 x 5 x 40	p-Type 90 cm ² /V·sec (RT), Point defect Opto-electronics	

412

Name(s): Morihiko KIMATA and Mitsuaki YANO
 Inst. & Address: Dep. Elect. Engin. Waseda Univ.
 4-170 Nishikubo, Shinjuku, Tokyo, Japan.

Telephone: 03-209-3211
 (264)

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
InAs GaAs InGaAs	MBE	10x10x ~10 ⁻⁴	purity ; $ N_D - N_A \sim 10^{16} \text{ cm}^{-3}$ orientation ; on (100) and (111) substrate dopant ; non, Mg, Mn.	
GaSb GaSbAs	MBE	10x10x ~10 ⁻⁴	properties studied ; by RHEED, X-ray, SEM and electrical measurement purity ; $ N_A - N_D \sim 10^{16} \text{ cm}^{-3}$ orientation ; on (100) and (111) substrate dopant ; non, Te. properties studied ; by RHEED, X-ray, SEM, electrical and optical absorption measurements	

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Name(s): katsuhiro Okuyama

Inst. & Address: Dept. of Electronic Eng., Faculty of Engineering, Yamagata
 University, 992 Yonezawa, Japan

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Tellurium	VP	100-2000 Å thick films	Evaporated polycrystalline films, fabri- cation techniques and electronic proper- ties. 1. "Effect of Au nucleation centers and deposition rate on crystallinity and electronic properties of evaporated Te films", J.Appl.Phys. 46(1975)105. 2. "Grain growth of evaporated Te films on a heated and cooled substrate", J.Appl.Phys. 46(1975)1473. 3. "Low-temperature grain growth of evapo- rated Te films", Japan.J.Appl.Phys. 15 (1976)1807. 4. "Effect of predeposited Ag,Sn,Al,Bi and Se nucleation centers on the grain growth of evaporated Te films", Japan. J.Appl.Phys. 15(1976)1881. 5. "Annealing effect in evaporated Te films", Japan.J.Appl.Phys. 16(1977)1571. 6. "Tellurium films evaporated on oriented Au islands", Japan.J.Appl.Phys. 16(1977) 1753. 7. "Behavior of metal contacts to evapo- rated Tellurium films", Thin Solid Films 30(1975)119. 8. "Mobility studies of evaporated Tel- lurium films", Thin Solid Films, 33 (1976)165.	

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Name(s): Tetsuro Ishida, Takashi Matsumoto, Takamasa Kato

Inst. & Address: Department of Electronic Engineering, Yamanashi University, Kofu, Japan

Telephone: 0552-52-1111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
ZnS(Zinc Sulfide)	GE	10x10x0.02	Grown on (111), (100) Ge and (100) GaAs substrates by Zn-S-H ₂ CVD.method. Japan. J. appl. Phys. <u>17</u> 227 (1978)	-
CdSnP ₂ (Cadmium Tin Diphosphide)	Sol	10x5x1	Prism-like crystals with [111] prism axis. Luminescent and electrical properties.	-
InGaP(Indium Gallium Phosphide)	Sol	10(dia)x5	Hall mibility of Te-doped In _{1-x} Ga _x P at 300K. Japan. J. appl. Phys. <u>13</u> 1481 (1974) Luminescence of Zn diffused In _{1-x} Ga _x P in the direct transition region. Japan. J. appl. Phys. <u>15</u> 2265 (1976)	-
	LE	5x5x0.1	Grown on (111) GaP and (100) GaAs substrates. Luminescent and electrical properties.	

414a

Name(s): H.SATOMURA, K.TSUSHIMA* and K.UEMATSU

Inst. & Address: Department of Physics, Waseda University, Ohkubo Shinjuku-ku, 160 Tokyo, Japan Telephone: 03 209 3211

* Broadcasting Science Research Laboratories of NHK, Kinuta Setagaya-ku, 157 Tokyo, Japan *03 415 5111

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Co ₂ GeO ₄ COBALT GERMANATE	Cz	15(d) , 10	Fe or Ni < 0.01 %, growth direction : parallel to the c-axis, magnetic properties studied, a = 8.3175 Å, J. Crystal Growth 39 (1977) 241.	-

415

Name(s): G. Aguilar S., J. Boldú O., E. Muñoz P., H. Murrieta S., and J. Rubio O.

Inst. & Address: Inst. de Física, UNAM, P. O. Box 20-364, México 20, D.F.

Telephone: 548-81-92

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
RbCl : Eu ²⁺	Br	10x10x10	Purity of starting material: reagent grade Dopant: EuCl ₂ 0.5 wt.%; in crystal ~ 130 ppm Eu ²⁺ Property studied: EPR spectrum of Eu ²⁺ Characterization: single crystal cubic Publication: Electron paramagnetic resonance of Eu ²⁺ in orthorhombic sites of RbCl single crystals, by J. Rubio O., E. Muñoz P., and G. Aguilar S., J. Chem. Phys. 61(12), 15 Dec. 1974.	-
KBr: Eu ²⁺ NaBr: Eu ²⁺	Cz "	10x10x10 "	Purity of starting material: reagent grade Dopant: EuCl ₂ 0.5 wt.%; in KBr crystal ~ 115 ppm Eu ²⁺ , in NaBr ~ 100 ppm, by at. ab. spectropho- tometry Property studied: EPR spectra of Eu ²⁺ Characterization: cubic single crystals Publication: Studies by electron paramagnetic resonance of K- and Na-bromides doped with Eu ²⁺ , by G. Aguilar S., H. Murrieta S., J. Rubio O., and E. Muñoz P., J. Chem. Phys. 62(4), 15 Feb. 1975.	+ +

416

Name(s): G. Aguilar S., J. Boldú O., E. Muñoz P., H. Murrieta S., and J. Rubio O.

Inst. & Address: Inst. de Física, UNAM, P. O. Box 20-364, México 20, D.F.

Telephone: 548-81-92

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NaI: Eu ²⁺ RbI: Eu ²⁺ KI: Eu ²⁺	Cz " "	10x10x10 " "	Purity of starting material: reagent grade Dopant: EuCl ₂ 0.5, 0.2, 0.06, and 0.003 wt% in the melts of all three hosts Property studied: EPR spectra of Eu ²⁺ Characterization: cubic single crystals Publication: Electron paramagnetic resonance of Eu ²⁺ in the alkali iodides, by J. Rubio O., H. Murrieta S., E. Muñoz P., J. Boldú O., and G. Aguilar S., J. Chem. Phys. 63(10), 15 Nov. 1975.	- - -

419

Name(s): ANDRZEJ WOLKENBERG

Inst. & Address: Institute of Electron Technology CEMI, 02 668 Warsaw, Poland

Telephone: 435401-224

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Si - Silicon	Sol	normal wafers or specially prepared samples	Electronic grade; 100, 110, 111; concentration, oxide quality, trap populations, surface state charge determination by electrochemical and electronic methods; barrier height measurements	-
SiO ₂ - silicon dioxide				-
CdF ₂ - cadmium fluoride				-

420

Name(s): J. Z. Damm, T. Morawska-Kowal

Inst. & Address: Institute for Low-Temperature and Structure Research, Polish Academy of Sciences, Department of Lattice Defects Physics, Plac Katedralny 1, 50-950 Wrocław, Poland

Telephone: 22-10-71

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NaCl, sodium chloride	Br	~ 25 dia, ~ 100 long	High-purity (OH ⁻ free, ~ 10 ⁻⁶ mole fraction Me ²⁺) or doped with Me ²⁺	
KCl, potassium chloride	Br	"	Materials used in examination of defect-sensitive optical, electrical, and mechanical properties	

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Name(s): R. Horyń, E. Pietrzyk

Inst. & Address: Institute for Low-Temperature and Structure Research, Polish Academy of Sciences, Special Materials Laboratory, Plac Katedralny 1, 50-950 Wrocław, Poland

Telephone: 22-10-71

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Pb ₅ Ge ₃ O ₁₁ , lead germanium oxide	Cz	15-20 dia, 20-30 long	Ferroelectric and electrooptic material Starting materials (PbO, GeO ₂ , SiO ₂) spectrographic grade purity; the crystals contain ~ 0.01-0.001 wt% Pt.	
Pb ₃ GeO ₅ , lead germanium oxide	Cz	"	Growth direction: c-axis	
Pb ₅ Ge ₂ SiO ₁₁ , lead germanium silicate	Cz	"	Crystals used in x-ray structure, thermal expansion, dielectric permittivity, and electric conductivity measurements	

422

Name(s): A. Modrzejewski

Inst. & Address: Institute of Nuclear Research, 05-400 Świerk, Poland

Telephone: 798644

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Pb, Lead	Br	∅100, h 200 200x90x15	4N, random, basic research 4N, at request, flat neutron monochromators	+
Zn, Zinc		∅100, h 200 200x90x14	4N, at request, basic research 4N, at request, flat and focusing neutron monochromators	
Al, Aluminium		∅80, h 150 150x90x10	4N, random, basic research 4N, at request, flat and focusing neutron monochromators	
Cu, Copper		∅110, h 200 180x90x10	4N, at request, basic research 4N, at request, flat and focusing neutron monochromators	
Sn, Tin		∅70, h 100	4N, random, basic research	
Mg, Magnesium		∅70, h 100	4N, " , " "	
Ag, Silver		∅50, h 70	4N, " , " "	
Bi, Bismuth		∅150, h 200	4N, " , " " , neutron filters	
Ni, Nickel		∅50, h 80	3N, " , " "	
Co, Cobalt		∅40, h 60	2N7 " , " "	
α Brass		∅30, h 60	4N " , " "	
Ni-Fe Alloys		∅50, h 80	Ni-3N, Fe armco, random, basic research	
Co-Fe 8 at. %		∅60, h 80	Co-2N7, Fe armco, random; oriented plates for neutron polarization	
Ni-Co Alloys		∅50, h 70	Ni-3N, Co-2N7, random, basic research	
Other Alloys at request				

423

Name(s): Janusz BERDOWSKI

Inst. & Address: Department of Physics, Silesian Technical University, Gliwice Telephone: 31-36-68
ul. B. Krzywoustego 2, Poland

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
NH ₄ (1-x)D ₄ xH ₂ (1-x)D ₂ AsO ₄ Deuterium Ammonium Dihydrogen Arsenate /DADA/	Sol	paraelec- tric phase 16x12x80 antiferro- electric phase 15x12x140	The paraelectric DADA crystals contained from 0 to 98% deuterium. The antiferroelectric DADA crystals con- tained 98% deuterium. Studies an electric, electrooptic, acoustooptic properties.	+

424

Name(s): Z. Bojarski, M. Surowiec, K. Wokulska, Z. Wokulski

Inst. & Address: Institute of Physics and Chemistry of Metals, Silesian University Telephone:
40-007 KATOWICE - POLAND

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
α- Fe whiskers	VP	0,5x0,5x40	1/Acta Phys. Polonica A39/2 /1971/ 189	+
Cu whiskers	VP	0,2x0,2x40	2/XI Intern. Congress of Crystallography August 3-12, Warsaw 1978	+
Ag whiskers	VP	0,05x0,05x10	Collected Abstracts S223, ibid S250	+
Co whiskers	VP	0,2x0,2x10	3/Sc. papers of Silesian Univ. 1 /1976/ 21	+
Cd whiskers	Sub	0,02x0,02x5	4/Kristal i Technik 4 /1976/ 399	+
Zn whiskers	Sub	0,02x0,02x5	5/Acta Phys. Polonica A39/2 /1971/ 189	+

425 Name(s): Andrzej Ciebarczyk Inst. & Address: Uniwersytet Śląski, Instytut Fizyki, 40-007 Katowice, Poland Telephone: 588211				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GdMn ₂ - GdAl ₂	M	globule diameter about 5mm	Purities of the constituents Gd, Mn, and Al were 99.5, 99.99 and 99.999% respectively. X-ray analysis of the polycrystals obtained did not reveal the presence of any foreign phase. The X-ray structural analysis has revealed that the samples were homogeneous. The character of the magnetic ordering of GdMn ₂ -GdAl ₂ results from the influence of the free-electron concentration and ordering on crystallographic sites (magnetic measurements and X-ray analysis). X-ray emission band spectra (SXS) of manganese, aluminum, and gadolinium have been investigated. The transfer of the 3d electrons of Mn to the Gd atom was observed. This suggests an ionic character of chemical bonding in the compound investigated. Publications: J. Less-Common Met. 57, 567 (1978); 59, 1 (1978) Phys. Stat. Sol. (a) 36, K171 (1976); 46, K29 (1978) Inst. Phys. Conf. Ser. No. 39, 393 (1978) Phys. Stat. Sol. (will be published) All polycrystalline samples were prepared by arc melting in an argon atmosphere and subsequently annealing at 1100K	

426 Name(s): E. Burzo Inst. & Address: Central Institute of Physics, P.O.Box 5206 Bucharest-Romania Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Rare-earth intermetallic compounds	M		Characterization by magnetic measurements, Mössbauer effect, magneto-resistivity, EPR, FMR.	-
Metallic solid solution of transition elements	M			-
Magnetic glasses	M			-
Binary systems based on V ₂ O ₅	M			-
Superconducting materials	M			-

427 Name(s): I. Ursu Inst. & Address: Central Institute of Physics, P.O.Box 5206 Bucharest-Romania Telephone:				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Uranium compounds			Characterization by EPR, NMR, FMR	-
Paramagnetic centers in ionic crystals (irradiated and unirradiated)	Cz			-
Ferroelectrics	S			-
Intermetallic compounds	M			-

428				
Name(s): Prof. J.S. Vermaak and Co-workers				
Inst. & Address: Department of Physics, University of Port Elizabeth, P.O. Box 1600, Port Elizabeth, 6000, R.S.A.			Telephone: 27961	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
GaAs/GaAlAs Single and Double Hetero- structures	LE	10 x 10 x 0.5	Dis. dens. $< 10^3/\text{cm}^2$; (001) substrate orientation; Various p- and n-type dopants; Electrical, optical and mechanical properties; TEM, SEM, X-rays, Auger, Optical and I.R. Birefringe microscopy, photolumi- nescence characterization: LED, Laser, solar cells; J. Appl. Phys. <u>48</u> , 3008 (1977); <u>49</u> , 1149 (1978); <u>49</u> , 2975 (1978); <u>49</u> , 1173 (1978); J. Cryst. Growth <u>42</u> , 132 (1977)	-

429				
Name(s): YEONG DER YAO				
Inst. & Address: Institute of Physics, Academia Sinica, Taipei, Taiwan, R.O.C.			Telephone: 7614170 Ext. 284	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\text{Ni}_{1-x}\text{Bi}_x$ ($x \leq 50$ at.%)	M	2x2x20	to be published	+
$\text{Bi}_{1-x}\text{Sb}_x$ ($x \leq 50$ at.%)	Z	2x2x20	to be published	+
$\text{Ni}_{1-x}\text{Cu}_x$ ($x = 0, 4.6, 9.3, 14.0$ and 18.8 at.%)	M	2x2x15	to be published	+
$\text{Ni}_{1-x}\text{Ti}_x$ ($x = 0, 0.5, 0.8, 1.0$ and 1.5 at.%)	M	3x3x30	Chinese Journal of Physics <u>15</u> , 1 (1977)	+
Al-B (containing 60 vol% uniaxial boron fibres)		2x2x15	Fibre Science and Technology <u>10</u> , 313 (1977)	-
$\text{Pd}_{1-x}\text{Ag}_x$ ($x = 30.0, 34.8, 40.0, 44.3$, and 49.9 at.%)	M	1x1x20	Physical Review B <u>15</u> , 2429 (1977)	-

430				
Name(s): Sung-Shan Jao				
Inst. & Address: Institute of Nuclear Energy Research, P.O.Box 3-11, Lungtan, Taiwan, R. O. C.			Telephone: (02)3814014 X. 2231	
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. NaCl, KCl, KBr, $\text{KBr}_{0.33}$ + KCl _{0.67}	<u>Cz</u>	130 * x 50 100 * x 80 75 * x 50	Publication in "Nuclear Science Journal" Vol. 13, No. 3, Sept. 1976	-
2. Ge, Si	<u>Cz, Z</u>	20 * x 60 25 * x 80 30 * x 70	Publication in "Nuclear Science Journal" Vol. 13, No. 3, Sept. 1976	-
3. Al, Cu	<u>Br</u>	30 * x 90 25 * x 50 20 * x 50	Publication in "Material Science Quarterly" Vol. 8, No. 3, Sept. 1976	-
4. SiO ₂	<u>Hyd</u>	40x30x10 40x25x5 40x15x10	To instruct a graduate student as a thesis for the degree of master of science	-

431 Name(s): Y. C. Chiou, S. Chang, S. Y. Lee, L. C. Chen, H. J. Yu, W. S. Feng, Y. N. Lai Inst. & Address: National Taiwan University, Dept. of Electrical Eng., Taipei, Telephone:(02)3415251 Taiwan, Rep. of China				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
1. Thick-film		30x50x1	Properties studied, characterization, application.	
2. GaAs _{1-x} P _x on GaAs, GaP & Si studies	VPE	15x10x0.1	Purity, properties studied, characteriza- tion.	
3. Ga _{1-x} Al _x As on GaAs studies	LE	10x10x0.1	Purity, dopants, properties studied, charac- terization.	

432 Name(s): C. Y. Sun Inst. & Address: Union Industrial Research Laboratories, Industrial Technology Telephone:(035) 245141 Research Institute, Hsinchu, Taiwan 300				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Gallium Phosphide Single Crystals GaP on GaP wafers	Cz LE	5 cmφ x 10 cmL	five 9's, {111}, Sulfur doped.	- -

433 Name(s): Jugoslav Todorović Inst. & Address: "Boris Kidrič" Institute of Nuclear Sciences, Belgrade, Vinča Telephone: Yugoslavia				
MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Mn _{1.98} Cr _{0.12} Sb	Cz	34x9x5	a=4.08 Å, c=6.52 Å Mn _I (2a), Mn _{II} (2c) z=0.295, Sb(2c) z=-0.28 T<318K, Antiferromagnetic (a,a,2c) Magnetic moments parallel to the (001) plane T>318K, Ferrimagnetic (a,a,c) Magnetic moments parallel to the c-axis Magnon soft mode (Solid State Comm. 21, 919(77) Antiferromagnetic incommensurate su- perlattice at 290K<T<318K (to be publ.)	+

434

Name(s): M. Stubicar

Inst. & Address: Institut za Fiziku Sveucilista, Bijenicka Cesta 46, Postanski Pretinac 304, Telephone:
41001 Zagreb, Jugoslaviya

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
Aluminum (Al)	Strain Anneal	50 x 1 mm (diam.)	J. Inst. Metals 98, 64 (1970)	

435

Name(s): H. Pollak § Ph. Le Docte

Inst. & Address: Université Nationale du Zaïre, Campus de Kinshasa
Département de Physique, B.P. 190 Kinshasa XI (Zaïre)

Telephone:

MATERIALS PREPARED OR STUDIED (Formula & Technical or Common Name)	METHOD*	THREE MAXIMUM DIMENSIONS (mm)	COMMENTS: Purity, orientation, dopants, properties studied, characterization, applications, publications, etc.	Material Avail- ability + -
$\left[\text{Fe} \begin{smallmatrix} (\text{RCOO}) \\ 3 \end{smallmatrix} \right] \begin{smallmatrix} (\text{RCOO}) \\ 6 \end{smallmatrix} \begin{smallmatrix} (\text{RCOO}) \\ 3 \end{smallmatrix}$ with R = CH₃ C H CH 6 5 2 Cl CH 2 Cl CH 2 F C 3 Cl C 3 Br C 3	SOL	2x2x3 mm	° X-ray studies ° Mössbauer-studies : quadrupole splitting as a function of inductive effects (Taft's σ)	+

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